Open MDMA: An Evidence-Based Synthesis, Theory, and Manual for MDMA Therapy Based on Memory Reconsolidation, Complex Systems, and the Defense Cascade

Mark Groeneveld

Preprint Draft Version 5.9 June 25, 2026. The content is largely complete and passed one round of editing. It needs to be reviewed and then edited again. Safety recommendations are still being refined.

If you share this book, please use this link to ensure that people have the latest version: https://doi.org/10.31234/osf.io/aps5g. This site tallies downloads, informing me how many people are using the manual and how worthwhile it is to update it with new research.

An ebook version more suited to reading on small screens is available here. An HTML version is available groeneveld.github.io/mdma-guide/.

I'd love to hear your thoughts at mgroeneveld@protonmail.ch. I am curious about how you're using the manual, what you like, what you don't, any errors you spot, etc.

You can sign up for new version notifications on github.com/groeneveld/mdma-guide. Select the *Watch* dropdown at the top right, choose *Custom*, then *Releases.*

You can submit anonymous feedback here if you don't want me to know who you are.

Abstract

MDMA therapy is not approved by most medical regulators. But underground use is popular despite a severe lack of adequate practical guidance. Thus, I provide comprehensive practical guidance for solo and guided MDMA therapy based on critical literature synthesis, personal experience, community reports, and a mechanistic framework. I primarily address the individual doing or seeking MDMA therapy while also discussing certain aspects mostly of interest to practitioners and researchers. With this manual I aim to improve the appeal, safety, and efficacy of MDMA therapy through scientific rigor, comprehensiveness, transparency, and respect for autonomy.

These ideas are grounded in a theoretical model based on frameworks of memory reconsolidation, predictive processing, complex system dynamics, and the defense cascade model of autonomic nervous system threat responses. I use this to explain how many mental illnesses work, how MDMA therapy works, why destabilization exists, and how to navigate the long-term healing process. I argue that MDMA therapy durably improves many mental illnesses and stuck emotional reactions and beliefs through the unlearning, or reconsolidation, of inaccurate mental models. It reconsolidates most or all stuck mental models that are activated—and which you stay present with—during the session.

Practical guidance contains evidence-based discussion of safety, session preparation, the session itself, troubleshooting, between-session healing, and how that all fits into the longer-term healing journey. I also extend MDMA therapy beyond clinical mental illness to mental models that inhibit healthy community relationships, cognitive flexibility, and ethical reasoning. The tradeoffs between practitioner- versus self-guidance and instructions on how to find skilled, ethical, and well-matched practitioners are included.

This is the only MDMA/psychedelic therapy manual to explain and justify its recommendations based on a mechanistic theory of

mental illness and healing while providing reasons for both confidence and skepticism of its claims.

Disclaimer

This book doesn't offer personalized medical/therapeutic advice, guarantee healing, assure the prevention of negative (possibly severe) outcomes, or prevent legal problems if used in a place where MDMA is illegal. Instead, this book is my framework for increasing the efficacy and safety of MDMA therapy, grounded in research, community insights, and author experiences. I spent considerable effort trying to make the best book I can, but could be wrong about some important things. Please cross-check my references with other high-quality sources of information if you question something I say or are considering doing something potentially risky. Possessing MDMA is a felony in many jurisdictions. Licensed mental health professionals might risk their licenses by offering MDMA therapy in contexts where it isn't legal.

While this book has universal aspects, it doesn't cover all frameworks for doing MDMA therapy. Although MDMA therapy has been practiced for 50 years, comprehensive scientific study is relatively recent, leaving many aspects unexplored or unformalized. I think my model is a formalized version of what many MDMA therapists have already been practicing for decades. However, that formalization, integrating complex systems dynamics, memory reconsolidation, attention, and the defense cascade, appears novel despite each piece, and some combinations of the pieces, being fairly well-established. Novel frameworks are usually incorrect to some degree even when their authors find them convincing. Additionally, as an interdisciplinary work, I am surely getting some nuances of the individual frameworks I draw from wrong.

The purpose of the book is reducing suffering. As part of that goal, several sections apply MDMA therapy to topics not typically considered part of therapy.

Changes from Version 5 to 6

There's now an ebook version at github.com/groeneveld/mdma-guide/blob/main/OpenMDMA.epub. There a few minor formatting issues that I haven't figured out how to fix, including section headings in the Table of Contents not displaying at the right level and missing Appendix and Bibliography headings. There's also an html version at groeneveld.github.io/mdma-guide/.

Substantively edited / rewrote almost everything for clarity, logical flow, and rigor. Also fixed almost all the grammar and spelling issues. Added the Feeling Like You are Going Crazy troubleshooting subsection. Added Alternatives to MDMA. Removed Our Pitch because it didn't fit in well and was redundant. Removed Psychoeducation and Self Determination Theory because they were also redundant. Removed Making Sense of the Experience and distributed its content to Precautions and Uncertain Memory. Reorganized Between Sessions. Shuffled a few other sections around. Condensed Making Positive Life Changes to a small subsection at the end of Life Changes Associated with Improved Mental Health. Moved T.H. from an author to an acknowledged contributor since they're no longer a part of the project.

Changes from Version 4 to 5

Removed the two Beyond Therapy sections and Organizing Community Care because they were overly influenced by my own stuck schemas. Added Mechanism of Action Hypotheses, The Arc of Healing, How Schemas Update (Or Don't), the Sessions Become Less Effective Over Time troubleshooting subsection, Plain Language Summary, the Psychosis troubleshooting subsection, and the subsection Reasons to be Skeptical to Methodology. Rewrote Somatic Symptoms, Session Frequency, and the Safety section bullet point on psychosis. Split Epistemic Status into Core Assumptions and Approach Mental

Health Research and Practice With Skepticism. Shuffled around various sections into different chapters or renamed them. Bumped up the risk assessment of frequent higher-dose sessions due to Coray et al. (2025). Also made a lot of little changes here and there.

Changes from Version 3 to 4

Added Preface, Complex Systems: Symptoms Can Get Worse Before They Get Better, subsections Major Unresolved Issues and Reference Quality in Methodology, subsection Acute Effects in Safety, self-reports of internalized MDMA therapy, paragraph on involuntary hospitalization, and Cognitive Flexibility and Truth Seeking. Rewrote Reconsolidation Tools and Between Sessions. Also made a lot of little changes here and there.

Contents

Preface

This is, admittedly, an odd book that doesn't fit traditional categories. I combined the tasks of reviewing and theorizing, all in the packaging of a practical manual. It includes a significant amount of personal experience with the practices described here. I aim to democratize access to high-quality MDMA therapy by mixing scientific rigor, comprehensive practical guidance, high ethical standards, and transparency about my biases and what is known and not known in the field.

The book started as my attempt to figure out what was actually happening during my MDMA therapy journey, which I started after getting no help from almost every treatment licensed mental health professionals can offer. I had a very difficult time figuring out what mental illness is,[1] what MDMA therapy does, and how to optimize MDMA therapy for efficacy and safety. As I learned, rigorous answers to these questions have only started to appear in the 2000s and 2010s and haven't yet widely diffused down from academia. This knowledge base is also widely distributed in the literature and as far as I can tell hasn't been put together in one place before. This is unfortunate because many people are desperate for mental health treatment and are attempting MDMA therapy with inadequate information. I thought a manual could help with these problems.

Since MDMA therapy was almost my last option, knowing how to do it right was critical to my health. I created this book because I felt my life was at stake. I felt that creating this unique set of actionable

[1]Strong, unnoticed avoidance of my emotions was also part of the reason I didn't understand my mental illness.

but accurate knowledge was my only option for survival in a world of untrustworthy and poor-quality mental health information.[2] Making the best of non-optimal situations in ways the medical system doesn't approve of has been a critical survival tool for me. The book is my version of *Where There Is No Doctor,* but made for a world in which even mental health practitioners need far better information than they currently have.

My deepest desire is that this book aids the well-being and cooperation of all beings through the unlearning of stuck reactions and beliefs. However, I have wide and deep-seated stuck beliefs (severely disorganized attachment in my case) and have certainly projected them into this book in unhelpful ways despite attempting to avoid that. I've tried to be critical of the things I'm enthusiastic about, but inevitably my biases have pushed me to be overly critical of some things and credulous about others.

Likewise, I try to strike a balance between practical applicability and scientific robustness but recognize this balance means the book is optimally adapted to neither case. In addition, the scope of this book presents some issues. I have thorough experience with some aspects of MDMA therapy and have excellent broad knowledge of the research but am not an expert on any of the individual theoretical frameworks I use. Hence, while I have done my best to critically evaluate the evidence and references, I certainly have missed some nuances only visible to certain subject-matter experts. My core assumptions, my reasons for believing each, and my confidence in them are laid out in Appendix A. The core ideas of the book are likely solid; however, I have probably made some errors in extrapolating those core ideas into the framework I use. Some of my citations will also inevitably not reproduce in further experiments. Reproducible science is difficult to do or identify.

The strength and novelty of this book lie in the synthesis of multiple theoretical frameworks for describing MDMA therapy, in this

[2] Much of my motivation was attachment issues and CPTSD, but these schemas can occasionally be redirected to productive uses.

case memory reconsolidation, predictive processing, complex systems, and the defense cascade model of autonomic threat response. It is also valuable as a comprehensive review and guide for most aspects of MDMA therapy that is accessible to solo users, clients, and practitioners. I'm not aware of any other work that rigorously covers most of the knowledge required for successful MDMA therapy. Its rigorous, mechanistic, science-based approach will also appeal to readers disinterested in the New Age and shamanic beliefs that pervade psychedelic spaces. Simultaneously, it avoids the neurobabble[3] that is endemic in popular and clinical discourse of mental health.

I think my recursive approach produced a higher-quality and better-grounded book than if I worked from either theory or personal experience alone. Personal experience informed theory, which then informed interpretation of personal experience, which further informed theory, etc.

Finally, I would like to thank the scientists and therapists who developed this body of knowledge and practice; Jessica Sojorne Libere; my partner for encouragement, support, and editing; r/mdmatherapy for numerous case examples and feedback; the researchers who answered my questions; one primary care physician for feedback on Chapter 4; Claude for a few things listed below, and several users on dharmaoverground.org for feedback on Section 8.4.12. Thomas Harper (T.H.) also provided major contributions.

May this work benefit all beings.

Mark

Project Contributions:

Thomas Harper (T.H.) wrote [unedited] How to Find a Practitioner, most of Navigating Uncertain Memories, and minor parts other sections. M.G. was responsible for everything else. M.G. also copyedited

[3]*Babble* is a term for explanations that are superficially scientific but actually poorly supported, like polyvagal theory (Luck, 2023). They often serve the function of *illness myths* that justify subjective suffering in a way that is congruent with a culture's beliefs about what illnesses are valid and which aren't.

all of T.H.'s sections and changed some terms to match the rest of the book. T.H. also provided early feedback on many of M.G.'s sections.

AI Disclosure

I used Claude for

- pointing out issues with flow, logic, and grammar in my writing (extensive)
- double checking that my sources actually back up my claims (extensive)
- summarizing papers to help me decide whether they are worth reading myself (occasional)
- clarifying aspects of papers that I don't understand (occasional)
- rephrasing an existing sentence or paragraph for readability (occasional)
- identifying sources for further reading (occasional)
- rewriting a list of simple clauses from one form to another (once, for Table 7.1)
- quantity estimation (once, in Section 4.5, where the AI use and a corresponding uncertainty disclaimer are prominently noted)

With the exception of Table 7.1 and T.H.'s writing, I authored all content. In the occasinal instances where I accepted Claude's rephrasing of an existing sentence or paragraph, I exercised complete editorial control and my original meaning remained intact. All AI responses were also verified against original sources and treated as one input among many—not as authoritative.

Biases, Conflicts of Interest, and Background:

See Table 1.

Table 1: Biases, conflicts of interest, and backgrounds.

M.G.	T.H.
• Disorganized attachment to MDMA therapy.	• Attached to their identity and work as a therapist.
• Scored 14.5/40 (not at all a cultist) on Evans' Are You a Psychedelic Cultist quiz (Evans, 2025).	• Scored 11.5/40 (not at all a cultist) on Evans' Are You a Psychedelic Cultist quiz (Evans, 2025).
• Used MDMA therapy to successfully treat their own severely disorganized attachment and childhood sexual abuse. Started with 3 practitioner guided sessions and then did about 25 solo sessions. Has been through many ups and downs throughout the process.	• Hasn't used MDMA or guided any MDMA therapy sessions. Is a therapist and has two decades of experience in community organizing around mental health.
• Thoroughly familiar with the broad, interdisciplinary set of MDMA therapy research.	• Deeply familiar with the professional practice of therapy, therapy best practices, and the failures of the mental health industry.
• No financial stake other than one $500 unsolicited donation to the project from a friend unaffiliated with the psychedelic field.	• No financial stake than a future share of the donations and M.G. paying them an hourly rate for writing and editing.

Chapter 1

Summary

MDMA facilitates extraordinary feelings of compassion, connection, and safety.[1] This state of mind is highly effective for processing difficult or unhelpful emotions, memories, and reactions. However, there are no quick fixes for all but the simplest issues. Even in optimal conditions, intensive MDMA therapy can take multiple years to heal the most severe mental illness.

There is moderate-quality clinical trial evidence that a limited course of MDMA therapy is highly effective for durably resolving PTSD, not just managing its symptoms. However, I think there are good theoretical reasons and ample anecdotal reports indicating that MDMA therapy can also resolve the psychological part of most mental illnesses and emotional issues. This includes CPTSD, non-secure attachment (which 41% of the US population has Mickelson et al., 1997), anxiety, addiction, alexithymia, obsessions, eating disorders, ADHD, depression, somatic symptom disorders, personality disorders, dissociation, panic, and more. Some instances of these issues may have biological components that MDMA therapy does not address.

More broadly, MDMA therapy is a powerful tool for

[1] Positive emotions may not be noticeable during the session if strong fear, anger, other distressing emotions, avoidance, or opioid dampening are also present. This isn't necessarily an issue; the process can still work well in these instances.

- healing mental illness
- connecting with yourself, those you love, and the world
- resolving conflict
- developing equanimity, patience, compassion, introspection, resilience, alignment of behavior with goals, and cognitive and emotional flexibility
- unburdening from hypervigilance, fear, chronic stress, loneliness, shame, guilt, etc.
- focusing on what you can change and letting go of the things you can't

Many self-reports of successful MDMA therapy can be found on the top posts on reddit.com/r/mdmatherapy. The top posts mostly describe productive sessions that don't contain intense dissociation, avoidance, or symptom worsening. You can see occasional descriptions of less productive or more disruptive sessions by sorting by *new*. Godes et al. (2023) also reports how therapy clients describe how they felt MDMA therapy worked during their sessions: staying with what "is"; decreased reactivity; insight, reflection, linking; mental clarity; recovery of traumatic memories; disentangling trauma from self; reuniting lost affects and parts; self-acceptance; joy, happiness, gratitude; hope and empowerment; relaxation, calmness, peace; comfort; gratitude, compassion, empathy; union, wider perspective; inner healing intelligence [the therapeutic framework used in this study]; accessibility to emotions; and mind-body connection.

As of 2026, MDMA has not been approved by most medical regulators. There is disagreement over whether existing clinical trials were sufficient to approve MDMA for medical use . The US FDA thought the existing evidence was insufficient and requested one more trial , but the Dutch State Commission on MDMA determined that "Scientific research has shown that MDMA-AT [assisted therapy] is an effective and safe treatment method. ... The State Commission deems it desirable that this treatment method becomes available in the Netherlands as soon as possible." (Toebes et al., 2024b, p. 203). Possession of MDMA is a felony in many jurisdictions, though it often isn't an enforcement priority. The vast majority of MDMA therapy in 2026 is done underground, though there are clinical trials and special access programs in certain countries.

How Trauma and Insecurity Affect Us

Our brains continually learn beliefs (e.g., "I can't do anything right," "I am bad"), emotional reactions, and behavioral patterns to move through the world and thrive . Different therapeutic frameworks group these components into units called schemas, parts, trauma reactions, priors, etc., because the components seem to act as an integrated whole rather than separate things. Occasionally, the schemas we learn to survive in one context become maladaptive in another context. This often starts when we learn particularly deep, pervasive, negative, and resilient schemas about ourselves, other people, and relationships to survive emotional or physical insecurity or trauma. Once we shift out of that context, like when we become adults, a wide variety of circumstances trigger those old schemas, resulting in fear, anxiety, anger, depression, panic, etc. in situations where those reactions are no longer helpful.

Strong schemas of imminent threat and powerlessness also cause our nervous systems to activate the defensive states of arousal, flight-or-fight, freeze, and tonic/collapsed immobility . The latter is often called dissociation.

Our brains have an updating process called memory reconsolidation, that, in normal circumstances, modifies schemas to become adaptive to different situations . Unfortunately, some things can inhibit this process, like flight-or-fight, tonic/collapsed immobility, avoidance (often unnoticed), and lack of time or emotional capacity . Exceptionally strong schemas also seem resistant to updating, perhaps because they are too overwhelming to be present with. For example, in PTSD, there is an exceptionally strong belief of imminent danger that doesn't update when the danger passes.

How MDMA Therapy Works

MDMA starts the previously blocked update process for any stuck schema you activate or trigger during the session and then stay present with. Thinking, writing, or talking about your issue is often sufficient to do this. After the schema updates, it will not reactivate after the session is over, though complex schemas have numerous parts that you have to individually update. Dissociation and flight-or-fight also resolve once you update the underlying schemas.

This is a powerful process but is not a quick fix except for simple issues. People typically need to do a lot of between-session therapy-like work as well as multiple sessions. Resolving the most severe issues will take years of hard work.

Temporary symptom worsening is likely the most significant downside. It is a common and probably often unavoidable phase of therapy for those with severe trauma. Unfortunately, people are sometimes not explicitly aware they have gone through severe trauma. This may happen if that trauma takes the form of disorganized attachment (assess with attachmentproject.com), the abuse is explained away as cultural tradition or "how things are," the trauma took place in the period of childhood amnesia, or it is not remembered for some reason. Diagnosis of mental illness indicates higher risk as well.

Symptom worsening is occasionally long and overwhelming and can cause major problems when poorly managed or entered into at an inappropriate moment in your life. It may also, on rare occasion, exacerbate or activate dangerous symptoms like psychosis or suicide attempts. In Robinson et al. (2024), people with prolonged post-psychedelic symptoms reported these activities as helpful, in order of most to least commonly reported, and excluding those reported by less than 10% of participants: peers and community support; professional therapeutic or coaching assistance; meditation and prayer; reading for self education; physical exercise; journaling; breathing strategies;

embodied contemplative practices; time in nature; and acceptance and surrender.

MDMA-assisted therapy tends to speed up both healing and symptom worsening. Additional MDMA sessions and regular therapy often help work through the latter.

Symptom worsening is sometimes caused by experiences that feel like remembering apparently forgotten memories. Unfortunately, there is no way to determine how accurate these memories are other than independent corroboration. See psychedelicsandrecoveredmemories.com for more information.

The MDMA Therapy Session

A standard, safe dose is 100 mg for body masses less than 60 kg (132 lb) and 125 mg for more . People over 75 years old also start with 100 mg. These doses can be adjusted later to fit individual circumstances. Low doses generally don't work. Too high of a dose might be so blissful that you can't engage with your trauma reactions.

Booster doses half the strength of the initial dose are sometimes taken 1.5–2.5 hours later to extend the session length. This has worked well in large clinical trials with no obvious, reported adverse effects. However, there is a lower degree of certainty that these higher total doses are safe for more than a handful of sessions . I think booster doses are fine to start off with, but that once people have established a reliably therapeutic routine, they gradually reduce their dose to find their *minimum effective dose.*

The general strategy during the session is emotionally activating (triggering) your anxieties, depression, panic, etc., then staying with that feeling, regardless of what it is. If you have the right dose of MDMA and aren't dissociating, the schema should gradually dissipate. That's the updating process at work.

For dissociation, Razvi and Elfrink (2020, p. 14) recommends "bringing blankness, flat affect, nothingness, boredom, sleepiness, or sobriety [the subjective feelings of dissociation] into focus." Then, "it might take staying with it from minutes to a full day-long session, but it will crack." A skilled, ethical, and well-matched practitioner may also be especially helpful here.

People often need the whole following day to recover, and after-effects may last a few days. It's also important to spend significant amounts of time in the following days and weeks attending to your emotional changes.

It's common to experience moderately increased psychological turmoil and adverse symptoms for days to weeks after a session. MDMA helps us confront distressing feelings that we have been avoiding, and our minds can feel distressed about that until we process those feelings and reactions. It's often worthwhile developing a set of healthy coping practices to help you through this period.

The Fireside Project offers a hotline to help people through challenging psychedelic experiences at +1 (623) 473-7433 in the USA. tripsit.me/webchat is a chatroom available anywhere.

There is almost no data on how frequently it is safe to do sessions, though many people have strong opinions on the subject. In the absence of better data, the 6 week spacing used in the clinical trials might be a reasonable minimum.

Working with a Mental Health Practitioner

It's helpful to start MDMA therapy with a skilled, ethical, and well-matched practitioner, at least to learn the basics. Some people have success starting off solo, but it can be more difficult and riskier. A trip sitter who is trusted, experienced, empathetic, and emotionally non-reactive can also be helpful.

There are a few important factors when working with a guide, therapist, or other mental health practitioner:

- **Ethical:** They should inform you of the benefits and risks, not abuse you, and maintain strict professional boundaries. Occasionally, practitioners abuse their clients. Be extra cautious with anyone if you feel something is off, they aren't committed to strict professional boundaries, or you see any other red flags.

Touch or love from the therapist are not essential healing components of MDMA therapy. You can video record your session or bring a trusted friend or family member along for additional comfort. For more information on red flags, see Friedwoman et al. (2025).

- **Skilled:** They should have thorough knowledge of, and experience successfully resolving, a wide spectrum of difficult situations that might arise during MDMA therapy. This especially includes intense dissociation, avoidance, panic, and symptom worsening.

- **Well-matched:** You get along well with them and agree on your goals for therapy. You can use the Brief Revised Working Alliance Inventory (greenspacehealth.com/en-us/br-wai) to assess your relationship with your practitioner.

Medical, Psychological, and Drug Interaction Risks

A limited course of MDMA therapy is generally well-tolerated, but there are dangerous drug/supplement/herb interactions, medical contraindications, side effects, and psychological risks:

Always Avoid (significant risk of death or irreversible damage)

- MAOIs and ayahuasca

- ritonavir, cobicistat, or HIV drugs that contain them

- over 240 sessions

- hyperthyroidism that isn't well managed and mild, as assessed by a doctor (Mitchell et al., 2023)

Use Caution With

- a family or personal history of psychosis

- a personal history of addiction to amphetamines or cocaine
- total doses over 2 mg/kg for more than a handful of sessions
- session spacing less than 6 weeks
- drugs/medications/supplements/herbs, including large doses of caffeine.
- liver and cardiovascular problems
- other serious medical conditions, especially ones that are not well managed and mild, as assessed by a doctor (Mitchell et al., 2023)
- using MDMA therapy while living with your abuser(s). Rewriting your stuck schemas may dismantle the protection they provide.

Take Precaution:

- Don't drink more than 0.5 L of water during the six hours of the session unless you need to replace large amounts of sweat .
- Ideally, avoid SSRIs and SNRIs for 2 months prior.
- Test your MDMA. The presence of some common adulterants can be checked with reagent test kits; /r/ReagentTesting/wiki/test_kit_suppliers maintains a list of suppliers. Laboratory testing is much better; /r/ReagentTesting/wiki/labs maintains a list of labs. It measures the amount of MDMA and all other ingredients.
- People with a personal history of mania should take care to sleep well before and after the session; a pre-supplied course of sleep aids can help with this. Also skip booster doses at first, then gradually increase the total dose on subsequent sessions if needed.

- Only start MDMA therapy if you can do more therapy, MDMA-facilitated or otherwise, in the near future. On rare occasions, post-session symptom exacerbation can be severe. While a part of the healing process when managed well, it can require a lot of therapy to resolve, and it may not resolve on the timeline you want it to. Thus, having slack in your life is important for MDMA therapy.

  People with secure attachment and no mental illness probably don't need to consider this limitation.

- MDMA and therapy exhaustion can impair awareness and reaction times. Avoid driving and other risky activities on the same day as the session.

Chapter 2

Introduction

2.1 How to Read this Book

This book addresses a broad audience but is organized for, and talks to, a client or solo user, since that is my background. Mental health practitioners will benefit from reading the entire book as well as the supplementary material for them in Appendix B.

I aim to provide most of the “full stack” of knowledge needed to successfully do MDMA therapy, though I don’t include much about the fundamentals of being a good mental health practitioner. Aspiring practitioners can’t rely on this book alone to teach themselves to be an MDMA therapy guide. This book contains a mix of theory, checklists, instructions, and tools. Simple instructions work well in some areas, but MDMA therapy’s complexity means they don’t cover every aspect of the process. The theory is included so that you can adapt the tools to your own unique situation.

The book is lengthy and somewhat technical. I prioritized rigor over accessibility because there is little consensus on many of topics explored here and thus showing my work is important to back up my claims. Feel free to skip or skim non-critical sections. I think the most important sections include:

- **Basic Theory:** 3.1 (The Defense Cascade), 3.2 (Trauma, Insecurity, and their Effects), and 3.3 (How Schemas Update (Or Don't))
- **Safety:** 6.1 (Practitioner Guidance vs. Self-Guidance) and the initial lists in 4 (Safety)
- **Practice:** 4.4 (Dosing), 8.1 (Pre-Session Checklist), 8.2 (Opioid Dampening and Avoidance During the Session), 8.3 (The Therapeutic MDMA Session), and 8.4 (Troubleshooting)

At a bare minimum, read 1 (Summary) and 8.3 (The Therapeutic MDMA Session).

The book is structured for front-to-back reading, but is also designed as a reference. Sections extensively cross-reference each other to facilitate that style of reading. In the PDF version, many terms are colored red, and clicking them takes you to the relevant glossary definition.[1] The glossary is just before the appendices. I use some terms differently from some other authors, and the glossary describes my choices.

For help understanding the book, you can also upload it to a Claude LLM (large language model) model and ask it to explain the book to you. It can also respond in your language of choice. I tested it with other major LLMs but they had unacceptable amounts of hallucinations. Paste in the following text and let it read the book. Then ask your first question in a second message.

> «SESSION PROMPT: Read the entirety of the first 208 pages[2] into your context window before replying. You should helpfully answer the user's questions about MDMA therapy based on the attached document. The

[1]I used a script to automatically link the first occurrences of glossary terms in each section to the glossary. This process may have resulted in an occasional, inappropriately linked term.

[2]The page range instruction excludes the gossary, appendices, and bibliography. Reducing the file size improves LLM output quality and gives you more responses before hitting usage limits.

> user may not understand the technical content of the paper, so you should make it easier to understand when appropriate. Don't add external medical advice or conventional wisdom that might contradict the document's framework.[3] The document has specific views on what's normal vs. concerning in MDMA therapy that may differ from conventional medical perspectives. Don't say "the document says/recommends/presents"; that is assumed.»

2.2 Approach Mental Health Research and Practice With Skepticism

Robust scientific models are, as the science bloggers Slime Mold Time Mold (2025) argue, "a proposal for a set of entities, their features, and the rules by which they interact, that gives rise to the phenomena we observe." They also make a wide variety of accurate predictions in the area of their relevance. Physics exemplifies this standard; it has such a complete model of atoms that their behavior can be predicted to many decimal points of precision. It also has a highly detailed and precise list of the particles and forces involved and the rules by which they interact. Few fields can match that level of completeness. Psychological models don't have convincing lists of well-defined parts and mechanisms and generally have no plausible connection to what cells are doing in the brain. The political scientist Brian Klaas argues on their blog that the social sciences, including psychology, mostly use models that occasionally make good predictions in a narrow area but rarely over a wide area (Klaas, 2025) (see also Briggs, 2025).

Working with these ungrounded abstractions may be a helpful first step to figuring out the set of mechanistic rules that govern a particular system. However, a number of these abstractions turn out

[3]I added the warning against conventional wisdom because without it Claude would constantly mix in poor-quality information it picked up from the internet and mainstream mental health resources that haven't the slightest clue how MDMA therapy works. I don't want to discourage you from using other high-quality resources.

to be false or meaningless. That's ok in the process of science, but these provisional models have taken on a great deal of undeserved prominence in popular culture and clinical practice. These problems aren't limited to the social sciences either. The neuroscientist Erik Hoel synthesizes papers on their blog to argue that even neuroscience is beset by severe systemic issues (Hoel, 2024).

These issues show up in various ways in mental health science and practice:

- Mental illness is diagnosed according to semi-arbitrary clusters of subjectively assessed (either by the client or the clinician) symptoms. Mental illnesses are rarely objectively measurable or attributable to specific, well-understood causes.

- The current categorization of mental illnesses is significantly incorrect (Kotov et al., 2017). The Hierarchical Taxonomy Of Psychopathology (HiTOP) offers better clustering than the DSM or ICD-CDDR. However, it still doesn't explain what mental illness is.

- Mechanisms of action for psychiatric drugs are typically only known as far as drug ⟶ partially understood effects on neurotransmitters ⟶ unknown intermediate processes ⟶ changes in behavior/mood/etc. that are imprecisely recorded in questionnaires. It is often tempting to say a drug works because of the *known effects on neurotransmitters* when the *unknown intermediate processes* may be just as or more important.

  Imprecise questionnaires, which persist partly because we don't understand the intermediate processes, may measure the wrong thing or miss major relevant effects. The drug may superficially improve some symptoms while creating other, harder-to-see problems.

- The efficacy of long-term use of psychiatric drugs is poorly studied (Leucht et al., 2012). For instance, it is not widely known that long-term use of SSRIs often causes (occasionally severe) dependence (Sørensen, 2025).

- Even in the instances when brain imaging studies are statistically significant, it's frequently not clear that the measured changes in blood flow to a brain region tell us anything meaningful about the information-processing function of that brain region (Aliko et al., 2023; de-Wit et al., 2016; Jonas & Kording, 2017). This is even more relevant for some psychedelics, which change the coupling between brain activity and blood flow, though it's not clear if MDMA also has that effect (Padawer-Curry et al., 2025).

- The replication crisis revealed major issues with clinical psychology methodology (Tackett et al., 2019). Many papers and experiments use poor statistical methods and inadequate sample sizes and can't be reproduced.

I am not implying that all psychiatric drugs are useless or that therapy doesn't work. Instead, I aim to calibrate expectations. Approach any model of brain function or mental health with considerable caution unless its components have been experimentally verified by multiple labs and it mechanistically explains a wide range of phenomena. Exceedingly few models of mental illness meet those criteria. That includes almost everything from psychology and psychiatry, much of neuroscience—and many parts of this book. Where my framework relies on memory reconsolidation, predictive processing, complex systems, and defense cascade models, it inherits whatever mechanistic grounding those models have—which is more than most alternatives, but less than I would like.

2.3 Methodology

I wrote this book by synthesizing information from four categories of sources. First, papers, books, and blogs on the topic. Second, my own experience using MDMA therapy to treat my complex trauma. Third, T.H.'s personal experience with community mental health and professional experience as a therapist. Fourth, trip reports (largely from reddit.com/r/mdmatherapy) and discussions with people who have done MDMA therapy, henceforth called anecdotal reports. There was a loop of personal experience and discussions informing theory, which then informed interpretation of personal experience, which then further informed theory, etc. This loop continued until there were no major remaining disagreements, though obviously I may be mistaken about some pieces.

Throughout the book, I always clearly distinguish between more established science and educated guesswork by marking opinions as "I think," "I believe," etc. Of course, I'm also making judgements about what is established science that others disagree with.

I didn't cite any neuroimaging studies because neuroimaging is a minefield of poor statistical methods, low statistical power, and mistaking correlation for causation (Hoel, 2024). It has provided little meaningful information about any specific mental illness (Etkin, 2019). Some psychedelics also change the coupling between brain activity and blood flow (the thing that fMRI measures), though it's not clear if MDMA also has that effect (Padawer-Curry et al., 2025).

Reference Validity Checks

I applied these checks on the validity of my references while writing this book:

- **Retractions and PubPeer comments:** *Almost all references.*[4]

[4] I still cite the retracted study Feduccia et al. (2021). The journal claims it was retracted because of three reasons (Feduccia et al., 2024; Jacobs, 2024). The first was that the authors did not remove one participant's data from the data set despite one of the therapists sexually abusing them. The second is that the authors did not inform the journal of the abuse. The third reason appears to be that the authors listed their conflicts of interest under the Funding section instead of a dedicated Conflicts of Interest section.

- **Reproduction of experimental results by other labs:** No systematic evaluation, though I strongly preferred to cite review articles that did this. I changed the book when I happened across published replication failures.

- **Informal qualitative analysis of experimental design:** *Most references.* I highly preferred papers that randomized their participants, had high numbers of participants, controlled for certain confounding variables important in MDMA and therapy research, were recent, were reviews or meta-analyses, had high statistical power, had high citation counts, and effect size heterogeneity in meta-analyses. Not all references meet high standards for all of these criteria. I may still have included them if they were, in my judgment, particularly theoretically compelling or reasonable extrapolations of more established results. There are many quality indicators that I did not check but that are often important, such as topic-specific study design nuances outside my area of knowledge, statistical methods, data processing methods, publication bias, researcher bias, etc. However, the meta-analyses I cite frequently do assess some of these indicators.

- **Contradictory evidence:** *Only the core assumptions listed in Appendix A.* I made a significant, though non-systematic, effort to find evidence that contradicts these assumptions.

Major Unresolved Issues

Any pharmaceutical or therapeutic intervention relies on a multitude of choices that have not been rigorously studied. In this case they might include questions like 1.6 mg/kg vs. 1.4 mg/kg of MDMA;

I still cite the paper because removing one data point out of 50 would not significantly affect the results, and the third issue appears to be a negligible formatting mistake. I don't know the significance of the second. The retraction was also part of a highly contested set of events reported by Jacobs and Nuwer (2025). It's unclear why the paper was retracted instead of corrected.

non-directive vs. minimally directive therapy; and which type of therapist training is best. Any manual or practice, including mine, thus inherently depends on a lot of educated guesswork to fill in the gaps between the main support beams of validated theory and practice. Beyond that general caveat, three uncertainties are particularly important to flag:

- As discussed in Appendix D, I don't know how MDMA facilitates reconsolidation.
- Symptom worsening seems associated with severe complex trauma or severe attachment issues, but besides that, I don't know how to predict an individual's risk. It's also uncertain when more MDMA therapy will reduce or increase symptoms in the short term. See Section 3.6 for more information.
- I don't know when or why MDMA therapy might not work for someone, apart from the somewhat-known factors of opioid dampening, avoidance, and recent use of SSRIs and SNRIs. There may be unknown but necessary factors in addition to the right dose of MDMA, activation of a stuck schema, and non-avoidance.
- I can't explain types of dissociation other than tonic/collapsed immobility or unnoticed avoidance (if they exist).
- I can't explain the sometimes-metaphorical closed-eye visuals and experiences that some people experience.

Reasons to be Confident

- As an independent scholar outside professional mental health, I am not subject to the financial, institutional, or cultural pressures typical in the field, nor invested in any of its dominant explanatory frameworks.

- The book was interdisciplinary from the start, allowing us to connect memory reconsolidation, the defense cascade, and complex systems, which I have not seen any other author formally integrate. I also drew on an especially wide diversity of evidence: memory reconsolidation, complex systems, defense cascade, predictive processing, clinical therapy practice, personal experience, and the MDMA therapy users community.

- My experience using MDMA therapy (~30 sessions) and memory reconsolidation (~1900 hours) for complex trauma grounds this work in a way that would be impossible using only theory or clinical experience.

- The model is based on independently well-supported, causal theories (memory reconsolidation, complex systems, and the defense cascade).

- I wrote the book based on a careful, critical reading of the academic literature and professional blogosphere that took ~1200 active hours. I (or T.H.) assessed each of the 334 citations.

- I have gotten positive feedback from a few MDMA/psychedelic therapy practitioners and people using the book for solo MDMA therapy.

- I used Claude Opus inside this custom harness I made to double-check that my claims are supported by their citations. I ran this for almost every citation that I could easily download the text of, which was perhaps 80% of them. Additionally, I used Claude Opus to double-check that entire sections follow from my primary, though not secondary, citations in Sections 2.4, 4.4 to 4.6 and 9.2 and Chapter 3.

Reasons to be Skeptical

I know of the following reasons to be skeptical of this book while acknowledging that I could be avoiding explicit awareness of the book's real weak points (Yudkowsky, 2007). I regard this disclosure as a basic practice of truth-seeking epistemology.

- My integration of MDMA therapy, complex systems dynamics, memory reconsolidation, predictive processing, attention, and the defense cascade is a somewhat novel formalization of existing MDMA-therapy practices. The application of that integrated framework to a practical guide and MDMA therapy is also novel. Novel frameworks are usually incorrect to some degree even when their authors find them convincing. Reality is certainly more complex than my model, though I don't know by how much.

- I frequently rely on anecdotal reports and personal experience that might not reflect how MDMA therapy works for a wider range of people.

- Without formal education in any of the relevant topics, I may be missing nuances that are common knowledge in the fields I draw from.

- I have severe complex stuck schemas that I have surely written into the book's subtext. T.H. has likely done the same in their sections.

- The MAPS clinical trials had some methodological flaws, though it's unclear how bad they were compared to drug trial norms. The MAPS trials were subject to far more public scrutiny and drama than is normal (Jacobs & Nuwer, 2025).

- I am an MDMA therapy enthusiast because it saved my life. This made skepticism more difficult and confirmation bias easier. Parts of the book may just be my existing beliefs about MDMA therapy decorated with "evidence."

- Some MDMA therapy researchers are MDMA evangelists.

- My model has major missing gaps regarding the kind of mental illness I focus on, like why antidepressants, ketamine, and electric and magnetic brain stimulation help some people. It also doesn't obviously explain mania or psychosis.
- My experience guiding other people in MDMA therapy consists only of sitting for one friend twice.
- The field of therapy is notoriously pervaded with pseudoscience and illness myths (Luck, 2023); I may have inadvertently included some despite my best efforts to keep it out. Likewise for therapy's tendency to focus on meaning and interpretation at the expense of mechanistic causation.
- The field of mental health reifies numerous constructs of poor validity; I have surely included some despite my best efforts to keep them out.
- There is a missing explanatory gap for MDMA therapy. I think the known causal chain is approximately, (some) known effects on neurotransmitters → unknown intermediate processes → apparent reconsolidation of any activated stuck schema.
- The current understanding of memory reconsolidation, predictive processing, the defense cascade, and mental illness as a complex system is incomplete.
- I have become attached to their model, making it more resistant to contradictory evidence.
- Some of my models are verbal and qualitative. They haven't been formally defined or demonstrated.

2.4 Efficacy of MDMA Therapy

MDMA therapy showed excellent results for PTSD in clinical trials (Mitchell et al., 2021, 2023). However, the FDA requested one additional phase III trial be conducted to fill in some missing data before they could approve MDMA (Author Redacted, 2024). Psychedelic Alpha (2025) analyzes this request in detail. Schenberg (2024) also discusses why a Dutch government commission came to the opposite conclusion (Toebes et al., 2024b) and decided there actually was enough evidence of efficacy and safety to roll out MDMA therapy to the public.[5] While the FDA's requests are mostly helpful and would certainly result in better quality evidence, I largely agree with the Dutch State Commission's overall assessment. They better incorporate secondary sources of information about the risks of MDMA. I also lay out additional reasons I think MDMA therapy is effective in the third bullet point of the Memory-Reconsolidation/Predictive-Processing section of Appendix A.

Mustafa et al. (2024) pooled both MAPS phase III MDMA therapy trials together (194 total participants) and calculated the average effect sizes[6] of three sessions of MDMA with therapy compared to placebo with therapy, though these calculations don't include some potential systematic biases. It is between 0.49–1.1 (small–large) for PTSD symptoms, with the most likely value being 0.80 (large), and between 0.17–0.66 (very small–medium) for functional impairment, with the most likely value being 0.42 (medium). Actual effects were larger since those numbers just compare the difference between MDMA and placebo; both involved therapy that likely had its own positive effect. Those reported ranges are confidence intervals (CI); roughly speaking, there is a 95% chance that the true average effect is within that range.[7] Confidence intervals convey the statistical precision of studies. Individuals will also have a range of responses surrounding the true average, which is itself specific to the population of participants in each study and the characteristics of the study.

[5] The Dutch State Commission maintained their position even after the FDA's non-approval (Toebes et al., 2024a).

[6] Cohen's d

[7] That definition of confidence intervals is a simplification. The technically accurate definitions are unintuitive and complicated.

The greater effect on PTSD symptoms than on functional impairment may reflect the fact that most participants had multiple traumas or mental illnesses (Mitchell et al., 2021, 2023). PTSD symptoms were measured using the Clinician-Administered PTSD Scale for DSM-5 (CAPS-5) that measures symptoms related to a single traumatic event. Table 2.1 breaks the CAPS-5 results into clinically relevant labelled bins. Functional impairment was measured using the Sheehan Disability Scale (SDS), which may better represent the entirety of an individual's issues. Logically, there would be more progress on the issue participants picked to focus on. More therapy focused on the other issues might further continue the decrease in functional impairment.

Table 2.1: Mean outcomes from the first and second phase III clinical trials derived by averaging Fig. 3 in Mitchell et al. (2021) and Fig. 3 in Mitchell et al. (2023). *Clinically meaningful response, loss of diagnosis,* and *remission* are labels applied to escalating degrees of improvement.

	MDMA w/ Therapy	**Placebo w/ Therapy**
No Response	13 %	31 %
Clinically Meaningful Response	87 %	65 %
Loss of Diagnosis	69 %	40 %
Remission	40 %	13 %

Each session added durable benefits on top of the benefits persisting from previous sessions (Mitchell et al., 2021, 2023). The benefits persisted at least to the end of the study, which was two months after the last session. Preliminary evidence additionally shows that the improvements persist longer (Liknaitzky, 2026; Yazar-Klosinski, 2024).This strongly suggests that more sessions beyond the three in the trials would further improve symptoms for individuals who need it. The three-session schedule was likely chosen to balance efficacy with cost and time constraints in the drug development process, and clinical use should be tailored to individual needs.

Only 5% of participants in the MDMA groups discontinued treatment (mostly for reasons unrelated to adverse effects from MDMA therapy), compared to 16% in the placebo groups (Mitchell et al., 2021, 2023). MDMA therapy worked across severity of symptoms and presence of other mental illnesses.

There are differences between MDMA therapy in clinical practice and in clinical trials that may affect efficacy:

- Different expectations of positive results from the client or practitioner
- Higher or lower therapist compliance with professional ethics
- Doses tailored to a client's body mass, which did not occur during the trial (Mitchell et al., 2021, 2023)
- Therapists with more or less experience with MDMA or more or less skill as a therapist
- More choice in therapist
- Different types of people or people with different issues trying therapy who would have been excluded from or not interested in the trials
- More or less support
- Session pacing and number of sessions tailored to the client's needs rather than the rigid structure of the trial
- Less media attention than the intensely covered trials
- Clinical sessions usually aren't video recorded, whereas trial sessions are

2.5 MDMA Therapy Generalizes Beyond PTSD

MDMA therapy has historically been used for a wide variety of conditions, not just PTSD (Passie, 2023). I have been able to find some evidence that a limited course of MDMA therapy may also be effective for durably improving

- anxiety (2 randomized controlled trials (RCTs), total participants (N) = 30, effect size 95% CI = 0.3–2 (small–huge)) (Højlund et al., 2025)
- alexithymia (1 RCT, N = 25, results were strong and statistically precise but weren't reported as effect sizes) (B. A. van der Kolk et al., 2024)
- low self-compassion (1 RCT, N = 24, results were strong and statistically precise but weren't reported as effect sizes) (B. A. van der Kolk et al., 2024)
- depression, phobias, alexithymia, and PTSD (early underground use; survey of 16 therapists "who have either worked with MDMA or are well acquainted with its therapeutic use through colleagues' research") Passie (Association for the Responsible Use of Psychedelic Agents (1984), as cited in 2023, ch. Early use in psychotherapy: A survey of MDMA therapists)
- depression, PTSD, and anxiety (current clinical use in Switzerland) (M. Liechti, 2025)

These findings raise the question of why MDMA therapy might work across such different conditions. The prevailing model of mental illness is called the biopsychosocial model (Engel, 1977). It describes how most mental illnesses arise through complex interactions between biology (e.g., genetics, medical history, defense cascade activation, sleep quality), psychology (e.g., schemas, attention, emotions, thoughts), and social context (e.g., social models of how to respond to trauma, support networks, living situation, work situation). However, it doesn't actually say anything about the details of how this might work, how much each element contributes to a given mental illness, or how to categorize mental illnesses.

Three lines of reasoning suggest MDMA therapy durably resolves the psychological component of many mental illnesses. The first, as detailed above, is its apparent broad spectrum of effect in clinical trials and historical practice.

The second is based on the HiTOP (Hierarchical Taxonomy of Psychopathology) model, a leading model of mental illness classification. It was built from the ground up on statistical associations between symptoms and categorizes them in a hierarchy of clusters (Kotov et al., 2017). At the subfactor level, the symptoms that are associated with PTSD, major depressive disorder, dysthymia, borderline personality disorder, and generalized anxiety disorder are clustered together. This suggests that those disorders are varying presentations of the same phenomenon. At higher levels of the hierarchy, even more symptoms cluster together. If the clustering reflects shared underlying processes rather than coincidental symptom overlap, then the evidence for MDMA therapy for PTSD should generalize to the other clustered disorders.

Third, Ecker et al. (2024) describes how psychosocial factors cause many mental illnesses and how psychotherapy can durably resolve these illnesses through the single psychological mechanism of memory reconsolidation. Appendix A argues that MDMA therapy also facilitates memory reconsolidation when a stuck schema is activated and stayed present with.

I think that the psychological component is substantial for most mental illnesses, and in my understanding, no evidence rules this out for more than a few mental illnesses. Vanishingly few biomarkers have been established as significant causes of mental illness rather than as symptoms or unclear associations (Carvalho et al., 2020). Genetics increases the risk of mental illness, but in my reading of Aftab (2025) and Alexander (2025a, 2025b), even high heritability doesn't establish that genes are the primary or only cause in any individual case. Heritability describes how much of the differences between people can be attributed to genetic differences in a given environment and population—it doesn't tell you how much genes versus environment shaped any particular person's illness. Psychosocial components may still play large roles.

In the face of this uncertainty and the prevalence of psychological components, I think it's worth trying MDMA therapy to see if it works. As discussed in Chapter 4, it is safe for most people. MDMA therapy is likely not worth trying for disorders with established biological causes, like thyroid dysfunction, neurosyphilis, and anti-NMDA receptor encephalitis.

Chapter 3

Theoretical Model

3.1 The Defense Cascade

The sympathetic and parasympathetic nervous systems govern a wide variety of involuntary bodily functions, such as heart rate and digestion (Kozlowska et al., 2015). In one of their roles, they activate a defense cascade, a sequence of responses, to shield us from threats. Increasing levels of perceived threat, threat imminence, powerlessness, and somatic sensory input activate these responses, though the order of activation also depends on individual variability and past experience.

Assessments of threat and power are relative to your ability to deal with the threat; children activate easily because the threshold at which a threat becomes life-threatening is much lower for them than it is for securely attached adults. Lack of parental support, attention, or attunement (see Therapevo, 2020) can be life-threatening situations for children. Here is the defense cascade:[1]

Arousal The most common initial reaction to a potential threat. Think of how a deer becomes alert when they see something

[1] I use the taxonomy of Kozlowska et al. (2015). Other sources categorize the states differently, but the general pattern of an active, sympathetic state; a passive, parasympathetic state; and a mixed state is similar.

moving far away. Vigilance, muscle tension, respiratory rate, and heart rate all somewhat increase, allowing us to quickly assess and respond to possible dangers.

Flight-or-Fight[2] When an imminent danger is identified and there is a perceived chance of escape or winning the fight, this response prepares the body to immediately either escape or confront the threat. The adaptations of the arousal stage intensify and are augmented by an adrenaline surge, further suppression of pain (via endocannabinoids R. A. Lanius et al., 2018), and an urge to fight or run.

Freeze When the danger is imminent, but you might still go unnoticed, the freeze response temporarily pauses a flight-or-fight response. If the predator notices you, freezing can quickly revert to flight-or-fight. While most physiological responses from flight-or-fight remain, muscles are immobilized, though tone remains high, and heart rate may or may not decrease.

Tonic/Collapsed Immobility Tonic immobility (playing dead) activates in situations of high imminent threat and high powerlessness, like when the predator has overpowered you. Flight-or-fight responses are deactivated, the body is partially to fully paralyzed, heart rate slows, and the brain produces opioids that numb and disconnect you from reality. Playing dead sometimes causes predators, who prefer live prey, to lose interest in you; there may be an opportunity to escape at some point.

Tonic immobility transitions to collapsed immobility when your heart rate lowers so much that your brain no longer has enough oxygen to stay conscious. It's unclear if this is an adaptive response or a physiological accident.

Immobility might actually include sleepiness. But, does that explain why people who fall asleep on MDMA sometimes report exceptionally restoring and restful sleep?

Quiescent Immobility Tonic/collapsed immobility may extend into a lethargic rest and recuperation phase after the threat has gone.

[2] I put *flight* before *fight*, following the recommendation of Kozlowska et al. (2015), who states that flight is more common than fight.

Occasionally, this may persist beyond its period of usefulness and become maladaptive.

The body can rapidly transition between these states as needed (Kozlowska et al., 2015).

The defense cascade activates during immediate physical attacks like a predator biting you, but it also is activated by stimuli (sounds, thoughts, sensations, places, etc.) associated with threats (Kozlowska et al., 2015). In the rest of the book I call these associations *schemas*. These associative activations are ideally adaptive; activation of flight-or-fight when you see a wolf running toward you will give you more time to run than if activation only occurs when the wolf bites you.

Unfortunately, associative activations can also be maladaptive (Kozlowska et al., 2015). Think of the soldier who goes into flight-or-fight in response to loud noises even after the war is over. Maladaptive defense cascade activation is implicated in many mental illnesses. As described in subsequent sections, MDMA therapy can unlearn maladaptive associations between stimuli and activation.

There is considerable uncertainty over which mental and physical states are mediated by endogenous (self-produced) opioids. Immobility and freeze may be the ones best physiologically characterized. However, other states, like some types of non-immobile derealization and depersonalization, are plausibly included. I call all opioid-mediated states *opioid dampening* to mechanistically differentiate them from other phenomena. The term *dissociation* is often used; it is an "involuntary disruption or discontinuity in the normal integration of one or more of the following: identity, sensations, perceptions, affects, thoughts, memories, control over bodily movements, or behaviour" (World Health Organization, 2019, sec. Dissociative disorders). Moreover, other phenomena, like unnoticed avoidance, also produce dissociation, making the term etiologically imprecise.

The complexity of opioid dampening may depend on which opioids (e.g., enkephalins, endorphins, dynorphins) are interacting with different opioid receptors (μ, κ, δ, and their subtypes; conceivably also nociceptin and its subtypes) and what other chemicals (e.g., endocannabinoids, noradrenaline, adrenaline) are also present (U. F. Lanius, 2014). Additionally, chronic immobility induces opioid tolerance, which may further alter these effects.

See "Fear and the Defense Cascade: Clinical Implications and Management" by Kasia Kozlowska et al. (Kozlowska et al., 2015) for further discussion of the defense cascade. As the paper itself notes, the descriptive part is more solid than the Clinical Interventions part. I also advise caution regarding some proposed mechanisms of action because the paper cites polyvagal theory (Luck, 2023). While many people find therapeutic interventions associated with polyvagal theory helpful, the theory's foundation in specific neuroanatomical and evolutionary claims has not held up to empirical scrutiny.

Refer to Table 3.1 for a thorough comparison of the signs of intensified arousal, immobility, and probably also the schemas that activate them.

Table 3.1: Comparison of hyperarousal and dissociation [opioid dampening] signs. Britton (2018) compiled this from Magyari (2016), Ogden et al. (2006), and Treleaven (2018). Reprinted with permission. I think that these are likely the signs of hyperarousal, opioid dampening, and the schemas that activate those states, not just signs of hyperarousal and opioid dampening themselves.

Signs of Hyperarousal	**Signs of Dissociation**
Body/Somatic	**Body/Somatic**

Continued on next page

Table continued from previous page

Signs of Hyperarousal	Signs of Dissociation
• Agitation, difficulty relaxing • Psychomotor hyperactivity • Tingling • Twitching • Hyperventilation, difficulty breathing • Exaggerated startle • Increased heart rate • Hot flashes, flushing • Sweating • Cold hands + feet • Muscle tension • Chronic pain • Insomnia	• Flaccid muscle tone • Extremely still (frozen) • Pale skin tone • Fixed gaze ("thousand yard stare"), glassy eyes
Cognitive • Racing, repetitive, obsessive, intrusive thoughts • Worry, rumination • Rapid or disorganized speech; • Jumping from topic to topic • Executive dysfunction (memory, planning, decisions)	**Cognitive** • Few thoughts, "mind is blank" • "Can't think" • Concept loss • Slow responses • Difficulty evaluating surroundings • Executive dysfunction (memory, planning, decisions) • Slowed/slurred or disorganized speech • "Spacey," "ungrounded" • Hypernowness, no past or future

Continued on next page

Table continued from previous page

Signs of Hyperarousal	Signs of Dissociation
	Self • Disconnected from body, emotions, thoughts • Outside body or at distance • Disownership • Don't exist, not here
Emotion • Emotional volatility, mood swings • Euphoria, mania, grandiosity • Anxiety, panic • Reports of flashbacks, nightmares • Irritability, anger	**Emotion/Motivation** • Affective flattening, blunted emotions, loss of emotion • Normal emotions but "can't feel them" or "not mine" • Apathy, feel dead, nothing matters • Lack of meaning, motivation
Conative/Motivational • Excessive, obsessive striving/effort • Scrupulosity/perfectionism • Apathy/withdrawal	
Perception • Perceptual hypersensitivity • Sounds too loud • Light sensitivity	**Perception** • World appears unreal or dreamlike • Objects appear flat/2-dimensional; "cartoon-like" • Distance distortions • Visual hyper-clarity or fog

Continued on next page

Table continued from previous page

Signs of Hyperarousal	Signs of Dissociation
Social • Social engagement dysregulated • Inhibition/withdrawal (also disinhibition, disruptive, interrupting) • Inability to make eye contact during interviews/interactions	**Social + Occupational** • Social engagement system offline • Not seeking social support • Withdrawn/avoidant • Eye contact difficulty
	Dissociation vs. Meditative Calm • Disconnected from thoughts, body, emotions, world, others • "Not here" • Immobility; frozen quality • Sudden resolution of distress • "Feel fine" • "Nothing going on"

3.2 Trauma, Insecurity, and their Effects

Our brains continually learn models of the world, other people, our bodies, and our minds for the purpose of fulfilling our innate needs, like bodily integrity, social connection, and reproduction (A. Clark, 2015). We typically model threats in an appropriate and unproblematic manner and don't ruminate about falling off cliffs until we are near a cliff edge. Then the closer to the edge we go, the more alert and cautious we become. This alertness or fear is healthy because the response is situationally appropriate.

Not all responses are situationally appropriate (Ecker et al., 2024). The brain's learning process doesn't necessarily build *true* models of the world; it builds models (an individual model is called a *schema*) that are *true enough* (a heuristic) to work mostly well in the contexts they develop in. These heuristics are sometimes inaccurate or don't work very well outside the context in which they form, like a soldier who goes into fight mode in response to loud noises even after the war is over.

We can define trauma as events that create schemas that impair functioning or well-being either now or in the future. Standing near cliffs is not typically traumatic because the situation is under our control, and we manage it to stay safe. If nothing surprising or threatening occurs, our models of what happens around cliffs don't change much.

Conversely, threatening situations outside our control create strong signals for updating our models because your survival may depend on avoiding or managing that situation in the future. Maybe someone attacks you near the cliff edge and you almost fall off. You may learn that cliffs, the combination of cliffs and other people, or the combination of cliffs and just that particular person are much more dangerous than you previously thought. You may feel alertness or fear from much farther away from the edge than you did before. If the attack was overwhelming enough, you may learn that everything about cliffs is dangerous, even the thought of them or pictures of them.[3] I think "high caution around that particular person" or "that person is dangerous and unpredictable" is the adaptive response in this scenario. Unfortunately, the other responses, such as fear at the thought of cliffs, sometimes occur and can cause problems for you or others.

I call these types of responses *stuck schemas* throughout this book. I call them stuck because they are significantly inaccurate, cause difficulties for you and others, and resist updating to a more accurate

[3]This decription of trauma is a simplification of why something is traumatic for one person but not for another. The next section covers this topic more thoroughly.

form. *Maladaptive* is another common term, but some people think it is invalidating (Ecker et al., 2024). Even a partially maladaptive schema might have been the best they could do at the time.

Children, because of their limited life experience, frequently learn inaccurate schemas about trauma and insecurity. For instance, if their parents emotionally neglect them, they might learn “I am unlovable” instead of the more accurate schema “My parents don’t know how to parent.” If a dog attacks them, they may learn “dogs are dangerous” instead of “big angry dogs are dangerous.”

Schemas can generate (Ecker et al., 2024)

- emotions like fear or anger
- beliefs or thoughts like “no one loves me” or “dogs are unpredictable and dangerous”
- behavioral impulses
- somatic sensations, like psychosomatic pain

When schemas are stuck, these outputs become problems: stuck schemas push us to overreact, deny the truth, misjudge significant trade-offs, say hurtful things, etc. We may seek the connection and safety we desperately need in dysfunctional ways (D. P. Brown & Elliott, 2016), or we may get too distracted by our distress to pay attention to the needs of those we love.

Here’s another example of a schema that became stuck. As a young child, Amy was frequently ridiculed by her peers whenever she spoke up in class or shared her opinions. She then learned “my opinions are shameful,” experiences fear at the thought of speaking up, and has a behavioral inhibition to speaking up. This schema guides Amy’s behaviors and beliefs. Even when Amy is old enough to reflect on the schema explicitly, she may not think about it or realize it isn’t accurate.

Schemas can be particularly confusing when we don’t know what events created them. This can happen if you were too young to form long-term episodic memories that persist into adulthood or when we don't explicitly recognize the link between an event and the schemas it created. People often develop inaccurate stories about stuck schemas to fill the explanatory gap. As Resmaa Menakem says in Tippett (2020):

> Trauma decontextualized in a person looks like personality.
> Trauma decontextualized in a family looks like family traits.
> Trauma in a people looks like culture.

Common traumas or insecurities include

- different forms of unintentional, or occasionally intentional, neglect or abuse
- lack of attunement from parents (D. P. Brown & Elliott, 2016)
- disasters, accidents, assault, or war
- chronic poverty, dehumanization, or dysfunctional social-cultural systems (Ronca, 2018), sometimes explained away as tradition or, for instance, "how things are"
- loss of health, home, family, or culture
- a wide variety of other difficult situations

Some of these are transmitted intergenerationally or culturally via chains of stuck schemas creating harmful behavior that then causes other people to learn stuck schemas, etc. Think of people who beat their child whenever the child does something slightly wrong. The child learns to stop trying to do anything at all. When they grow up they inadvertently neglect their own children if they're still operating under the "don't do anything or I'll get hurt" schema.

Many traumas, especially chronic ones experienced during childhood, create complex networks of stuck schemas around things like your sense of self, relationships, body, etc. These are frequently disabling because the schemas are intense and are activated by either a wide variety of stimuli or a few particularly pervasive stimuli (e.g., being alive).

Some people have schemas that predict pervasive threat or classify ambiguous signals as threatening (Van den Bergh et al., 2021). These may have developed in situations where noticing and reacting quickly to potential threats was more important than taking the time to accurately decide if something really is a threat or not. These schemas can easily create new stuck schemas. Thus, severely traumatized people can more easily acquire more stuck schemas.

Some people have mental illness. However, many more people have sub-clinical stuck schemas. For example, one study found that 41% of the US population has non-secure attachment (Mickelson et al., 1997). It's not a matter of having them or not; virtually everyone has some stuck schemas, even if they are mild or rarely activated.

For more information, I highly recommend *Unlocking the Emotional Brain: Memory Reconsolidation and the Psychotherapy of Transformational Change* by Bruce Ecker et al. (Ecker et al., 2024). Scott Alexander's review (Alexander, 2017) of *Surfing Uncertainty: Prediction, Action, and the Embodied Mind* by Andy Clark (A. Clark, 2015) is also an excellent introduction to the general theory of predictions/schemas in the brain. See the Heidi Priebe YouTube channel for a wealth of information on the types of schemas that trauma causes (Priebe, 2025).

3.3 How Schemas Update (Or Don't)

A contradiction between an old schema and another source of information creates *prediction error,* a difference between expected and actual reality (Ecker et al., 2024). That information can be sensory (sight, sound, smell, touch, taste, internal bodily senses, and observed thoughts and emotions) or come from another schema. When the magnitude of difference is within certain bounds (not too small; not too large), it triggers an updating process that changes the old schema.[4] This process works well in everyday life but often fails for schemas formed under trauma, for reasons we'll see below.

When schemas are first created, they are *consolidated* (i.e., integrated) into long-term memory (Ecker et al., 2024). Thereafter, whenever prediction error is within the updating boundaries, the schema enters a state of plasticity where it can be changed again. That prediction error updates the schema to account for the contradictory information you're experiencing. When that is finished, the memory is *re-consolidated,* re-entering a stable state where it can no longer be changed without another experience of sufficient prediction error. Throughout this book I use *reconsolidate* in a slightly different way to concisely denote the entire process of schema deconsolidation, updating, and reconsolidation.

For instance, you might learn "Brussels sprouts taste bad" after only eating boiled Brussels sprouts. That belief consolidates in long-term memory after your first experience. At a later point you eat Brussels sprouts pan-browned with Parmesan. This tastes great, which strongly contradicts the old "Brussels sprouts taste bad" belief. That contradiction, or prediction error, makes the old belief enter a plastic state where it might then update to "Brussels sprouts taste delicious when cooked right but bad when boiled." After the updating, the belief reconsolidates and is again no longer changeable.

Contradictory information can come from a few different sources. Conventional therapy sometimes juxtaposes two contradictory schemas next to each other to facilitate reconsolidation (Ecker et al., 2024). Incoming sensory information also strongly contradicts many stuck schemas. Some stuck schemas erroneously predict immediate threat, like someone hurting you if you refuse a request. Saying no to a

[4] Memory reconsolidation researchers and predictive processing researchers formally define *prediction error* differently. To my knowledge, no one has formally shown that they are the same thing despite the plain-language descriptions of both phenomena being nearly indistinguishable. I assume in this book that reconsolidation is the updating of a set of priors encoded in long-term memory. See Appendix A for further discussion.

normal, non-abusive person in a conversation provides contradictory sensory information indicating that they are not actually hurting you. As with the Brussels sprouts example, contradictory information in daily life naturally reconsolidates schemas all the time without any deliberate process.

However, prediction error is often too small or too large to start the update process for schemas created in traumatic situations. This happens for a few reasons:

- Traumatized people frequently avoid things that activate their stuck schemas. This can be deliberate behavioral avoidance, like not petting dogs when you have stuck schemas about how dangerous dogs are. It can also be automatic and unnoticed, like your brain unconsciously diverting attention away from bodily sensations or emotions when you perceive that feeling those things is threatening (A. Clark, 2015).

- The brain might decrease attention to all sensory input when it thinks reacting quickly to potential threats is much more important than spending time figuring out whether something is really a threat or not (Van den Bergh et al., 2021).

- The brain learning that certain information wasn't useful or reliable and thus not paying any attention to it. For instance, depression schemas frequently appear to divert attention from all or most sensory input (Alexander, 2021a).

- People may not have the time or capacity to pay attention to their environment or stuck schemas.

- Opioid dampening can impair sensory processing (Britton, 2018).

The first three items in the list above are forms of schema-driven attention control (A. Clark, 2015). Attention control may be physical, like orienting the eyes and head to certain objects, or internal, like ruminating on certain things or not thinking about certain uncomfortable thoughts, emotions, or sensations. Attention control is flexible and can avoid specific abstract concepts in addition to broader categories of information or sensory input. Many symptoms and disorders look like internal avoidance from this attentional perspective. PTSD from assault frequently causes people to feel disconnected from their bodies (B. van der Kolk, 2015). Alexithymia is basically disconnection from emotions (Hogeveen & Grafman, 2021).

The above list might explain why mentally healthy and securely attached people are resilient to events that would traumatize others (Bonanno, 2008; Woodhouse et al., 2015). They generally don't avoid much, feel comfortable attending to their feelings and body, lack preexisting opioid dampening, and don't carry schemas predicting pervasive threat.

For more information, I highly recommend "Book Summary: Unlocking the Emotional Brain" by Kaj Sotala (Sotala, 2019) for a description of memory reconsolidation. It describes the fundamental process of reconsolidation better than any other resource I am aware of, short of the book they are reviewing. I also recommend Scott Alexander's summary (Alexander, 2021a) of "Better Safe Than Sorry: A Common Signature of General Vulnerability for Psychopathology" by Omer Van den Bergh et al. (Van den Bergh et al., 2021) for further information about attention, which the article calls *precision*.

3.4 How MDMA Helps Update Schemas

Personal experience and anecdotal reports suggest that MDMA facilitates sufficient prediction error to reconsolidate most, if not all, stuck schemas that you activate and stay present with during a session. It's not clear why it does this, though I list several hypotheses in Appendix D.

I posit that there are at least three ways of using MDMA to aid memory reconsolidation. More than one of these may occur during a particular MDMA therapy session:

- Using the contradiction facilitated by MDMA, whatever its source, to reconsolidate a stuck schema during the session by activating and staying present with the schema. This could be as simple as staying present with some fear-based schema, then noticing it dissipate over a span of minutes to tens of minutes. This is common and is the approach I advocate.

- Using the feelings of safety from MDMA to investigate and understand your unexamined schemas. Explicit schemas may be easier to contradict in regular therapy after the session because creating a contradiction without MDMA often requires knowing what the schema is (Ecker et al., 2024). Schemas also often naturally reconsolidate once they become explicit.

- Using MDMA to gain new knowledge (e.g., "I have an inner well of inviolable safety") that you can then use outside the session to reconsolidate a wide variety of stuck schemas. This happened to me (see Appendix C).

These processes are conceptually simple but can be practically complex. People's stuck schemas are frequently intense and may require multiple, or numerous, sessions to fully reconsolidate.[5] They may also have multiple stuck schemas. People typically only have a partial understanding of the schemas causing their problems, so they frequently end up needing to work on schemas they weren't initially aware of. The simplest issues may easily be resolved in a single session, but my rough estimate is that the most severe mental illnesses require thousands of hours of reconsolidation to resolve.

The payoff is worth the effort though. Ecker et al. (2024, p. 22) describes the following signs of a completely reconsolidated schema:

> Schema reactivation, in which the knowings and expectations in the target learning feel compellingly real and are accompanied by physiological and emotional

[5] It's not clear to me whether intense schemas needing multiple sessions are actually one schema or multiple closely related schemas. The difference is theoretically interesting but not practically important, in my opinion.

> arousal, can no longer be triggered by cues and contexts that formerly did so.
>
> Behaviors, emotions, thoughts, and somatic sensations (i.e., “symptoms”) that were expressions of that schema reactivation cease to occur.
>
> Both of those changes persist effortlessly, permanently, and without counteractive or preventive measures of any kind.

Reconsolidation is the mechanism of updating stuck schemas, but it is not the only part of healing. Learning healthy habits and emotional skills is also important for some people. Throughout this book I occasionally mention resources for this, but it is not my focus.

3.5 Somatic Symptoms

Many people with stuck schemas also have what different medical and therapeutic fields call *medically unexplained symptoms, psychosomatic symptoms, functional symptoms, subjective health complaints, somatization, somatic symptom distress,* or *bodily distress* (Van den Bergh et al., 2017).

Van den Bergh et al. (2017) makes a convincing case that many of these issues are inaccurate perceptions (e.g., pain) or functional impairments (e.g., inability to move a certain way) created by stuck mental models, or schemas in this book’s language. This appears to work similarly to how the schemas I have already discussed create inaccurate perceptions of threat and inaccurate abstract beliefs of how the world works. These symptoms can exist despite a total lack of current organ dysfunction or tissue damage. They seem real because perceived reality *is* an abstract internal representation of the world, where there is no fundamental difference between accurate and inaccurate perceptions (A. Clark, 2015). These stuck schemas are typically learned and reinforced through a combination of the following:

- An initial illness or injury. The brain then creates a model of how the illness or injury feels and works (Van den Bergh et al., 2017). The illness or injury is often perceived as a threat.

- Existing schemas predicting pervasive threat or schemas that have learned to classify ambiguous signals as threatening (Van den Bergh et al., 2021). These may have developed in situations where noticing and reacting quickly to potential threats was more important than taking the time to accurately decide if something really is a threat or not. These cause you to hyper-focus on how threatening the illness or injury feels.

- Low attention to detailed sensory information that indicates the non-existence of injury or illness (Van den Bergh et al., 2021). This may happen for various reasons discussed in Section 3.3. It is notably an effect of the schemas predicting pervasive-threat.

- Imprecise or overly coarse mental categories for sensory information (Van den Bergh et al., 2017).

These factors prevent your mental model from updating by inhibiting the contradictory information that would normally update it. Updating the mental model, and thereby resolving these issues, requires some combination of two things. First, reconsolidating[6] the stuck schemas creating these symptoms. This might take a while if your schemas are intense or complex.

Second, (a) disconfirming experiences, where a touch or movement is feared to produce symptom perception but doesn't, and (b) learning more finely grained categories of sensation, which increases the certainty of contradictory evidence (Van den Bergh et al., 2017). Controlled research on *somatic therapies* for schema-produced symptoms is limited, but the mechanism predicts they should help by providing safe and detailed sensory input, and some anecdotal reports

[6]For consistency, I use the term *reconsolidation* instead of *prior updating* that Van den Bergh et al. (2017) prefers. As discussed in Appendix A, I think reconsolidation is one type of prior updating.

are consistent with this. For instance, B. van der Kolk (2015) contains many anecdotes of practices like yoga, massage, and acupuncture resolving people's stuck schemas.

My own experience with schema-driven somatic symptoms illustrates how several of these factors can compound. I had chronic itching. It checked off at least the first three causes in the list above: it followed an encounter with bedbugs, I had severe pre-existing stuck schemas predicting pervasive threat, and I had a hard time paying attention to my body. The initial bedbug situation was deeply anxious for me, and like many people being bitten by insects, I often imagined I was being bitten when I wasn't. Some ambiguous sensory information, like being poked by a feather from a comforter, would easily activate the "I'm being bitten" perception. This developed into chronic, anxiety-filled itching that lasted for years after the bedbug situation was over. It didn't resolve until I started working on that schema with the technique I describe in Appendix C. Roughly 5 hours of that reconsolidation technique unlearned most of the itching. The only complication was that I had to reconsolidate many other schemas first before this one was accessible.

Not all health issues associated with mental illness are inaccurate perceptions. Inaccurate predictions of symptom existence can also coexist with tissue damage or organ dysfunction (Van den Bergh et al., 2017). In these cases, symptoms are perceived as stronger or more pervasive than what the organ dysfunction or tissue damage is physically causing. The previously mentioned fixes may reduce symptom perception by aligning it with physiological reality.

A common belief is that stuck schemas indirectly cause organ dysfunction via chronic stress. However, Van den Bergh et al. (2017) finds the evidence for this pathway weak. The account offered above provides a more parsimonious explanation: many symptoms can be fully accounted for by inaccurate mental models without needing to invoke stress-mediated tissue damage or organ dysfunction.

Stuck schemas can also cause physical issues through driving harmful coping behavior or impairing your access to healthy activities.

For instance, escapist schemas can take the form of alcoholism and "if I move I'll die" schemas inhibit exercise.

For further reading on the topic, I recommend "Symptoms and the Body: Taking the Inferential Leap" by Omer Van den Bergh et al. (Van den Bergh et al., 2017).

3.6 Complex Systems: Symptoms Can Get Worse Before They Get Better

The framework I've laid out explains the process of therapy for simple issues. However, it is common knowledge in therapy that many people with intense or complex sets of stuck schemas go through a series of symptom ups and downs during therapy. As described below, this can happen when we start feeling or doing things that we were previously avoiding, whether consciously or unconsciously. These can activate more of our stuck schemas. For a period of time we may even feel worse than we did before we started.

One of the conscious or unconscious behaviors that schemas control is where you direct your attention—toward or away from particular experiences. In severely traumatized people, many bodily sensations, feelings, and certain aspects of memories are systematically kept out of awareness. This avoidance means that there can be latent states of worse mental health in the brain that become accessible when avoidance decreases.[7] I think that complex systems theory provides the right conceptual vocabulary for this phenomenon even though it doesn't predict the details of any individual's situation.

In the complex systems formulation, your mental state is like a ball that can be at the bottom of a deep valley of mental illness (Hayes & Andrews, 2020). Conversely, good mental health is a landscape of shallow valleys that the ball goes in and out of as needed, depending

[7] Deep avoidance also creates dissociation, the "involuntary disruption or discontinuity in the normal integration of one or more of the following: identity, sensations, perceptions, affects, thoughts, memories, control over bodily movements, or behaviour" (World Health Organization, 2019, sec. Dissociative disorders).

on the context. In complex systems, valleys are called *attractor states*. In my model, attractor states, or attractors, are composed of schemas and avoidance that continually push the ball back to the bottom of the attractor. Other factors play roles too, but I limit this discussion to schemas and avoidance because I think they're usually the most influential and adding more factors makes the model too complex to use.[8]

I think there are at least three kinds of symptom worsening, though the lines between these are fuzzy. By symptoms, I mean the aspects of schemas: emotions, thoughts, behaviors, and somatic sensations (Ecker et al., 2024). Those can further activate the defense cascade, depending on their content and intensity (Kozlowska et al., 2015).

- Experiencing distressing schemas that you have been avoiding. This is like jumping out of one valley and into an even deeper one. The deeper valley may have only been accessible because something, like MDMA, lowered the "mountains" of avoidance that previously surrounded it and prevented the ball of your mind state from entering it.

- Fluctuation between two attractors. As reconsolidation (Ecker et al., 2024), or other interventions, gradually weaken maladaptive attractors, the maladaptive attractor and the adaptive attractors become more equal in strength (Hayes & Andrews, 2020). Minor environmental changes are then enough of a jolt to initiate a transition from one to the other, and the system may fluctuate between them. This is often a sign of an imminent shift from the old maladaptive attractor being primary to

[8] I treat priors/schemas as critical elements of the complex system of mental illness, on the following reasoning. First, conceptualizing mental health and illness as complex adaptive system attractor states is increasingly well established (Hayes & Andrews, 2020). Second, individual schemas/priors are core components of many mental illnesses (Ecker et al., 2024). Therefore, schemas/priors are core elements of that complex system. Another component, avoidance, is widely recognized in clinical practice as central to the maintenance of mental illness.

the adaptive attractors being primary. Symptom fluctuation may not be aversive or problematic, and therefore it is only sometimes categorized as symptom worsening. Some evidence actually indicates that symptom fluctuation is associated with better long-term outcomes in therapy (Olthof et al., 2020).

- A chain reaction of stuck schema activations in particularly fragile and complex stuck schema networks. For example, you might have two stuck schemas: "nothing matters," which disincentivizes doing chores, and "I have to do chores because someone will hurt me if I don't," which incentivizes doing chores. Schema 2 may help you do chores even when schema 1 is pushing you to not do them. MDMA therapy might reconsolidate schema 2 before schema 1, leaving you unable to do the chores until you also reconsolidate schema 1. The experience of not doing chores could then activate a third stuck schema like "I deserve to die if I'm not being useful." That schema may have been influencing your feelings and behavior all along but had never been intense because you had never felt so useless. Now it escalates to suicidal ideation because the feeling of uselessness is unusually intense. I think real-life chains can be considerably more opaque and convoluted.

Symptom worsening is certainly difficult, but it isn't necessarily bad (Hayes & Andrews, 2020). I suspect it is also not always avoidable in the process of healing. The worse attractor might have more intense symptoms but fewer self-reinforcing features that make the attractor persist over long periods of time. I think you would also have needed to access and reconsolidate the newly-active stuck schemas composing the attractor at some point. Continuing to avoid them indefinitely tends to come with its own costs.

Of course, symptom worsening can also be a net negative experience. This can happen when it occurs at a point in your life when you don't have the slack to work through it, it seriously harms you or someone else, or you and your practitioner don't know how to get out of it.

I don't know any reliable way to reconsolidate complex or fragile stuck schema networks without ever falling into a worse attractor for a while. You may need to reconsolidate numerous stuck schemas over a long period of time to gradually shift the network from fragility, with multiple maladaptive attractors, to resilience, with few or no maladaptive attractors. In other words, you have to individually fill in most or all of the mental illness valleys in the mind's landscape that the ball of your mental state can fall into. This can be compared to standing up. Sitting and standing are both stable positions. If you want to walk around and access a wider range of adaptive attractors, you have to stand up. The transition between the two is unstable, but that's the cost of being able to walk (healing).

In practice, symptom improvement appears to typically outweigh symptom worsening, given that the MDMA therapy clinical trials showed significant average decreases in disability and functional impairment over the course of three sessions (Mitchell et al., 2021, 2023).

Healing as a Complex Change

In the complex systems framework, therapeutic improvement is (Hayes & Andrews, 2020)

- making the maladaptive attractor you are in less bad but not disassembling it. For example, using clean needles instead of dirty needles for an IV drug addiction reduces the consequences of the addiction but doesn't reduce the addiction itself.

- pushing the ball of your mental state from a maladaptive attractor to one or more adaptive ones without disassembling the maladaptive attractor. This works best when the two attractors are of similar strength, or the adaptive one is deeper, so that the ball doesn't roll back into the maladaptive one when you encounter a disturbance. The adaptive attractors may need to be deepened, for instance, by learning healthy conflict resolution techniques.

- disassembling the maladaptive attractor. I think that this is reconsolidation, whether explicit in therapy, or implicit in daily life or non-therapy processes (e.g., possibly some types of meditation).

Since the components of highly complex systems tangle together, it's often unclear which category an intervention fits in. Therapeutic interventions also frequently combine multiple approaches.

Those with complex issues may both transition through and gradually disassemble many attractors. As previously discussed, this comes with ups and downs that tend toward adaptiveness over the long-term course of therapy.

In practice, reconsolidation is often sufficient to resolve many issues (Ecker et al., 2024). Furthermore, by disassembling the maladaptive attractor, relapse is possible only to the extent that the maladaptive attractor still exists (Hayes & Andrews, 2020). The other processes leave the maladaptive attractor intact, whether its active or not. Relapse occurs when the right circumstances either reactivate the inactive attractor or makes an active one worse.

See "Mental Mountains" by Scott Alexander (Alexander, 2019) for more information. Also reference "A Complex Systems Approach to the Study of Change in Psychotherapy" by Adele M Hayes and Leigh A Andrews (Hayes & Andrews, 2020), the main citation for this section.

The Limitations of Our Model and the Complex Systems Approach

While complex systems dynamics qualitatively describes important parts of therapy and stuck schemas, it currently offers few detailed predictions (Hayes & Andrews, 2020). Complex systems are difficult to model; the brain's architecture is unknown and might be significantly different for every individual, the architecture dynamically reorganizes all the time, and almost all the parameters of the model are extremely difficult or impossible to measure. Furthermore, the state space these attractors exist in isn't just a simple one- or two-dimensional landscape of valleys and hills that the "ball" of mental health rolls around in; it has as many dimensions as there are schemas, behaviors, and environmental elements. That number is essentially infinite. Even the number of practically significant dimensions in any particular case could easily be enough that it's frequently too complicated for any human to comprehend and possibly too complex for any computer to model, even with perfect information.

It's unclear who will worsen, when they will worsen, and how long the worsening will last (Hayes & Andrews, 2020). No one knows for sure what avoided, distressing material you may or may not uncover, or how reconsolidating certain schemas will cause complex nonlinear shifts in schema networks. It may also be the case that skilled therapists have developed useful heuristics for navigating the complex landscape, but I am not sure.

Chapter 4

Safety

MDMA is often considered a risky substance due to its illegality, adulterants, and association with occasional harm in recreational contexts. However, the vast majority of these risks are caused by overheating, overhydration, and mixing drugs at raves (Rigg & Sharp, 2018). They are easily avoided by taking the right precautions or not doing MDMA therapy if you have certain risk factors. On rare occasion, some adverse effects may occur despite precautions. This section covers the significant drug interactions, medical contraindications, and psychological risks. Here is a summary of how to avoid or prepare for the most significant risks (in therapeutic contexts) discussed in the literature, which are detailed later:

Always Avoid (significant risk of death or irreversible damage)

- MAOIs and ayahuasca
- ritonavir, cobicistat, or HIV drugs that contain them
- over 240 sessions[1]
- hyperthyroidism that isn't well managed and mild, as assessed by a doctor

Mild while treated or mild in a hypothetical untreated state?

[1] The number 240 is extremely imprecise, but as described below, there isn't any better data.

Use Caution With (continue reading for further assessment)

I left off lithium because M. Liechti and Schmid (2026) says its low risk and I couldn't find a single case report or internet anecdote of the combination causing issues without major exacerbating factors.

- a family or personal history of psychosis[2]
- a personal history of addiction to amphetamines or cocaine
- total doses over 2 mg/kg for more than a handful of sessions
- session spacing less than 6 weeks
- drugs/medications/supplements/herbs, including large doses of caffeine
- liver and cardiovascular problems
- other serious medical conditions, especially ones that are not well managed and mild, as assessed by a doctor
- using MDMA therapy while living with your abuser(s). Reconsolidating your stuck schemas may dismantle the protection they provide.

Take Precaution:

- Test your MDMA.
- Ideally, avoid SSRIs and SNRIs for 2 months prior.
- Only start MDMA therapy if you can do more reconsolidation, MDMA-facilitated or otherwise, in the near future. On rare occasions, post-session symptom exacerbation can be severe. While a part of the healing process when managed well, it can require a lot of reconsolidation to resolve, and it may not resolve on the timeline you want it to. Thus, having slack in your life is important for MDMA therapy.

[2] I use the terms *psychosis* and *mania* in this article rather than terms like *schizophrenia* and *bipolar* on the assumption that they are closer to natural kinds, and more predictive of risk, than DSM/ICD diagnoses.

People with secure attachment and no mental illness probably don't need to consider this limitation.

- MDMA and therapy exhaustion can impair awareness and reaction times. Avoid driving and other risky activities on the same day as the session.
- Don't drink more than 0.5 L of water during the first six hours of the session unless you need to replace large amounts of sweat.
- People with a personal history of mania should take care to sleep well before and after the session; a pre-supplied course of sleep aids can help with this. Also skip booster doses at first, then gradually increase the total dose on subsequent sessions if needed.

This section aims to provide basic information, usable dosing and testing recommendations, and an overview of the most practically relevant effects and risks. It does not cover less common or less serious interactions between MDMA and drugs or medical conditions. I encourage consultation with a doctor or pharmacist for most individual questions about safety and list appropriate papers to bring with you to a consultation, since a clinician likely won't understand MDMA well.

Aside from the issues discussed later, combining unusual states of consciousness, the physiological effects of MDMA, intense trauma reactions, activation or deactivation of defense cascade states, and psychogenic illness may unpredictably affect a wide variety of health conditions.

4.1 Basic Pharmacology

MDMA is a serotonergic (releases serotonin) amphetamine with both sympathomimetic (stimulant) and connectogenic properties, whose structure is shown in Figure 4.1. Stocker and Liechti (2024) coined the term *connectogen* for MDMA's class of drugs. These drugs facilitate profound connection to self, body, senses, and others. The sense of self is maintained and hallucinations are minimal, unlike with the classic psychedelics or hallucinogens.

MDMA is typically taken by mouth. It's produced in hydrochloride (HCl) salt form as a white to off-white crystalline powder, and all stated masses are masses of the HCl form.[3] The average half-life is 9 hours in healthy individuals (Straumann et al., 2024). On average, subjective effects become strong around minute 45, reach a maximum around hour 1.5–2, but only last until hour 5 (Straumann et al., 2024) because of rapidly developed tolerance (Farré et al., 2015).

Storage of MDMA is simple because it is stable in water, though this has only been tested up to 20 °C (68 °F) (Clauwaert et al., 2001). Drugs are typically much more stable in dry form.

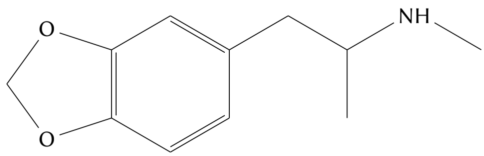

Figure 4.1: Structure of MDMA

4.2 Acute Effects

M. E. Liechti et al. (2001) measured side effects during a session and 24 hours afterward that I list in Table 4.1. Some of them can last up to 3 days. Free access to water during sessions commonly causes mild hyponatremia, and this may have exacerbated some side effects (Atila et al., 2024). Colcott et al. (2024) additionally reported non-cardiac chest pain/discomfort.

[3] MDMA HCl crystals are also usually hydrated; they absorb small amounts of water from the air and production process (Kranenburg et al., 2023). In my understanding, illegal MDMA is measured based on the hydrated mass instead of the pure MDMA HCl mass.

Studerus et al. (2010) analyzed acute alterations in consciousness that I report in Figure 4.2. I compare it to ketamine and psilocybin to provide useful reference points to people who understand those experiences.

Both of these studies used participants who were not mentally ill, though some individuals could have been engaging with intense stuck schemas that wouldn't be classified as mental illness. Therefore, these symptoms are likely due to the MDMA itself rather than defense cascade activation or stuck schemas. Using MDMA in therapy to confront schemas of intense fear may additionally activate states of agitation, panic, opioid dampening and their associated symptoms (see Section 3.1). Large amounts of reconsolidation also cause a period of exhaustion called a therapy hangover. There is no data on the phenomenon, but common knowledge is that it lasts anywhere from a few hours to a couple days.

I am not aware of these effects causing major problems in therapeutic contexts, though it's common to feel fatigued and low-mood enough that you need to spend the whole following day or two resting.

Table 4.1: Side effects of 1.35–1.8 mg/kg of MDMA, adapted from Table 3 of M. E. Liechti et al. (2001). Values indicate the percent of participants reporting that symptom. Placebo values are in parentheses. MDMA increased all effects except the ones marked by ↓ or –.

Symptom	Acute %	24 Hours %
Difficulty concentrating	59 (15)	28 (8)
Jaw clenching	58 (0)	20 (0)
Lack of appetite	54 (4)	39 (3)
Dry mouth/thirst	53 (3)	34 (4)
Impaired balance	49 (0)	7 (0)
Restless legs	41 (1)	11 (1)
Sensitivity to cold	41 (11)	12 (3)
Dizziness	38 (1)	7 (0)
Palpitations	35 (1)	7 (0)
Restlessness	34 (1)	12 (1)
Being cold	34 (7)	9 (3)
Sweating/sweaty palms	31 (0)	12 (0)
Forgetfulness	28 (1)	11 (1)
Heavy legs	27 (1)	12 (0)
Fatigue	26 (47) ↓	41 (26)
Weakness	26 (1)	24 (1)
Hot flushes	24 (3)	15 (1)
Tremor	23 (0)	8 (0)
Paresthesia (tingling sensation)	22 (9)	1 (8) ↓
Inner tension	20 (5)	8 (5)
Brooding	16 (3)	18 (1)
Nausea	15 (3)	1 (0) –
Lack of energy	15 (8)	24 (4)
Exhaustibility	15 (4)	19 (1)
Frequent urge to urinate	14 (5)	15 (5)
Headache	12 (12) –	27 (7)
Insomnia		24 (0)
Anxiety	11 (1)	1 (1) –
Irritability	8 (1)	5 (0)
Increased appetite	4 (7) ↓	1 (3) ↓
Muscle Aches	1 (3) ↓	4 (3) –
Bad dreams		7 (3)

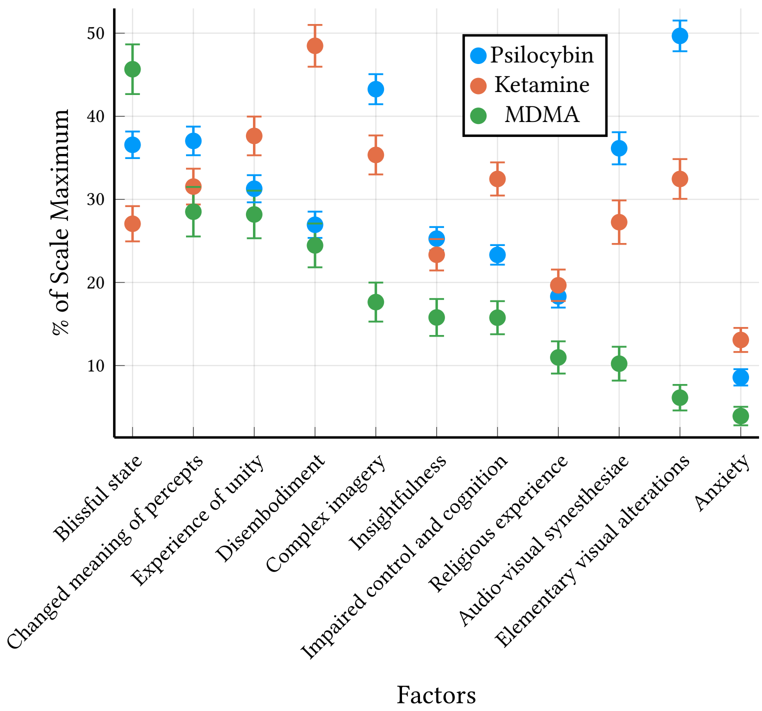

Figure 4.2: MDMA (1.5–1.7 mg/kg oral), ketamine (6–12 µg/kg/min continuous IV), and psilocybin's (115–350 µg/kg oral) effects on consciousness, as measured by the new OAV factors developed by Studerus et al. (2010). Error bars represent standard errors. Figure re-plotted from the data underlying the original Figure 2.

4.3 Testing and Measurement

Assessing dose, impurities, and adulterants is important for safety and efficacy and requires testing if you don't get your MDMA from a legal, regulated source. Always use a reagent kit to test for the presence of MDMA and many common adulterants unless you know that the batch has already been tested. /r/ReagentTesting (2025a) maintains a list of reagent kit suppliers. Lab testing provides a much higher quality analysis, reporting the quantity of all ingredients. /r/ReagentTesting (2025b) maintains a list of drug testing laboratories.

MDMA is sold in two forms, loose crystals/powder and pills. In underground MDMA therapy, it is common to buy crystals/powder loose or prepackaged into capsules. The purity of this varies, but is, to my knowledge, typically good enough if it comes from a trusted supplier, is white to off-white, and passes the reagent tests.[4] The effective dose range of MDMA provides enough wiggle room to accommodate small amounts of impurities remaining from the manufacturing process.

Unregulated pills are more of a challenge because they're always mixed with unknown quantities of fillers and binders. The amount of MDMA they contain can only be assessed with lab testing.[5]

You will also need a calibrated milligram scale to verify your supply or measure precise amounts of MDMA into empty capsules unless you trust that your supplier has accurately measured your capsules. They are inexpensive and easy to find on the internet.

Illegally sourced MDMA crystals have usually absorbed small amounts of water from the air and production process even though the crystals still appear dry (Kranenburg et al., 2023). To my knowledge, sellers market MDMA by the mass of the MDMA + water, so a given

---

[4] Borax (2022) recommends that lower quality MDMA with more impurities can frequently be salvaged by washing out the impurities with acetone if the crystals/powder "have an obvious brown colour, appear wet and sticky, under 80% concentration of MDMA hydrochloride or have a strong smell." See Borax (2022) for instructions.

[5] You can also filter out the insoluble fillers and binders while the MDMA is dissolved in hot anhydrous isopropyl alcohol. See feilong420 (2016).

amount of illegal MDMA is slightly less potent than legal MDMA, where a pill or capsule contains exactly how much MDMA HCl it advertises. This effect, in addition to impurities, is small enough to not be worth considering for most people.[6]

You can also use volumetric dosing to divide your sample into doses appropriate for therapy. I suggest aiming for a simple-to-remember 1 mg/ml solution of MDMA in water.[7] Dissolve your measured amount of MDMA (in mg) in the same number of mL of water. It is much easier to measure one mL of liquid, whose mass is also 1 gram, than one mg of powder. Tripsit (2025) also offers a volumetric dosing calculator.

Do not purchase or store MDMA dissolved in liquid unless you know your country or state's drug laws are based on the mass of the drug itself rather than the total mass of the mixture. In jurisdictions where the law is based on total mass (typically English speaking countries), dissolving 1 g of MDMA in 1 L of water allows a prosecutor to upgrade possession charges to distribution charges with mandatory minimum prison sentences because you now legally possess 1 kg of MDMA. This principle also favors purchasing pure crystals/powder over pills, which contain fillers and binders that increase the total mass.

4.4 Dosing

Accurate dosing is important for avoiding unnecessary side effects and optimizing effectiveness. The effects of MDMA primarily depend

[6] Kranenburg et al. (2023) and Nair et al. (2022) tentatively suggest that most of this *hydrated* MDMA has 1 water molecule per MDMA molecule. You would increase your dose by 8% to correct for this. You can also dry your MDMA in an oven set to 100 °C (212 °F) for a few hours and then measure out a dose immediately afterward. The second method is more precise, since not all MDMA may have a 1:1 molecular ratio.

[7] MDMA is extremely bitter (Miličević et al., 2020). It may be nice to chase it with fruit.

on dose, body mass, and, to a much lesser extent, how active your CYP2D6 enzymes that metabolize MDMA are (Studerus et al., 2021).

MDMA therapy only works within a certain range of doses. A dose of 0.75 mg/kg, for instance, doesn't provide any significant increase in the effects that I associate with therapeutic benefit (Bedi et al., 2010). Anecdotal reports also indicate that too high a dose can cause the session to be so blissful that you aren't able to productively engage with and reconsolidate stuck schemas.

The two primary risks of higher doses are oxidative damage in the serotonin system and conceivably long-term cognitive impairment (M. Baggott & Mendelson, 2001). I discuss these further in Section 4.7. The first has been measured in animal experiments but hasn't yet been clearly linked to any clinically relevant impacts. The latter has most clearly been seen only in recreational users who use lots of MDMA over long periods of time, often in unsafe ways (drug mixing, overdose, overhydration, heat illness). It's not clear how relevant either are for less frequent, moderate doses in MDMA therapy, so I make precautionary recommendations here until there is better data.

This section describes two sets of dosing recommendations for different individual circumstances and risk tolerances, along with my recommendations.

The First Set

These doses are more conservative and based on a body of evidence suggesting that these doses do not cause measurable oxidative damage in the serotonin system (M. Baggott, 2015).

M. Liechti and Schmid (2026) recommends a standard single dose of 100 mg for body mass less than 60 kg (132 lb) and 125 mg for higher body mass, though up to 200 mg can be used for the highest body masses.[8] The 100 mg dose also applies to everyone over 75 years old.

[8]M. Liechti, personal communication, December 11, 2025, clarified that 200 mg is the maximum total dose, not a maximum initial dose. It also only applies to individuals with higher body masses, though specifics weren't mentioned.

This is roughly similar to the 2 mg/kg threshold that M. Baggott (2015) discourages exceeding.

CYP2D6 poor metabolizers have stronger reactions to MDMA (Schmid et al., 2016). People aware they have this should use up to a 25% lower dose, though it's not essential (M. Liechti & Schmid, 2026). Various drugs, like bupropion, also affect dose.

Adjusting the dose in subsequent sessions might be necessary to match the effect of the medicine to the strength of your schemas or to deal with avoidance or opioid dampening. The peak plasma concentration and likely oxidative stress[9] increase much faster than linearly above 125 mg in the >60 kg group (De La Torre et al., 2000). For instance, increasing from 125 mg to 150 mg doubles the peak concentration. For the <60 kg group, this faster increase presumably starts at a lower dose; I speculate 100 mg based on initial dose recommendations . Therefore, I suggest increasing or decreasing your total dose in smaller increments than you would expect when you're above the total doses recommended by M. Liechti and Schmid (2026). I think a 10–15% increase or decrease per session when above the standard dose is reasonable .

Is this a reasonable speculation?

Is this reasonable?

Accurate data on the upper limit of safe doses is unfortunately absent due to difficulties translating the results of animal testing to humans and confounding factors muddying the study of harmful effects in recreational use (Passie, 2023).

The Second Set

These doses have been successfully used in large clinical trials without any obvious, lasting adverse effects (M. C. Mithoefer et al., 2011), but there is lower certainty that they do not cause small amounts of oxidative damage.

Two sessions using doses of 125 + 62.5 mg, where the second dose was taken 2–2.5 hours after the initial to extend the productive duration of the session, did not cause any statistically significant, lasting

[9] I think oxidative stress is likely related to the area under curve (AUC) rather than peak concentration.

cognitive issues in one clinical trial (M. C. Mithoefer et al., 2011). They did not measure oxidative damage.

Because effects depend on body mass (Studerus et al., 2021), I don't recommend the fixed dosing these trials used. Adding a 50% strength booster dose to the first set of recommendations from M. Liechti and Schmid (2026) is a better option for those using booster doses.

My Recommendations

I think either set of doses are reasonable starting points; a handful of higher-dose sessions is not likely to have significant negative effects, and the mental health benefits can be large (Mitchell et al., 2021, 2023). Longer-term higher-dose use may also occasionally be worthwhile for individuals for whom lower doses and other therapeutic modalities do not work well.

As a general principle for everyone, especially those planning more than a handful of sessions at the higher total doses, I strongly recommend finding your personal minimum effective dose. First, establish an effective MDMA therapy routine and gain a good sense of how your sessions feel and produce durable therapeutic improvement. Then you might reduce your total dose by 10% each session until you notice sessions becoming less helpful. Judge this a few weeks after the session, after the afterglow is finished and you have stabilized to a new baseline. This small reduction is unlikely to waste a valuable session by making it completely ineffective. You can also always return to your previous dose in the current session with an additional booster of the right amount.

You could also start by cutting the booster dose and then continuing with a 10% decrease on the single remaining dose on subsequent sessions. In my experience, a single initial dose often provides enough working time to get me to the therapy hangover limit. If therapy hangover is the limiting factor, then a higher dose or booster dose may not add much benefit.[10]

There are a few other preventative measures for oxidative stress that some people use. Most of these have not been tested in humans, and therefore the dose and practical utility are unclear. I recommend primarily focusing on reducing your MDMA dose in the face of these uncertainties, though people doing more than a handful of higher-dose sessions could consider trying some of these.

- **Caffeine:** High doses of caffeine exacerbate neurotoxicity in rats given very large doses of MDMA, so caution may be warranted (Vanattou-Saifoudine et al., 2012). Withdrawal is also undesirable, so it may be worthwhile to taper off moderate-high doses of caffeine for the months–years of MDMA therapy.

- **Hot Ambient Temperature:** High ambient temperature exacerbates neurotoxicity in rats given large doses of MDMA (M. Baggott & Mendelson, 2001). It may be prudent to avoid hot ambient temperatures during sessions. It's plausible that the combination of high temperatures and high humidity is even worse given that it reduces the cooling effect of sweating.

- **Antioxidants:** High doses of certain antioxidants, including alpha-lipoic acid, ascorbic acid (vitamin C), and acetyl-L-carnitine, prevent oxidative stress in rats given extremely high doses of MDMA (Aguirre et al., 1999; Alves et al., 2009; Shankaran et al., 2001). See M. Baggott (2015) for more information.

- **SSRIs:** Fluoxetine and likely other SSRIs taken 3–4 hours after the MDMA prevent neurotoxicity in rats (M. Baggott & Mendelson, 2001).

[10]It's conceivable that a booster dose could push someone through their single-dose therapy hangover limit, but that would also result in even higher levels of post-session therapy hangover.

4.5 Session Frequency

Specific recommendations for session spacing are difficult because individuals have different risk/reward tradeoffs and neurological responses to MDMA. There are two main limitations relevant for minimizing risk (M. Baggott, 2016; M. Baggott & Mendelson, 2001). First, the brain's antioxidant capacity needs time to replenish after a session. MDMA causes oxidative stress, but the brain's antioxidant buffer is normally capable of neutralizing this for the doses of MDMA recommended for therapy. However, using more MDMA before it has recovered may cause oxidative damage in the brain.[11] To my knowledge, no human studies, including the clinical trials, have measured this time span.

Three expert panels independently ranked MDMA, or substances sold as MDMA, among the least harmful drugs on their scales, including on criteria that would plausibly capture neurotoxic effects if those were producing clinically apparent outcomes (Bonomo et al., 2019; Crossin et al., 2023; D. Nutt et al., 2010). While their criteria are oriented toward observable clinical effects rather than subclinical neural changes, the absence of any signal strong enough to elevate MDMA in these rankings is consistent with oxidative damage from typical non-therapeutic use patterns being, at most, a minor contributor to population-level harm. The typical therapeutic protocol—low-to-moderate dose, proper hydration, infrequent administration, no hyperthermia, no drug mixing—would be expected to produce even less.

That being said, even mild-to-moderate amounts of harm are worth minimizing when possible. Occasionally, people may also need to do many MDMA sessions to treat severe CPTSD, and gradually

[11] I don't think using antioxidant supplements to do more frequent sessions is wise because the oral dose of antioxidants humans need to prevent oxidative damage is unknown (M. Baggott, 2015). It's not even established that it's possible or safe to get antioxidant plasma concentrations high enough using oral dosing. The studies demonstrating antioxidant protection injected rats with massive doses, though that was to protect against similarly massive doses of MDMA.

cumulative effects may become more noticeable. To myour understanding, there are no public estimates on how long it takes for the human antioxidant buffer to recover after using MDMA. In the absence of any better line of reasoning, I used the research function of the Claude 4.7 Opus LLM model to find and read scientific papers on the topic and then generate a span of likely values.[12] Its semi-educated guess was that it takes 2–10 days to replenish the antioxidant buffer after complete depletion, and there is some evidence that total doses under 2 mg/kg are unlikely to exceed the buffer (M. Baggott, 2015). I tentatively propose using twice[13] the upper end of that range as a "don't go below" spacing for oxidative stress, while noting that it is likely inaccurate due to the nature of LLM agents and the limited evidence available.

I recognize these estimates are tentative at best. I'm open to removing them, but I thought it would be helpful to provide some estimate rather than leaving readers to make wild guesses.

---

[12]I used this prompt:

> Give me two 95% confidence intervals for how long the brain's antioxidant buffer takes to replenish after a single dose of MDMA, assuming no hyperthermia. The first CI is for a dose large enough to use up all the buffer but not any more. The second CI is for doses that exceed the buffer. The research on MDMA oxidation is scarce, so you'll have to extrapolate some from other research. I've attached a primer [Jomova et al. (2024)] on the endogenous antioxidant system as a reliable starting place.

The report is available here.

It reported 0.5–7 days for full replenishment after complete buffer depletion and 3 days to 8 weeks to recover from buffer-exceeding doses that cause damage. The latter interval doesn't necessarily imply that all damage has been reversed, just that the antioxidant buffers have been restored. Furthermore, Claude bumped up the first interval to 2–10 days when I asked it to investigate the differences between the Opus 4.6 and 4.7 reports.

These are not real confidence intervals; they are the LLM's guesstimates. I did try a few different prompts with the research functions of five different LLMs (Gemini 3.0 thinking, Claude 4.5/4.6/4.7 Opus thinking, and ChatGPT 5.2 prompted to think hard). All the reports suggested spans from a few days to a few weeks. The numbers reported above came from the run that I judged was the most rigorous.

[13]Using twice the upper end adds protection for people with exacerbating circumstances, like poor CYP2D6 metabolic capacity, other interfering drugs, and poor nutrition.

The second main limitation is downregulation of the serotonin system during an MDMA session, which causes short-term tolerance (M. Baggott, 2016; M. Baggott & Mendelson, 2001). Using MDMA again before tolerance has returned to baseline will result in lower efficacy. Using higher doses to overcome that will increase the risk of oxidative damage if the dose becomes too large for the antioxidant buffer to handle. Neither the time required for tolerance to return to baseline nor the limits of the antioxidant buffer are known, so I strongly recommend spacing sessions far enough apart that you don't notice any lessening of effect. I don't think anyone has measured how long short-term tolerance lasts, though the six-week spacing of the phase III trials (Mitchell et al., 2023) produced progressively increasing benefit after each session, suggesting that this time span may be sufficient.

Doing sessions more frequently than those limits may also risk long-term tolerance, sometimes called "losing the magic" (M. Baggott, 2016; M. Baggott & Mendelson, 2001).

I think the six-week spacing used in the phase III trials (Mitchell et al., 2023) is likely a reasonable spacing to start at, though longer is always ok too.[14] From there, I recommend establishing an effective MDMA therapy routine and gaining a good sense of how your sessions feel and produce durable therapeutic improvement. Then, if you feel that a shorter spacing would benefit you, you might reduce your spacing by one week every one or two sessions until you notice sessions becoming less helpful. Judge this a few weeks after the session, after the afterglow is finished (see Section 8.5), and you have stabilized to a new baseline. This reduction is unlikely to waste a valuable session by making it completely ineffective.

That procedure only measures short-term tolerance. You may not be able to notice when you go below your antioxidant system

[14] The sparsity of data means that well-informed people with different risk tolerances and starting assumptions will disagree on what a reasonable starting spacing is.

recovery timespan, and thus I recommend the two-week spacing as an absolute minimum.

See "Mechanisms of MDMA Tolerance and Loss of Magic" by Matthew Baggott (M. Baggott, 2016) for further discussion of tolerance and oxidative stress.

Additional Downsides of Frequent Sessions:

- More frequent medication pausing or tapering if your medication is incompatible with MDMA.
- People with risk factors for psychosis may want to avoid frequent sessions, since frequent use of a wide range of psychoactive drugs is another well-established risk factor (Fiorentini et al., 2021).
- Frequent sessions are emotionally taxing.

Benefits of Frequent Sessions:

- Compressing a given amount of therapy into a shorter period of time delivers faster therapeutic progress with no increase in adverse events (Dell et al., 2023; Ehlers et al., 2014; Foa et al., 2018). This was found in non-MDMA therapy for PTSD, but I have no reason to think it would be different for MDMA therapy and different combinations of stuck schemas.
- Many people who might benefit from working with a skilled, ethical, and well-matched mental health practitioner can't access one because of stuck schemas involving trust. Relatedly, many people already have a therapist but don't benefit as much as they could. A critical mass of MDMA therapy sessions may provide a valuable on-ramp to greater amounts of therapeutic support and healing. This can lead to a strong upward trajectory, whereas the same amount of reconsolidation spread over a longer period of time might not achieve the same critical mass necessary.

- Another session after a few weeks might provide enough additional reconsolidation to resolve particularly overwhelming symptom worsening from a previous session that doesn't respond to other symptom management strategies.

- Van den Bergh et al. (2021) hypothesizes that the high-level schemas predicting pervasive threat that many people with mental illness have can easily categorize minor negative stimuli as new threats.[15] This process can readily create new stuck schemas or make existing stuck schemas activate in the presence of new stimuli. Put another way, relatively minor events can easily traumatize people with mental illness. I speculate that this might show up in therapy as "two steps forward, one step back," or needing to achieve some intensity of reconsolidation just to maintain baseline. Actual improvement would depend on higher intensities of reconsolidation.

- Similarly, from a complex systems approach, it's conceivable that certain complex stuck schema networks may reconstitute certain components over time after those components are partially or fully reconsolidated. High intensities of reconsolidation over short-to-long periods of time might be needed to reconsolidate multiple reinforcing components and break free of these traps.

4.6 Drug Interactions

MDMA has dangerous or undesirable interactions with various drugs, supplements, and herbs. If you regularly take another drug, I suggest consulting M. Liechti and Schmid (2026) and Malcolm (2025b) (more accessible) or Sarparast et al. (2022) (more technical) for recommendations on whether you should continue or discontinue it during MDMA therapy, avoid MDMA, or modify the MDMA dose. If your

[15] "The Precision of Sensory Evidence" by Scott Alexander (Alexander, 2021a) provides a more accessible summary of Van den Bergh et al. (2021).

medicine is essential for your health, I strongly recommend consulting your doctor or pharmacist for help managing this. They will likely not understand the effects of MDMA, so you may need to provide Sarparast et al. (2022) to them. That paper discusses pharmacokinetics, pharmacodynamics, and psychiatric drug interactions.

If you can't access a doctor or pharmacist, pausing a drug for 3 half-lives[16] before the MDMA dose and 24 hours (3 MDMA half-lives) after might work, provided a few conditions are met. The drug shouldn't be critical for your health, isn't one of the few high-risk interactions discussed in the section summary, and can be paused with tolerable effects. This doesn't work as well for drugs whose MDMA-relevant effects persist after the drug itself has been flushed out, like SSRIs and irreversible MAOIs. Certain health conditions also affect half-lives.

Most supplements and herbs can just be paused on treatment day to be safe.

In this section I list a couple of classes of interaction, then the most risky specific interactions and a few other selected interactions that frequently appear in therapeutic contexts. Combining MDMA with other prescription psychiatric or psychoactive drugs causes various changes to the intensity or duration of different effects, including changes to the efficacy of MDMA therapy (Sarparast et al., 2022).

The most risky classes of interactions:

- **Serotonin Syndrome:** Serotonin syndrome is a potentially deadly condition caused by extreme amounts of intrasynaptic serotonin (Malcolm & Thomas, 2022; Sarparast et al., 2022). It's generally caused by interactions between MDMA and MAOIs.

[16]One half-life is the time it takes for a drug's concentration in your body to decrease by half. Each drug has a different half-life, which can be found on DrugBank under Pharmacology → Metabolism. So the concentration would be 50% after one half-life, 25% after two, etc.

- **Alterations in the CYP450 Enzymes:** The liver enzymes CYP2D6, CYP1A2, CYP2B6, CYP2C19, CYP3A4 (all part of CYP450), and COMT metabolize MDMA and its metabolites (other molecules that the body converts MDMA into) (Sarparast et al., 2022). Drugs that enhance these enzymes may reduce the intensity and duration of MDMA effects by removing it from your blood at a faster rate. Drugs that inhibit these enzymes cause higher and longer blood concentrations, though the multiple enzymes provide redundancy if one pathway is blocked. Drugs that strongly inhibit multiple of these enzymes may be deadly to take with MDMA. Drugs that only inhibit CYP2D6 are ok to take with MDMA, though M. Liechti and Schmid (2026) recommends reducing the MDMA dose by up to 25%.

  MDMA itself also almost completely inhibits CYP2D6 (O'Mathúna et al., 2008). This inhibition returns to baseline with a half-life of 47 hours. Thus, about 2 days after a session, enzyme activity will be 50% of the way to baseline, 75% after 4 days, 88% after 6, 94% after 8, and 97% after 10. This effect may counteract tolerance or lead to unexpectedly strong reactions to subsequent doses of MDMA. Inhibited CYP2D6 slows the metabolism of MDMA and many other drugs, especially ones that are not metabolized through any other parallel pathways that could take up the slack. Thus, dangerous concentrations of certain drugs could accumulate in your blood when those drugs are used within a few days after MDMA.

  Flockhard (2025) maintains a list of drugs that inhibit, enhance, or are metabolized by (called substrates) CYP450 enzymes.

Specific drug interactions, starting with the most risky:

(does not include low–moderate risk interactions with various drugs)

- **MAOIs:** Taking irreversible MAOIs within two weeks before an MDMA session or immediately after can cause severe serotonin toxicity and death (Edinoff et al., 2022; Malcolm & Thomas, 2022). Ayahuasca contains a shorter-lasting, reversible MAOI whose effects are gone within 2–3 days (Malcolm, 2023).

- **Ritonavir and cobicistat:** These strongly inhibit multiple CYP450 enzymes and can cause death when combined with MDMA (Bracchi et al., 2015; Sarparast et al., 2022).

- **SSRIs and SNRIs:** SSRIs and SNRIs highly inhibit the effects of MDMA (Sarparast et al., 2022) but are not dangerous. Long-term use of these drugs causes this effect to persist long after drug discontinuation (Feduccia et al., 2021). The therapeutic efficacy of MDMA therapy is reduced by half even after 25 days of discontinuation. Further discontinuation may bring further benefits. Discontinuation typically requires multiple additional weeks of tapering to manage withdrawal. See *Crossing Zero: The Art and Science of Coming Off — And Staying Off — Psychiatric Drugs* by Anders Sørensen (Sørensen, 2025) and *Antidepressant and Psychedelic Drug Interaction and Taper Planning Guide* by Benjamin Malcolm (Malcolm, 2025a) for more information and practical advice on tapering.

- **Caffeine:** Large doses of caffeine dramatically increase tachycardia and body temperature in rats given large doses of MDMA (Vanattou-Saifoudine et al., 2012). It also increases the risk of serotonergic neurotoxicity to an unclear degree. It's not clear how this applies to human MDMA therapy, but caution might be warranted.

4.7 Medical Risks

MDMA's interactions with certain serious health conditions are not well understood. As of 2026, MAPS clinical trial exclusion criteria are commonly regarded as the baseline for what medical conditions are incompatible with MDMA. However, clinical trial exclusion criteria are conservative and designed to reduce unknown variables, regulatory scrutiny, and actual, if uncertain, harm. I think the exclusion criteria indicate that additional caution should be taken but are not all absolute contraindications. In the phase III trials (Mitchell et al., 2023, Supplementary Information, p. 5) "any medical condition that could make receiving a sympathomimetic drug harmful due to increased blood pressure and heart rate" was cause for exclusion, along with additional details for cardiovascular conditions discussed below. Please consult your doctor if there is any question about whether that applies to you. It would be useful to review the MAPS pharmaceutical investigator's brochure (Multidisciplinary Association for Psychedelic Studies, 2022) with your doctor.

This document is too long to really be of use but I don't know what else to recommend.

The trial inclusion criteria did regard some serious medical conditions as compatible with MDMA therapy (Mitchell et al., 2023). Mitchell et al. (2023, Supplementary Information, p. 5) said that "individuals with medical conditions such as hypertension, asymptomatic hepatitis C virus, diabetes mellitus, hyperthyroidism, and glaucoma were eligible, providing the condition was well managed and mild." They couldn't have assessed every type of serious medical condition, so some conditions not on that list are presumably also compatible with MDMA therapy.

Preexisting health conditions:

- **Cardiovascular Disease:** MDMA increases blood pressure and heart rate, largely between hours 0.75–4 when a single dose is used (Straumann et al., 2024). Doses of 120 + 60 mg given roughly 2 hours later increase average blood pressure by 28/12 mmHg and heart rate by 23 b/min over placebo in therapeutic contexts (Mitchell et al., 2021).[17] This may be a risk for individuals with cardiovascular disease. Individuals with "uncontrolled hypertension, history of arrhythmia,[18] or marked baseline prolongation of QT or QTc interval" were

[17] Blood pressure and heart rate are averaged values from the 2nd and 3rd sessions. I excluded the 1st session because it used a different dose.

[18] I suspect that benign premature ventricular or atrial contractions are compatible since ~50% of people have them but I haven't been able to confirm this.

excluded from clinical trials for this reason in Mitchell et al. (2023, Supplementary Information, p. 5). Mitchell et al. (2021, sec. Methods) also states that "any medical condition that could make receiving a sympathomimetic drug harmful due to increased blood pressure and heart rate" was cause for exclusion. It's unclear exactly how much of a risk these actually pose, and clinical trial exclusion criteria are conservative.

People with anything more significant than well managed and mild cardiovascular illness, as assessed by a doctor, might want to make precautions in case there is a problem during the session.

- **Liver Disease:** In one case, an individual with advanced alcohol-induced liver cirrhosis tolerated 100 mg without issue (Kraus et al., 2025). It's unclear how this generalizes to other cases and what the boundaries of safety are. Approach with caution with medical support.

- **Untreated Hyperthyroidism:** Rats with untreated hyperthyroidism given MDMA have a much higher risk of dangerously high body temperature (Shokry et al., 2018). Individuals with hyperthyroidism were allowed in the clinical trials provided that the condition was "well managed and mild," according to Mitchell et al. (2023, Supplementary Information, p. 5).

Acute adverse effects:

- **Hyponatremia:** MDMA commonly causes mild hyponatremia (low plasma sodium concentration) in individuals who drink fluids as desired during the session (Atila et al., 2024). I extrapolated from M. J. Baggott et al. (2016) that drinking a maximum of 0.5 L of water during the six-hour session would easily prevent hyponatremia. It is also more than sufficient for preventing dehydration in the average person at a comfortable temperature (Valtin, 2002). Adding electrolytes has not been tested as a solution and is known to not prevent hyponatremia in athletic activities (Hew-Butler et al., 2008).

People worried about dehydration could fully hydrate two hours before taking MDMA (Matthew Baggott, personal communication, November 24, 2025). That would provide enough time for the body to excrete any excess water by the time the session starts.

On rare occasion, people at raves drink an extreme amount of water and die (Rigg & Sharp, 2018).

- **Heat Illness:** Prolonged, intense physical activity in high temperatures combined with dehydration can cause dangerous heat illness, as sometimes occurs at raves (van Amsterdam et al., 2021). Alcohol co-use significantly exacerbates this risk.
- **Seizures:** As with most problems associated with MDMA (Rigg & Sharp, 2018), seizures are very rarely reported. When they are, they are mostly associated with mixing intoxicants, extremely high doses, hyponatremia from drinking too much water, or heat stroke from dancing all night without adequate fluid intake (Freidel et al., 2024). It's possible that caffeine co-use increases the risk (Vanattou-Saifoudine et al., 2012).

Chronic adverse effects:

- **Valvular Heart Disease:** Extremely high lifetime use of MDMA and possibly most psychedelics causes valvular heart disease via serotonin 5-HT_{2B} receptor activation in heart valves (Droogmans et al., 2007; Tagen et al., 2023). In one observational study, 28% of chronic MDMA users showed signs of valvular heart disease (VHD) when evaluated with echocardiography, compared to 0% in a matched control group who reported no MDMA use (Droogmans et al., 2007). The chronic users with clinically significant VHD self-reported a mean consumption of 943 ± 1162 MDMA tablets, while the chronic users without clinically significant VHD reported a mean consumption of 242 ± 212 tablets. These results are so imprecise and assuredly confounded by abuse of multiple other drugs that they are inadequate as safety guidelines. Unfortunately, it's the only data I know of. I take the lower number as a "don't exceed" limit.

- **Oxidative Damage:** MDMA causes oxidative damage in the serotonin system when the oxidative load of MDMA's effects exceeds the system's antioxidant buffer capacity (M. Baggott & Mendelson, 2001). M. Baggott (2015) states that there is decent experimental evidence that doses below 2 mg/kg don't cause oxidative damage. Indicators of oxidative damage haven't been investigated at higher doses in humans.

  A dose of MDMA that is not neurotoxic in a single administration can also become neurotoxic when additional administrations are done before the antioxidant system has recovered (M. Baggott & Mendelson, 2001). Unfortunately, it's not known how long this recovery takes. It's also unclear whether this oxidative damage has any clinically relevant effects.

- **Long-Term Cognitive Impairment:** There is limited and contested evidence that many high-dose sessions cause long-term cognitive impairment (M. Baggott & Mendelson, 2001). Unfortunately, it's unclear what counts as high-dose or what the shape of the session-count–impairment curve looks like. The evidence for this comes from a combination of observing inconsistent and mild behavioral differences after giving rats and monkeys extreme doses of MDMA and observational studies of recreational MDMA users. The human observational studies show an association between cognitive impairment and frequent or high cumulative use in recreational contexts. However, those studies rarely adequately control for the fact that recreational MDMA users often use a wide variety of other somewhat dangerous drugs (often mixed), overdose, or overheat or get hyponatremia at raves. One of the best observational studies, Coray et al. (2025), narrowed down the association to MDMA in particular, rather than other drugs the users had been taking. However, even that study did not control for the other mentioned factors that are not present in therapeutic contexts. One small randomized study of MDMA therapy did not find any significant cognitive effects after two sessions spaced 3–5 weeks apart using up to 188 mg, but that doesn't rule out cognitive effects from a much larger number of sessions (M. C. Mithoefer et al., 2011).

Higher risk groups:

- **Adolescents:** To my knowledge, MDMA therapy has not been studied in adolescents. Practitioners will have to weigh the unknowns of how adolescents respond to MDMA against the risks of the clients not getting effective treatment. I suggest not using booster doses. The adolescent's home environment may also be a hostile place to deal with potential symptom worsening and increased sensitivity, given the frequent role of families in creating adolescents' stuck schemas.

- **Pregnancy:** There isn't any high-quality data about humans using MDMA while pregnant. The precautionary principle indicates that it should be avoided until it's rigorously demonstrated to be safe.

- **Breastfeeding:** There isn't any high-quality data about humans using MDMA while breastfeeding. The precautionary principle indicates that milk shouldn't be used while it contains significant amounts of MDMA. Bartu et al. (2009) recommends discarding all milk from the 48-hour period following the use of methamphetamine. Its structural similarity to MDMA and longer half-life might indicate a useful recommendation for MDMA.

4.8 Psychological Risks

There is uncertainty about which psychological disorders are compatible with MDMA therapy. These are the exclusion criteria for the phase III trials (Mitchell et al., 2023, Supplementary Information, p. 5):

> Individuals were ineligible to enroll if they were unable to give informed consent. Individuals were also excluded for a history of or current primary psychotic disorder, bipolar I disorder, dissociative identity disorder, eating disorder with active purging, major depressive disorder with psychotic features, personality disorders, severe alcohol or cannabis use disorder (also moderate if not in remission), any substance use disorder other than cannabis or alcohol within 12 months prior to enrollment ... [or] serious imminent suicide risk.

Clinical trial exclusion criteria are conservative and designed to reduce unknown variables, regulatory scrutiny, and actual, if uncertain, harm. I don't know their reasoning for each item, but have some informed speculation. MDMA therapy induces psychotic episodes on rare occasion, though risk is likely higher in those with a personal history. Amphetamines are a known risk factor for manic episodes. Personality disorders make forming a healthy relationship between a client and therapist difficult to achieve. MDMA therapy might produce overwhelming symptom worsening in people with dissociative identity disorder if it facilitates abrupt confrontation of extremely distressing dissociated schemas. The trial environment and staff were likely not equipped to cope with some of these conditions, even if they are addressable with MDMA therapy.

Mental illness treatment is highly individualized. Therefore, I don't think most of these conditions are absolute contraindications. Rather, each case should be assessed on an individual basis for its expected reward/risk ratio. Individual practitioners have different capacities for which symptoms they can support in their clients. Client factors include the amount of symptom worsening they can cope with at a particular point in their life, how capable they are at managing dysregulation, how much their basic functionality depends on avoiding certain feelings or memories, and how much healthy external support they have.

I know that some people have enough resources, skill, and slack to use solo MDMA therapy to successfully treat debilitating mental illness. Many more people can use solo MDMA therapy to reconsolidate relatively minor issues. Skilled, ethical, and well-matched support is especially desirable for people with significant risk factors for dangerous conditions like psychosis and mania. However, I recognize that people sometimes have to make the best of bad situations. Solo MDMA with risk factors is sometimes a better option than the available alternatives, though this is highly individualized and I can't make any specific recommendations.

Practitioners should be able to accurately identify each of the following conditions and have a plan to either manage them or get the client to an appropriate higher level of care. I rank these according to my impression of risk, starting from the most significant. See Chapter 9 for management recommendations and when and where to seek additional care.

- **Symptom Worsening:** Temporary symptom worsening is a common occurrence in therapy (see Section 3.6). If it is intense enough and not managed well, it can severely interfere with your life.

  I think that MDMA therapy tends to produce stronger symptom worsening and more rapid therapeutic progress than traditional psychotherapy. Severe trauma, diagnosis of mental illness, and severely disorganized attachment are risk factors. People are sometimes not explicitly aware they have gone through severe trauma. This may happen if the trauma takes the form of disorganized attachment, abuse is explained away as cultural tradition or "how things are," the trauma took place in the period of childhood amnesia, or it is not remembered for another reason. The therapeutic alliance (see Greenspace, 2023 for an assessment scale) is a moderate mitigating factor when working with a mental health practitioner (Flückiger et al., 2018).

I haven't been able to find any data on symptom worsening specific to MDMA therapy. The next best data I know of, Evans et al. (2023), surveyed people who experienced new, persistent negative symptoms after recreational, professional-therapeutic, and DIY-therapeutic psychedelic experiences. This data applies to all psychedelics, not just MDMA, and a significant part of it only applies to traumatic experiences caused by large doses of hallucinogens like LSD, ayahuasca, or psilocybin. Most symptoms dissipated with time, but 17% of respondents said theirs lasted more than 3 years. From most to least common, participants reported emotional (76%), self-perception (58%), cognitive (52%), social (52%), ontological (50%), spiritual (34%), perceptual (26%), and other (21%) difficulties. These symptoms could be due to (Calder et al., 2025)

- surfacing of existing stuck schemas and subsequent defense cascade activation, a necessary and healthy part of the therapeutic process if managed well. You may have been avoiding these schemas until the session.

  I think there is a high likelihood of this for MDMA therapy. It's conceivable that a skilled MDMA therapist could help you keep symptom worsening to small, easily dealt with chunks.

- trauma from life impairment or symptom worsening due to poorly managed surfacing of stuck schemas and trauma.

  This happens (Evans, 2024), though the risk can probably be reduced with assistance from a skilled, ethical, and well-matched (see Greenspace, 2023) mental health practitioner (Flückiger et al., 2018).

- trauma from the psychedelic experience itself.

I think this usually results from large doses of hallucinogens, unsafe settings, and abusive or incompetent practitioners. I think that traumatization risk is low for MDMA itself because of its intense feelings of safety and low hallucinatory and mystical effect (Studerus et al., 2010).

- difficult changes to your understanding of self and existence.

  I think this is uncommon compared to psychedelics since MDMA produces relatively little ego-dissolution and mystical experience (Holze et al., 2020); however, it does happen (Ingram, 2024; Martin, 2020). See Section 8.4.12 for more information.

- something else. I don't know if this exists, and if it does, what it is or how often it occurs.

Even in this subgroup of people who have experienced extended difficulties in the previously mentioned study, 90% agreed with the statement "I believe that the insights and healing gained from psychedelics, when taken in a supportive setting, are worth the risks involved" (Evans et al., 2023, Fig. 2). However, it is possible that a population of psychedelic users who experience debilitating effects was missed due to sampling bias.

- **Psychosis:** There is virtually no high-quality experimental data because people with a personal (though not family) history of psychosis were excluded from clinical trials (Mitchell et al., 2023). Like other mental illnesses, psychosis is a complex biopsychosocial phenomenon. Therapy often reduces the symptoms of psychosis (Sitko et al., 2020), suggesting that stuck schemas often play some role, though how strong that is compared to other factors likely varies by case. This implies that psychosis might start and stop at hard-to-predict points during the reconsolidation process and in life in general for people with some level of predisposition.

A variety of anecdotal reports are congruent with this complex framing (Goodman, 2003; /r/mdmatherapy, 2025a, 2025b): A few people state that a single MDMA therapy session triggered a psychotic episode. Four people state that an MDMA therapy session resolved an existing psychotic episode. A few people state that they have safely used MDMA therapy despite previous psychosis, even when psychedelics were major causes of the psychotic episodes.

Psychosis may be difficult to predict, but there are well-known risk factors. High doses and frequent use of a wide range of psychoactive drugs (especially cannabis) are well-established risk factors (Fiorentini et al., 2021), as is stress (van Winkel et al., 2008). This explains why case reports of MDMA-induced psychosis typically, though not always, report confounding factors like co-use with other psychoactive drugs, chronic abuse of other drugs, heat stroke, hyponatremia, extreme doses, or extreme frequency of use (Arnovitz et al., 2022; McGuire & Fahy, 1991; Patel et al., 2011; Sulstarova et al., 2025; Vaiva et al., 2001).

Psychedelic-induced psychosis sometimes transitions into schizophrenia, but it's not known how often this happens in MDMA therapy and whether it really causes schizophrenia vs. just accelerating its onset in those who would have otherwise gotten it (Sabé et al., 2025).

MDMA-induced psychosis is a risk for those with a predisposition, but some people with risk factors may still think it's worth trying. There are some precautions people with a history of psychosis should take if they try MDMA therapy. You could minimize the known risk factors of stress (van Winkel et al., 2008), cannabis (Fiorentini et al., 2021), and abrupt withdrawal of antipsychotics (Moncrieff, 2006). Skipping the booster dose of MDMA should also help, assuming a dose-dependent risk.

- **Suicidal Ideation and Behavior:** MDMA therapy with high levels of support decreases suicidal ideation on average about as much as placebo with the same level of support (Mitchell et al., 2021, 2023). When interpreting these results, it is important to understand that average improvements can mask the possibility that a minority of individuals can get worse while the majority improve. Of course, this applies to the placebo group as much as the MDMA group. Suicidal ideation is part of the biopsychosocial complex system of mental health. The psycho/schema component seems particularly influential since suicidal ideation almost always involves accompanying schema-like beliefs and justifications. Suicidality might get worse for a period of time, like many other schemas during the reconsolidation process.

- **Addiction:** Three independent panels of drug-misuse experts assessed non-therapeutic MDMA, or drugs sold as MDMA, as having among the lowest risk of dependence and harm among 20 of the most popular illegal or harmful drugs (Bonomo et al., 2019; Crossin et al., 2023; D. Nutt et al., 2010). Alcohol and cannabis scored much worse. Significant withdrawal was also not found in one rodent study, even at an extreme dosing schedule (Robledo et al., 2004). Some minor symptoms were noted, but the authors concluded that "chronic MDMA administration does not induce classical manifestations of physical dependence." When MDMA is abused, it is associated with partying or avoidance of difficult feelings rather than therapeutic engagement (Erowid, 2024).

  Context is a critical component of addictive potential. Many commonly abused prescription drugs do not cause addiction or harm when used appropriately. The same panels scored several classes of prescription drugs, including amphetamines (some ADHD drugs), benzodiazepines (fast-acting anti-anxiety drugs), ketamine (anesthesia), and opioids (painkillers), as far more harmful than MDMA in non-therapeutic contexts (Bonomo et al., 2019; Crossin et al., 2023; D. Nutt et al., 2010). This might indicate that, when used responsibly for therapy, MDMA also has a lower risk of addiction and harm than medical use of those prescription drugs. Congruent with this, no instances of MDMA dependence have been reported in clinical trials (Colcott et al., 2024).

This suggests that the addictive potential of MDMA therapy is minimal for most people. If you are particularly worried about your potential for addiction or have significantly impaired impulse control, I suggest only doing MDMA therapy in structured, practitioner guided contexts. This might include people who have, or are in remission from, serious amphetamine or cocaine addictions who might find the stimulant effects of MDMA close enough to the effects of their abused drug.[19]

MDMA therapy has shown tentative promise for improving other addictions. Structured MDMA therapy with high levels of support was well tolerated in individuals with current mild alcohol addiction or early remission of moderate alcohol addiction and did not lead to increased alcohol intake (Nicholas et al., 2022). People using MDMA therapy for recovery may be interested in Psychedelics in Recovery (2025), a 12-step program for integrating psychedelics into recovery.

- **Hallucinogen Persisting Perceptual Disorder:** Complex or compelling distortions of external reality on MDMA are rare or non-existent (M. E. Liechti et al., 2001). People more commonly have closed-eye visuals, possibly involving traumatic events they experienced. These visuals may be symbolic instead of a realistic reliving. Temporary and mild visual changes such as color and texture enhancement are also common.

[19] The addiction pharmacology of these substances might share enough similarities with MDMA to warrant caution (D. J. Nutt et al., 2015). Other classes of addictive substances have different addiction pathways.

Experiences involving a combination of intense fear and visual distortions, like some MDMA and psychedelic experiences, occasionally create persistent visual distortions or anxiety about existing but unnoticed visual distortions (Halpern et al., 2018). When this causes significant distress or impairment, it is called Hallucinogen Persisting Perceptual Disorder (HPPD). HPPD is strongly, but not exclusively, linked to pre-existing anxiety or dissociative disorders and often improves as those are treated. HPPD from MDMA is unrecorded in clinical trials, but some recreational users report it (Litjens et al., 2014; Vizeli & Liechti, 2017). Carhart-Harris and Nutt (2010, p. 289) found that when people do report persistent visual or auditory distortions (from MDMA or psychedelics), 73% say "they [the symptoms] don't bother me at all," 24% "I'd rather not have them, but I can live with them," 0% "they irritate me," and 1% "they drive me mad".

Given its origin in the combination of fear and sensory perception, I think HPPD is likely one of many possible somatic symptoms of mental illness in the model of Van den Bergh et al. (2017) that I expand on in Section 3.5. In that model, the initial experience creates a link between visual distortions and fear. The resulting HPPD would be either fear-driven hyperfocus on normal visual distortions that are typically filtered out of awareness or fear-driven hyperfocus on abnormal visual distortions that the brain recreates as top-down sensory fudging. Either could be treated with reconsolidation.

- **Mania:** There is virtually no high-quality experimental data because people with bipolar I were excluded from clinical trials (Mitchell et al., 2023). The MDMA phase III trials did not exclude individuals with bipolar II, and no manic episodes were reported. Bipolar I might have been excluded because psychological stress, sleep disruption, and dopaminergic amphetamines are linked to mania (Salvadore et al., 2010), though MDMA releases much less dopamine than other amphetamines (KankaanpÄÄ et al., 1998).

Unlike MDMA-induced psychosis, I couldn't find a single published case report of mania where MDMA was unambiguously involved in the recent past. I only found three plausible anecdotal reports of MDMA-induced mania on the Internet that didn't obviously involve major exacerbating factors (overdose, multi-drug abuse, etc.) that aren't present in therapeutic contexts (Bengt, 2006; Girl, 2004; I<3Hallucinogens, 2013).[20] Two of those people reported pre-existing bipolar, and the third was unclear.

This scarcity of evidence suggests that mania is not a significant risk in MDMA therapy. However, there are some risk-reduction precautions that people with a history of mania could take if they try MDMA therapy. Adequate pre- and post-session sleep is critical given sleep's role in mania. MDMA should be taken early in the morning so that it's easier to sleep that night. It would also help to have a short course of sleep aids supplied in advance in case there are post-session sleep issues. High doses of caffeine might be avoided on the day of the session and continue until short-term side effects have dissipated (Lara, 2010), though abrupt withdrawal would also increase risk. Skipping the booster dose of MDMA might also help, assuming a dose-dependent risk.

[20] I searched for the terms "manic" and "mania" on reddit.com/r/mdma, reddit.com/r/mdmatherapy, erowid.org, and bluelight.org and read every result, or in the case of bluelight.org, read every search result snippet. There were a few additional reports on bluelight.org where people said that MDMA worsened their manic-depressive symptoms, but it wasn't clear that the manic symptoms in particular were worsened.

4.9 When to Seek Emergency Care

Most of the interactions that cause severe acute harm are known and can be avoided with proper precautions or by not using MDMA (Malcolm & Thomas, 2022; Rigg & Sharp, 2018). However, individuals may not always follow proper precautions or may have undiagnosed health conditions. The direct causes of severe acute damage involving MDMA are almost always high body temperature, hyponatremia, or serotonin syndrome. Cardiovascular events may also rarely occur in people with pre-existing cardiovascular issues, especially when MDMA is combined with other drugs associated with cardiac function abnormalities (Makunts et al., 2023).

Many typical symptoms of heat illness, hyponatremia, serotonin syndrome, and cardiac events are also therapeutically appropriate symptoms of MDMA, flight-or-fight, freeze, tonic/collapsed immobility, and confronting extreme fear or anger. Thus, recommendations specific to MDMA therapy are important. Malcolm (2023) recommends seeking emergency care for any of these symptoms of psychedelic/MDMA-facilitated serotonin syndrome: **myoclonic seizures, fever greater than 38.5 °C (101.3 °F), fluctuating or unstable blood pressure and heart rate, delirium or coma, and muscle rigidity.**[21] To my knowledge, that recommendation also covers severe hyponatremia, which is only possible with very high fluid consumption, and forms of severe heat illness not caused by serotonin syndrome.

Is this right? I haven't been able to find any reasonable hyponatremia guidelines that distinguish it from normal MDMA symptoms.

In the absence of MDMA-specific advice, one cardiologist offers this general advice for what type of chest pain you should go to the emergency room for (Fromson, 2023):

Is there any better advice? I haven't been able to find any.

> It most often boils down to the severity of the pain and the heart attack symptoms I mentioned above. If the

[21]Malcolm (2024) further discusses these criteria, but the author lists a lower temperature threshold in this document for unexplained reasons. The 100 °F threshold of danger is implausibly low; a moderate dose of MDMA alone can raise body temperature that high, and it is not dangerous (M. E. Liechti, 2014).

pain is so severe that you feel like you can't function, or if you are experiencing central or left-sided chest pain—especially if you have nausea or a cold and clammy feeling [also side effects of MDMA (M. E. Liechti et al., 2001)] alongside it—it is always safest to go to the emergency room. With chest pain, it's best to be cautious.

4.10 Common Concerns

- **Loss of Control:**
  - *Dangerous Loss of Anger Inhibition:* Three independent panels of drug-misuse experts all judged that the risk of direct and indirect injury to yourself and others associated with MDMA is low compared to other common illegal drugs (Bonomo et al., 2019; Crossin et al., 2023; D. Nutt et al., 2010). All three scored it as significantly less than cannabis and far less than alcohol.
  - *Confronting Overwhelming Feelings:* While this can feel scary, confronting difficult feelings is necessary for reconsolidation. Personal experience and numerous anecdotal reports indicate that MDMA makes this feel more safe.
  - *Losing Touch with Reality:* MDMA does not produce complex open-eye hallucinations at therapeutic doses (M. E. Liechti et al., 2001). Psychosis is rare and mostly occurs when there are other risk factors (see Section 4.8).
  - *Post-Session Functional Impairment:* MDMA therapy increases functionality on average (Mitchell et al., 2021, 2023). Anecdotal reports also indicate that major functional impairment is rare. I think functional impairment is an exceptionally difficult instance of the common *it gets worse before it gets better* phase of therapy (see Section 3.6).
  - *Inappropriate Disclosure or Behavior:* MDMA notably increases openness. Choose a practitioner who you trust to

maintain professional boundaries, not pressure you, and maintain confidentiality (see Section 6.3). Solo sessions are also an option. That being said, I'm not aware that MDMA systematically facilitates inappropriate behavior during therapy sessions.

- **Drug Stigma/Discomfort/Misuse:** Discomfort with drugs is understandable, since many drugs are harmful. However, the legality of drugs has little relation to their potential for harm (Bonomo et al., 2019; Crossin et al., 2023; D. Nutt et al., 2010). Three independent panels of drug-misuse experts all estimated that even non-therapeutic use of MDMA, or drugs sold as MDMA, poses much less overall harm than alcohol or cannabis. They assessed that non-therapeutic MDMA has 8–13% of the harm of alcohol and 22–45% of the harm of cannabis. The different numbers reflect differences in how and where they measured harm.

- **MDMA Being Synthetic:** See Ruggeri (2025) for a nuanced discussion of what natural means. Most illegal MDMA is created by making several chemical changes to the plant substances safrole or piperonal (European Union Drugs Agency, 2025; United Nations Office on Drugs and Crime, 2014). Legal MDMA may use different processes but is not yet in mass production.

- **Unlearning Healthy Schemas:** I am uncertain whether the contradictory information that facilitates reconsolidation in MDMA therapy comes from other schemas or from sensory input. In either case, reconsolidation tends to create more accurate models of reality. Ecker (2015, p. 28) states:

  > When two mutually contradictory schemas are juxtaposed consciously, the schema that more comprehensively or credibly models reality, and therefore more usefully predicts how the world will behave, reveals the other schema to be false, and the

> falsified one is immediately transformed [reconsolidated] accordingly.

Based on, Ecker et al. (2024), I think reconsolidation updates schemas toward whichever representation more credibly models the person's experience, which generally improves accuracy but is not infallible; the "winning" schema is whichever feels more credible during juxtaposition, not necessarily whichever is most accurate in an absolute sense. I am also not aware of any unambiguous instances of MDMA therapy unlearning healthy schemas or creating false beliefs.

Chapter 5

The Arc of Healing

As described in Section 3.3, healing is fundamentally a process of aligning stuck schemas with current reality and integrating previously-avoided information.

MDMA-facilitated reconsolidation may cause a straightforward decrease in symptoms and increase in mental health for those with simple issues. However, as described in Section 3.6, those with disorganized attachment or other forms of severe complex trauma face a long process of unpredictable ups and downs that gradually tends upward. MDMA and reconsolidation help you confront previously avoided sensations and emotions. Confronting them may then activate other stuck schemas, producing new fear, anger, sadness, grief, dysfunction, etc. You then have to reconsolidate these newly activated stuck schemas, which might surface still more previously-avoided information. This cycle of reconsolidation → decreased avoidance → reaction → reconsolidation → decreased avoidance → reaction continues until there are no more major avoided information and stuck schemas. Note that even this cycle is a simplification of an even more complex and inscrutable process described in Section 3.6.

The simplest issues may be resolved in a single session, but my rough estimate is that the most severe mental illnesses require thousands of hours of reconsolidation to resolve. This process is necessarily spread out over a long period of time for those with complex trauma because therapy hangover limits active reconsolidation to 2 hours/day in my experience.

MDMA is a moderately scarce resource because of the unclear risk that frequent or high-dose sessions may cause long-term tolerance (Farré et al., 2015; Parrott, 2005), long-term cognitive impact (Coray et al., 2025), and valvular heart disease (Droogmans et al., 2007). Thus, I suggest achieving the bulk of reconsolidation with sober reconsolidation exercises like those described in Section 9.1 if possible.

If you want to progress as quickly as possible, I recommend maxing out on reconsolidation exercises whenever you have the time. There are two difficulties to consider, though. First, while you might be able to achieve 2 hours of active reconsolidation a day, that will also create multiple additional hours of therapy hangover. Second, reconsolidation exercises typically require some additional time for the overhead of figuring out what your stuck schemas are and finding mismatches, not to mention all the other tasks involved in practitioner-guided therapy.

If you can, try to develop techniques that require less overhead, like the first one in Appendix C. I've also noticed in my practice and that of one person I know that you don't have to experience a therapy hangover for as long when you reconsolidate right before going to sleep. The therapy hangover resolves while you sleep.

Unnoticed avoidance makes it difficult to assess how much material you need to work through. You could fill out the Buchanan et al. (2024) at least once to get a rough inventory of your stuck schemas, though it's not essential. It's the broadest scale I'm aware of and should catch a wide variety of common stuck schemas. You can fill out the scale again every few months to track your therapeutic progress. The scale is long, so it may be worth finding and using a shorter, more specific scale if your stuck schemas are limited to certain areas. That will save you time if you're filling it out frequently.

Unnoticed avoidance and lack of understanding about what improvements are possible also make it difficult to assess when you are done with the process. If you get to the point of thinking you are done or close enough, I suggest working through every item in Chapter 7 during MDMA sessions. Filling out Buchanan et al. (2024) may also help. These will help you uncover, understand, and reconsolidate many of the stuck schemas that even mentally healthy people typically avoid dealing with or thinking about. I think it's worth working through most of the items in those sections; just getting to a point of good-enough mental health leaves a great deal of individual improvement and capacity for connection and compassion on the table.

I recommend the following resources for guidance on the long-term process of healing:

- Most skilled, ethical, and well-matched therapists who have successfully helped their clients reconsolidate complex stuck schema networks should understand your journey even if they don't have experience with MDMA or use different terminology.
- *MDMA Solo* by Phoenix Kaspian (Kaspian, 2024) is the only other manual that addresses the big picture process of healing with MDMA. I recommend it for the techniques it developed rather than the safety recommendations or commentary.

When to Schedule Sessions

It may be especially productive to schedule sessions when you have a few hours of available time most days for the next 1–3 weeks. Some people report that reconsolidation exercises like therapy are more productive than normal in this period of afterglow (see Section 8.5). Spending a few hours a day paying attention to the emotions that came up during the session may also be particularly useful (see Section 9.1 (Increasing Attention)). I list some ideas for when to schedule sessions in Table 5.1.

Table 5.1: Example decision matrix for when to schedule MDMA therapy sessions, based on available time and session cost. This represents my informed thoughts and is not based on any rigorous research process. Individuals will need to consider other factors as well.

	Low Session Cost	**High Sessi**
Lots of Time	Whenever you want, within the safety constraints of Chapter 4. Experiment with different frequencies to see how they affect therapeutic progress.	When you have enc out on post-sessior every
Little Time	Ideally, when you can make time for post-session reconsolidation, but sessions may still be very helpful without that.	When you have the schedule for p reconsoli

Chapter 6

Working With or Without a Practitioner

6.1 Practitioner Guidance vs. Self-Guidance

MDMA therapy can be safe and effective in a wide variety of contexts, including therapist-guided sessions with pre- and post-session support (Mitchell et al., 2021, 2023), do-it-yourself couples therapy (Colbert & Hughes, 2023), and solo therapy (Hills, 2023). Unfortunately, to my knowledge, there is almost no evidence regarding what level of support is appropriate in any particular case. I rely on my experience and anecdotal evidence for the following information and recommendations. I start out discussing factors that favor either practitioner or self-guidance then make a set of recommendations.

Factors Favoring Practitioner Guidance

An ethical, skilled, and well-matched practitioner may provide

- an introduction to a sometimes-complex process
- an outside perspective on things that are difficult to figure out from a first-person view

- personalized education on trauma, healing, what healthy relational patterns look like, and healthy ways to deal with emotions
- in-session assistance for exercises or issues that are difficult to solve by yourself while in an altered state of consciousness (e.g., PSIP Razvi & Elfrink, 2020)
- expert personalized assistance through the occasionally intense ups and downs of your long-term healing journey. This includes recommendations for or direct assistance with grounding techniques, non-MDMA reconsolidation, session planning, and helping you solve related issues in your life.
- adequately pure MDMA, depending on the provider
- improved medical screening for conditions that might make MDMA particularly risky for an individual, depending on the provider's expertise
- in-session medical monitoring and intervention capacity, depending on the provider. This makes MDMA therapy safer for people with some medical conditions, like severe cardiovascular issues.

Factors Favoring Self-Guidance

As discussed in Section 6.3, finding a skilled, ethical, affordable, and well-matched practitioner is difficult.

Practitioner guidance is expensive when it's not covered by insurance. In 2026, in the US, a guided psychedelic therapy session including pre- and post-session support costs in the low $1000s (Althea, 2025; Psychedelic Passage, 2026; TheraPsil, 2026). MDMA itself only costs 6–50 Euros per gram (in Europe in 2022), which is enough for 5–10 sessions (European Union Drugs Agency, 2025).

Scheduling is an obvious advantage of self-guidance; you don't need to find a day when your practitioner is also available.

Self-guidance may be the only option for those who have been traumatized by mental health practitioners or who don't trust them.

I suspect that access to a high-quality trip sitter improves the risk and efficacy of MDMA therapy in many cases. Sitters aren't replacements for practitioners, but could make effective solo sessions significantly more accessible. The sitter should be trustworthy, empathetic, and emotionally non-reactive (Thal et al., 2022).

As discussed in Section 6.2, some licensed practitioners may call the cops on you if they think you are at high risk of imminent suicide, hurting someone, or are so psychotic that you might do one of the former accidentally. Involuntary commitment can be traumatizing and often actually *increases* the risk of suicide and hurting other people.

Practitioners who are ethical and skilled but whose style or personality are not a good match for you should be easy to approach about this mismatch. They might even recommend any colleagues who they think would be a better match. The risks inherent in working with an ethical and skilled practitioner who you don't match well with mostly involve wasted time and money. However, the risks of unethical or unskilled practitioners can include

- emotional, physical, financial, or sexual abuse
- unnecessary dependence on the practitioner
- increased risk of overwhelming symptom worsening
- demoralizing ineffective treatment
- false beliefs of abuse (Scoboria et al., 2017) or false beliefs about how trauma and mental illness work

Recommended Models

- Start off with a skilled and ethical practitioner who you align with and who is MDMA-trained or personally experienced with the complexities of MDMA therapy. Transition to self-guided sessions with as-needed therapy sessions with your provider when you and your clinician collaboratively decide that you sufficiently understand the process of MDMA therapy and can handle difficulties by yourself or with a sitter.

I think this is the ideal model for most people who can access a skilled, ethical, affordable, and well-matched provider. It offers dramatically lower costs than continually working with a provider, and there are numerous anecdotal reports of people (including myself) finding it safe and effective.

- Work with a skilled, ethical, non-MDMA-trained practitioner you align well with and collaboratively assess your readiness with them. You might read Section 4.8 together. If your reward/risk ratio is high enough, you then self-guide all your medicine sessions while maintaining as-needed therapy sessions with your clinician. You also use high-quality resources to educate yourself on the nuances of effective and safe MDMA therapy and use a high-quality sitter as appropriate, which includes at least the first several sessions.

  I think this model is more difficult and riskier than the previous option, but still reasonable for most people who can't access the right provider or already have a trusted non-MDMA-trained provider.

- Self-guide all your sessions and do all of your own between-session work, perhaps talking about your healing journey with emotionally skilled friends or friends skilled in safely using MDMA for healing. You also read high-quality literature on trauma healing and MDMA therapy (like this book) and use a trusted sitter for at least the first few sessions unless you have prior experience with psychedelics.

  I think this model is reasonable for people who have the capacity to work through intense and difficult emotional experiences.

- Continually work with a skilled, ethical/accountable, MDMA-trained practitioner you align well with.

  This model is the only one that has been investigated in clinical trials, where it was shown to work well (Mitchell et al., 2023). I think this model is important for people with a personal history of recurrent psychosis or mania that the individual cannot manage by themselves. A practitioner who has experience with these states can provide valuable reality testing and guidance in case MDMA therapy activates either of these symptoms. This model may also be necessary for people with a severe lack of impulse control and escapist habits or a serious addiction to cocaine or amphetamines who might abuse MDMA if they had their own supply.

I don't recommend self-guiding without high-quality reference material and an understanding of the nuances of safe and effective healing. The lack of an accurate framework to contextualize the variety of experiences in MDMA therapy impairs safety, efficacy, and your ability to fully access the possible benefits.

Possible subsection on transitioning from guided to solo. Do you want to continue doing therapist sessions because they add as much value as they cost or because of some other reason? Non-secure attachment, inaccurate fears, etc.

6.2 Involuntary Commitment

Licensed mental health providers are usually legally obligated to call the police on you if they think you are at high risk of imminent suicide, hurting someone, or are so psychotic that you might do one of the former accidentally (Alexander, 2018). In practice, it depends on their interpretation of local laws that they may not understand, how much they fear being sued or losing their license if someone is hurt, and how much they believe involuntary hospitalization will help you. If you are suicidal or fantasize about hurting your abuser, I strongly suggest asking your provider to elaborate on their decision criteria before you open up to them. Then you can decide if you can cope with the level of self-censorship necessary to not cross their boundary. Or, you may trust your provider enough that you can be completely open with them because you know they will only tell you to go to the hospital if you really need it. In that case you could write up a crisis plan involving which hospital you want to go to, which family members or friends you want notified, etc. If your provider does call the police on you, know that police typically have little training in mental health and will likely take your provider's word over yours. There is also a good chance that the police dragging you away against your will to a place you can't leave—where medical staff may do various invasive and non-consensual things to you—will traumatize you. Even worse, Emanuel et al. (2025) found that (at least in Allegheny County) involuntary hospitalization actually significantly increases the chance of a patient killing themselves or hurting someone else over the 6 months following admittance in cases where clinicians might disagree about admitting a patient (43% of evaluations in this study). It's not clear how well these results apply to situations where multiple clinicians would all agree on admitting a patient. Alexander (2018) is a good guide for navigating/avoiding the inpatient mental health system.

6.3 [unedited] How to Find a Practitioner

M.G.'s note: /r/mdmatherapy maintains a list of practitioner directories and referral services in the sidebar (on the new Reddit; not old.reddit.com). I have not vetted most of them. You may also be able to find practitioner by talking to people at psychedelic meetups. Also, in addition to the information in this section, "Psychedelic Safety Flags" by Leia Friedwoman et al. (Friedwoman et al., 2025) is a list of green/yellow/orange/red flags to look for in practitioners.

Which therapist you work with matters a lot. For instance, in a large study done by Firth et al. (2019), after adjusting for demographic factors like symptom severity, the best 3.9% of clinicians had 77.2% of their clients recover; the recovery rate for therapists in the average range was 58%; and the 3.9% of clinicians who had the worst outcomes only saw 41.4% of their clients recover. Note that a significant percentage of people recover even without therapy, so the worst therapists might have a negative influence on their clients.

T.H. hopes that the following recommendations, which largely emerge from personal and clinical experience,[1] can make finding a therapist less overwhelming and more hopeful for those who are struggling with the search.

Even under optimal circumstances, accessing mental healthcare can turn out to be a slog. These challenges are common knowledge for people who provide community mental health services or who have accessed them repeatedly: the financial and administrative costs are often daunting. Interacting with licensed mental healthcare professionals is inherently vulnerable for many, especially those who have witnessed or experienced carceral hospitalization or forced medication. Intake interviews frequently demand intimate details of your finances, sexuality, and medical and mental health. All of this is the price of accessing a clinical relationship that may or may not be very helpful. If it isn't, clients may feel they need to stick with it because they do need help, and finding someone else to help them seems like more than they can take on. That doesn't necessarily mean the clinician will stick around, especially if they are students whose clinical internships end only a few months after starting to work together. Online services, which have endeavored to bridge the gap between what is needed and what is easily available, are plagued by serious ethical and product-quality concerns (Arshad, 2024; Bartov, 2023; Better Business Bureau, 2024; Foisy, 2024).

The good news is that there *are* excellent clinicians out there, and not all of them are late in their careers. Early career therapists, like those who tend to staff more affordable clinics, provide just as good of care as their more experienced counterparts (Goldberg et al., 2016).

---

[1] T.H.'s experience includes some common knowledge of professional norms among the licensed mental health professions, much of which can be found in the ethics codes of the major licensed mental health professions. These include codes of ethics from the National Association of Social Workers, American Psychological Association, American Counseling Association code of ethics, American Association for Marriage and Family Therapy, etc.

Here are T.H.'s recommendations on how to find one that's right for you.

Bad Therapy is Worse than no Therapy

Remember that really bad therapy can really hurt you (Hook & Devereux, 2018), and as such it is worse than no therapy. Bad therapy can leave you stagnant for a long time with the impression that no real help for you exists. It can make your symptoms worse without subsequently improving them. In the worst scenario, it can leave you with additional trauma. See Section 6.3 (MDMA-Specific Complications in Obtaining Practitioner Care) for a discussion of the particular challenges of avoiding negative and damaging clinical experiences in the realm of MDMA therapy.

Good therapy is often very uncomfortable (Ecker et al., 2024). However, you should feel a sense of mutual trust, respect, collaboration, and consent regarding the goals you are working towards and the methods you use to get there (Greenspace, 2023). T.H. thinks a skilled and well-matched clinician will take the time to help you develop trust commensurate with the discomfort of what they are asking you to undertake. They should do this even if you don't understand the methods your clinician is using to help you.

Although T.H. cannot speak to the specific trade-offs of your situation, in general T.H. recommends continuing to search until you can access good therapy.

Trust Your Personal Experience

Trust your perceptions, because your personal experience of your clinician impacts the efficacy of your treatment (Horvath et al., 2011). You don't just need a good provider, but a good provider who is also a good fit with you. T.H. strongly recommends using the BR-WAI (Greenspace, 2023), an empirically validated tool for understanding how you and your therapist are connecting and what might be required to improve that connection.

In T.H.'s experience, it's a good idea to look for someone it feels like you could say anything to. If you cut your arm, any skilled emergency physician will be able to competently stitch you up. The same is not true with a comparable degree of psychiatric injury (Firth et al., 2019). In mental healthcare, the bedside manner is part of the intervention, and the same bedside manner doesn't work for everyone. Although the most skilled mental healthcare providers connect well with an extremely broad spectrum of clients, it is unrealistic to expect every provider to *click* with every client. Many excellent mental healthcare providers ultimately focus narrowly on particular populations of interest to them. If you aren't connecting with a particular clinician, it doesn't mean anything is wrong with them, and it doesn't mean anything is wrong with you.

Feedback

Good therapists love feedback and direction. It is likely that many therapists who could do great work with you if you are expressive about what you need and want would, in contrast, be bad therapists for you if you aren't expressive about those things (Delgadillo et al., 2022; Lambert et al., 2001; Macdonald & Mellor-Clark, 2015; Oanes et al., 2015). If you can, T.H. recommends asking clearly for what you want and talking clearly about how things feel. Additionally, there are several instruments that have been developed to help therapists measure and improve their performance. Examples include the BR-WAI, the ORS-SRS, the Core-OM, or the QR-45.2. If your therapist asks you to participate in one of these formal feedback mechanisms, T.H. recommends doing so, even if it feels awkward. These measures really appear to help therapists provide better services.

I've observed that it is sometimes helpful to imagine what therapy would be like if it went as well as you could possibly imagine and to share this, and your fears about the process, with your clinician. You can also ask if your ideal hopes align with their experience of their methodology, which may help you set expectations for your treatment. You can use the list above (of what a skilled therapist can provide) to identify ways you would like your therapist to support you. It may be helpful to identify which forms of support you are interested in receiving and to spend some of your initial sessions learning how your clinician feels about providing support in those particular ways.

If you have been working with a particular clinician for a while and your issues are within their main practice area, you are *well* within your rights to ask them to learn or implement a new-to-them intervention. An accredited therapist is required to complete continuing education hours anyway. There are many legitimate reasons they may say no, like cost, time, and accessibility of additional training, and that's OK too, but please don't be afraid to ask.

Sometimes things don't work out even after you've made some effort to recruit your clinician's help in fixing the situation. As a final service, they may be willing to spend some of your final few sessions helping you find and connect with someone who is a better fit.

Boundaries

Good therapists have good boundaries. Clients often come to therapy without an understanding of what healthy therapeutic boundaries are and why they might be important. That's OK; it is the clinician's job to have this knowledge, to share it with the client, and to assert boundaries as needed.

One of the most critical reasons boundaries are crucial in therapy is a phenomenon called adverse idealizing transference (AIT) (Devereux, 2016; Hook & Devereux, 2018). Idealizing transference is a phenomenon in which clients develop strong positive feelings towards their therapist. This can be totally healthy and extremely helpful to the course of therapy, supporting clients in their sense of safety and their ability to sustain focus and effort through the sometimes severe discomfort of healing. However, in some cases, it is possible for these positive feelings to be so strong and misdirected that they cause considerable harm, causing lasting distraction and disruption in the client's life, potentially continuing for decades. This situation creates a severe vulnerability that the therapist, if they are unscrupulous or unskilled, may exploit (intentionally or not) for emotional, sexual, or financial gain, creating severe trauma for the client and sometimes impacting others as well. In these cases, idealizing transference has become adverse idealizing transference. Just as helpful drugs sometimes have side effects for a small percentage of the people who use them, a small percentage of therapy consumers experience AIT.

AIT can happen even when a clinician is doing everything right (Devereux, 2016). However, both the emotions of AIT and the harm created by them can be greatly amplified when therapists fail to communicate and follow through on healthy boundaries. Here are some commonly recognized healthy professional boundaries for therapists:

- The clinician does not disclose details of their personal life to you unless that disclosure enhances your treatment and is motivated by a desire to promote your welfare.

- The clinician avoids dual relationships wherever possible. For example, if a client cannot pay for services and offers to do yard work or provide other professional services in barter, it would cause a dual relationship to accept this offer. The most commonly accepted exception to the dual relationship rule is in extremely rural practice, where access to services is very limited. For instance, a clinician might provide mental health services to someone who is also their children's pediatrician. However, in these cases, other professional boundaries should still be maintained on both sides.

- The clinician is very clear from the beginning of treatment about their policies. These include the location and timing of sessions, confidentiality practices, payment details, acceptable communication channels outside of therapy sessions, contact outside the therapy context, lateness, and missed sessions. The clinician follows through on these policies as stated and communicates any policy changes in a timely way.

- The clinician does not communicate to you in any way that they feel differently about you than they do about their other clients, or that they treat you differently than their other clients.
- The clinician does not permit or encourage the exchange/offering of significant gifts, especially financially significant gifts.
- The clinician does not provide advice outside the realm of their expertise; most clinicians minimize the time they spend giving advice even within their expertise, because supporting clients in the process of arriving at their own conclusions is more aligned with ethical standards and more effective towards lasting change.
- If physical touch is engaged in at all, it generally should be minimal, such as a brief hug at the end of each session or a single brief hug at the termination of treatment. Physical touch may be avoided entirely, and if present, must be for the client's benefit, not the clinician's.
- Even if the clinician is providing therapy on a very generous sliding scale basis that is essentially free for an individual with great financial need, it is a good sign if they insist on always charging a fee. A fee less than a dollar can still serve as a healthy reminder about the nature and boundaries of the relationship, helping both client and clinician maintain a mindset that minimizes the risk of AIT.

Therapists are, unfortunately, not explicitly educated on AIT at this time. T.H. strongly recommends that all clinicians read the two reference articles for this section. That said, therapists in all the major licensure categories should be familiar with and generally compliant with the boundaries listed above; understanding the importance of boundaries and respecting the vulnerability of clients are important topics in their clinical training.

Per item 3 on this list, if you are a client with a high number of risk factors for AIT (Devereux, 2016; Hook & Devereux, 2018), T.H.

recommends discussing preventative measures with your clinician. You might also heavily weigh boundary practices in your assessment of clinicians' fit for you. Note that these risk factors are not required to develop AIT, nor do they guarantee AIT. They simply correlate with an increased risk. The risk factors include

- a history of dependent/idealized relationships, especially with health practitioners
- an approach to therapy that is primarily seeking care, rather than insight
- unrealistic views of what therapy can provide
- being female, especially if working with a male therapist who is older than you
- having a therapist of a gender you are sexually or romantically attracted to
- being a sexual minority
- experiencing significant symptoms on a spectrum with borderline or narcissistic personality disorder

A final recommendation on boundaries and the prevention of AIT: although confidentiality is a critical therapeutic boundary, T.H. feels it is an excellent sign if your clinician seeks regular supervision and consultation as needed. They need to disclose some details to their supervisor or consultant, but it should only be the minimal amount necessary. Neither you nor your clinician should feel that anything is happening in the therapy room that they would be ashamed or embarrassed to disclose to a trusted friend in your case or a trusted HIPAA-compliant colleague in theirs. Therapy should feel private, but if it starts to feel like a secret, something may be off.

Targeted Therapy

In T.H.'s experience, in some circumstances it is important to seek therapy that is targeted to the challenges you are experiencing. One of the most common and arguably benign forms of bad therapy happens when clinicians offer a generally empathetic and supportive environment without bringing clients into a space of productive discomfort. This often results in therapy that feels pleasant but not very helpful and which perpetuates the damaging myth that working with a skilled mental healthcare provider is interchangeable with, but pricier than, having an empathetic friend.

Although the current diagnostic system is substantively flawed (Cohen & Öngür, 2023; Eaton et al., 2023), T.H. recommends that a best practice for seeking effective mental healthcare is first obtaining one or more accurate diagnoses. Second, research the most evidence-supported treatments for your diagnoses. Then, seek clinicians who are trained and experienced with those specific treatments and ideally, who are experienced using those specific treatments to address those specific diagnoses. According to T.H.'s experience and Thomas Insel, MD (Klein, 2021), this leads to radically better outcomes than a less targeted search for mental health treatment.

A less medicalized approach to this process recommended by a colleague (Sinback, 2024b) is to identify what kind of change you would like to make with the help of a practitioner. Then, do some research on how people seem to be working towards that change in various contexts and seek a practitioner who has a good reputation or training in that method. This may be especially appropriate if the help you are seeking is not specifically oriented around mental illness.

This whole sequence may or may not feel accessible to you. However, T.H. recommends that if you are experiencing therapy that feels pleasant but not as helpful as you need it to be, it is worth discussing it with your therapist. You can ask what specific interventions and modalities they are employing and what might work better. Often, in this situation, you as a client require a treatment method that will push you more.

Finding a Therapist Can Be a Long-Term Project

If you feel daunted by the process of finding a good therapist, T.H. strongly encourages you to recruit some support and treat finding a therapist as a long-haul effort. For example, if you have a supportive friend or partner, you could ask them to commit to providing you with your favorite takeout every time you complete ten or fifteen *actions* on your therapist search. These actions might include getting a list of clinicians your insurance covers, messaging therapists to see if they are taking clients, completing an intake interview, scheduling a session, and completing a session. Alternately, you might start a text thread with your closest supporters, where you can report your efforts and be rewarded with GIFs and emojis. A perspective shift that may be helpful when searching for a therapist is to regard each failure or action as an intermediate step towards finding the help you need and worth celebrating.

T.H. recommends completing three or more sessions with a given therapist before committing to giving treatment with them a try. If you luck out and find a great fit on your first or second try, the BR-WAI will help you have confidence that you've really found what you're looking for. If a particular clinician (or a string of clinicians) is not a fit for you, that's OK too; that doesn't mean you have anything to apologize for. It's good to trust the process and expect that it may take some time.

Accreditation and Safety

T.H. would feel remiss not to acknowledge that unlicensed[2] or minimally accredited mental healthcare providers are common (Aboujaoude, 2020), often billed as coaches, guides, shamans, pastors, or

[2]For the purposes of this section, unlicensed does not referring to pre-professionals and pre-licensed professionals who are operating under the license of an independently licensed professional as part of their training progression. Examples would include a resident counselor or psychologist. In terms of accountability and ethics enforcement, T.H. feels comfortable recommending this class of providers at the same level as independently licensed professionals.

simply healers. In this era when a mass-scale mental health crisis is met with a healthcare affordability crisis in the USA, individuals in need of healing seek assistance wherever they can. Indeed, talk therapy and psychiatry are modern inventions, and both common sense and through personal experience suggest that skilled and ethical healers of mental and emotional distress exist across many contexts and training levels.

That said, it can be extremely difficult to verify whether the individual you are considering working with is ethical and skilled, and working with an unskilled or unethical provider can be extremely harmful (Hook & Devereux, 2018; Miller, 2024). Although the protection offered by working with a licensed counselor is imperfect, T.H. feels it is a critical consideration, especially when working with MDMA, as described in Section 6.3 (MDMA-Specific Complications in Obtaining Practitioner Care).

T.H. perceives significant risks associated with using an unlicensed mental healthcare provider. They may

- provide, and charge for, interventions that are useless or harmful.
- not have been trained in differential diagnosis, and in any case are not legally permitted to diagnose you.
- provide services to you while they are impaired through the use of drugs or alcohol.
- not have received training about the importance of boundaries and of respecting the vulnerability of clients within the power dynamics of mental healthcare. If this is the case, they are less likely to express and enact boundaries and other practices that minimize the risk of adverse idealizing transference (AIT).
- enact harm that emerges not from the interventions themselves, but from other aspects of how they do business. For instance, unnecessary dual relationships or boundary violations can leave

clients feeling disempowered, violated, or humiliated across multiple domains of their lives.

- encourage you, during particularly vulnerable and suggestible times, to make decisions that are bad for you and your life; they may leverage their intimate knowledge of your trauma to exploitatively encourage you to make choices that benefit them at your expense.

- not experience any negative consequences for physical, financial, or psychological harm that may come to you in their care. Exceptions might include crimes they commit in a way that would be recognizable as criminal and punishable by law even outside a therapeutic relationship.

The protections provided by working with a licensed professional are mediocre. For instance, the field of mental health talk therapy is still relatively young, and it is widely accepted that talk therapy is as much art as science. T.H. has observed that most fully licensed professionals center their practices on interventions that could be covered by insurance, and this may provide some probabilistic guardrails against interventions that have no empirical support at all. However, even among fully licensed practitioners, there is little enforcement that compels clinicians to focus on the most empirically validated interventions or delivery methods. There is also little enforcement for matching a client's particular situation to the most empirically validated treatment for that specific situation. If these practices are important to you, T.H. recommends asking many detailed questions about them when you are searching for a well-matched therapist.

Additionally, T.H. has observed that various systems of power and oppression can and do play out in the therapy room if clinicians are not actively, vulnerably, and skillfully working to avoid this outcome. The prestige and respect generally afforded to therapists can sometimes foster hierarchical and non-collaborative dynamics. Therapists sometimes say extremely inappropriate, dismissive, harmful, or stigmatizing things. That can be even more harmful than normal because those statements were made by a therapist, someone who they perceive to be an expert and who is supposed to have the answers. Licensure does little or nothing to protect against or prevent these forms of harm.

Although cultural acceptance is improving, mental illness is still very stigmatized, and there are many reports of clinicians who turn on their clients, abusively labeling them "borderline" or simply crazy if those clients file a complaint against them (Hook & Devereux, 2018). The privacy of the therapy room, the power of stigmatized diagnoses, and the prestige of the therapist role mean that in "he said, she said" adjudications, a client is unlikely to be listened to. The situation is further complicated by the reality that clinicians really are misrepresented and attacked occasionally (Gutheil, 1991; Williams, 2000). This can emerge out of clients' profound attachment wounds, overwhelming trauma, and at times delusions, hallucinations, and paranoia. It is inherently very difficult to tell from the outside, and sometimes from the inside too, what has really gone on. As such, many therapists may be more likely to empathize with their peers (Hook & Devereux, 2018) than wronged clients when they hear about misconduct by colleagues. This unfortunately creates a robust haven for a minority of unethical clinicians,[3] extremely incompetent, negligent, or predatory.

Despite these shortcomings, T.H. feels it is important to highlight the advantages of working with a licensed clinician and encourage you to weigh the risks carefully. Some certificates for coaching or counseling can be obtained in a few weeks to a month for less than a thousand dollars. These kinds of certifications do not bring significant professional accountability. The loss of such a certification doesn't usually create significant occupational impairment, and certifications

[3]"Studies generally show remarkable consistency in age, gender, and practice characteristics in that the typical transgressor [of sexual relationships with clients] is a middle-aged male therapist in solo private practice who engages in a sexual dual relationship with one female patient" (Celenza, 2024).

this small are not typically backed up by a licensure organization with sufficient resources to keep track of practitioner misconduct or enforce consequences (Carr, 2011). Obtaining independent licensure as a mental health clinician demands years of study, additional years of supervision, and an often six-figure financial investment in education and accreditation. If a fully licensed professional practices therapy while they are impaired, crosses sexual boundaries with a client, or exploits clients for financial gain, the client can file a complaint with the clinician's licensure board (Barsky & Spadola, 2023; Vinson, 1987). If the clinician is then found to have committed the harm described in the complaint, they may, depending on their particular license and jurisdiction

- have their name published on a state registry that lists the misconduct they were found guilty of
- be required to take ethics classes or complete other professional development work
- be required to work under a supervisor for a period of time, and during that time announce to every single client at the start of treatment that they are working under the license of another professional and who that professional is
- lose their access to providing treatment through a hospital they had previously worked through, or they may lose the ability to have their work reimbursed through insurance
- have their license entirely revoked for severe or protracted misconduct or if they refuse to comply with rehabilitation requirements

Independently licensed clinicians must typically pass a criminal background check to become accredited (Dunlap et al., 2021). They may also lose their accreditation if they are found guilty of a felony in other areas of their lives (Barsky & Spadola, 2023). This is an important safeguard because a pattern of exploitation across various domains of their life is one of the earmarks of a genuinely predatory individual (Cooke & Michie, 2001). These consequences are not always commensurate with the harm caused, but at least one study indicated that the process of filing a board complaint against a harmful clinician tends to be a positive experience for survivors (Vinson, 1987).

To check whether your clinician is licensed or accredited in some way, try asking them for their license number and then looking it up on the website of their professional association. There are also several professional directories (e.g., Psychology Today in the United States) that will only list clinicians after verifying their credentials. In the United States, there are *many* qualifications and certifications that allow a practitioner to legally provide mental health counseling.

To further add to the confusion, T.H. has observed that even within the same license, training can vary widely. If you are committed to working with a licensed professional, T.H. recommends searching for providers based on the professional's experience, preferred client demographic, or treatment modalities, and then verifying their license. You can attain a clearer understanding of the details of a particular clinician's background with the particular issues you are having or the particular modalities and interventions you are interested in being treated with by asking them detailed questions. For example, how many clients have you worked with who have x diagnosis? What is your training and background in y intervention? They might answer with details including trainings or continuing education units they have completed, books they have read, classes that were part of their degree, relevant experience with those populations before they became a therapist, and much more. All these details are highly variable between individual practitioners, but the baseline safety protections that come from working with a licensed professional are always attached to the specific license they hold. Finally, please note that the fields of life coaching, guiding, or other unlicensed practitioner support often serve as havens for individuals who have lost a license due to misconduct.

If you choose to work with unlicensed practitioners, T.H. recommends exclusively working with practitioners who are very clear about their scope of care, participate in accountability and transparency practices like accountability pods[4] and publicly post their business ethics (as exemplified in Sinback, 2024a). Indeed, T.H. feels these practices, though imperfect, are a green flag from providers of any licensure level. T.H. hopes they will become normalized across mental healthcare, along with providers routinely seeking appropriate supervision and consultation and being transparent with clients about who is supervising or consulting with them. Finally, many of the suggestions from Section 6.3 may be adapted to reduce your risk profile, even if you are not working with MDMA. The caveats T.H. placed on that list apply here as well. No matter who you work with, if they choose to harm you through their behavior, that is not your fault.

To understand the scope of care when you are considering working with an unlicensed practitioner, it may help to ask detailed questions about which services they provide and how and where they learned the skills to provide those services. Unlicensed practitioners are forbidden by law from diagnosing or treating mental illness (Aboujaoude, 2020; Joseph, 2022). However, many of the interventions that are used to treat mental illness can be appropriately implemented by unlicensed practitioners and have many legitimate uses outside mental illness treatment. For instance, some coaches teach reconsolidation, thought records, mindfulness practices (see Brach, 2023), or myriad other skills and strategies that constitute legitimate mental illness interventions. These skills are not exclusively relevant to mental illness; a reasonable person might learn them simply to enrich their life and increase their personal growth. When unlicensed practitioners deploy them, they are also not doing so (or legally should not be doing so) in the context of a treatment plan wherein a therapeutic relationship is constructed and interventions are performed that will

[4] Accountability pods are a practice developed by the Bay Area Transformative Justice Collective. See Mingus (2016).

specifically address a specific mental illness. As such, it can reasonably fall within the domain of unlicensed care to teach these skills and to help folks identify some circumstances where it is helpful to use them. These kinds of services may significantly help you self-manage or self-treat your mental illness, especially (as is often the case) if you are unable to access high-quality licensed mental healthcare.

On top of this, T.H has observed that unlicensed practitioners are frequently helpful for bridging a gap between the needs of people with mental illness and the necessary level of support (often, more than meeting with a clinician once a week) that would allow them to achieve a significantly better quality of life. For example, a coach might call you several times a week when you are most vulnerable, to help you accomplish task initiation or avoid doom-scrolling, a service a therapist is unlikely to provide. Unlicensed practitioners might help you with more practical, seemingly superficial, yet crucial aspects of changing your life for the better: sitting with you while you fill out job applications, or declutter your house, or practice eating mindfully. Some unlicensed practitioners can help you learn and apply healthy relationship skills that will radically improve your life.

In contrast, T.H. believes the very best mental health practitioners can complete a detailed biopsychosocial assessment of your situation, accurately diagnose you, and offer you mental health interventions that are well-suited to you. They offer a deep understanding of how various constellations of symptoms tend to show up and some awareness of pitfalls you are likely to encounter along the way based on that. They will have an understanding of what level of care is appropriate to your situation, and if they do not have the particular expertise appropriate to your condition, they will help you find a provider who does. They may know much sooner than you do if it is urgently important for you to receive a higher level of care. For instance, early intervention for a first psychotic episode has a massively positive impact on the lifelong trajectory of individuals with psychotic spectrum disorders, and timely intensive treatment for eating disorders or substance use disorders can be lifesaving. Particularly if you do not have a case manager to take on this role, a mental health practitioner may help you work through what is stopping you and learn the skills to recruit and coordinate care from many sources. Examples include

- psychiatric prescribers
- specialist care providers like a dietitian or a trauma-informed OB-GYN
- peer support from friends and loved ones
- peer support from potential future friends and loved ones, as when joining an activity group that helps you stay consistent with positive coping strategies
- community programs, like a senior center, meditation center, or gym
- coaches, ecclesiastical leaders, or other appropriate unlicensed practitioners
- personal care assistants to assist with the activities of daily living, when appropriate

Although reconsolidation may ultimately heal most or all of your mental illness in a deep and durable way, meanwhile you must live with your symptoms and build the best life you can, despite your symptoms. The best mental healthcare providers are experts not only in addressing the root causes of mental illness but also in helping you reduce the incidence of your symptoms and in helping reduce the impact of your symptoms on your life.

MDMA-Specific Complications in Obtaining Practitioner Care

Finding practitioner support for MDMA therapy carries challenges over and above the challenges of finding mainstream mental health-care. The specific nature of MDMA therapy amplifies the vulnerability of seeking psychiatric care, including (T.H. surmises) vulnerability to AITs (Meikle et al., 2024). Some of these challenges emerge from the legal status of MDMA therapy. Finally, these factors have combined to create an existing culture of underground MDMA therapy that can be painfully exploitative (Hall, 2021). An effective process for securing practitioner support must take all of these challenges into account.

Challenges caused by the vulnerability of seeking psychiatric care

As discussed in Section 6.3 (Accreditation and Safety), there is a severe power imbalance between providers and consumers of mental healthcare. When a fully licensed professional enacts behavior that is clearly abusive, it creates the case in which T.H. would expect the maximum possible structural support for accountability within conventional criminal-legal and administrative systems. However, even in these cases, proving misconduct and enforcing appropriate consequences for it is not always possible (Biaggio et al., 1998). On top of this, I've observed that many individuals consuming mental health care feel that they must put up with a certain degree of discomfort to access care they may desperately need. T.H. finds they are often understandably unskilled at detecting the difference between the healthy discomfort of effective treatment (Ecker et al., 2024) and discomfort related to mistreatment or misconduct.

Challenges caused by the nature of MDMA

As detailed below, MDMA can increase client vulnerability. MDMA can create experiences of great mental and sometimes physical intensity, of much longer duration than a traditional therapy session. Long sessions, sometimes a risk factor or red flag for harmful boundary violations (Strom-Gottfried, 1999), are necessary in MDMA therapy. MDMA can generate sexual feelings (McElrath, 2005), and T.H. thinks the same enhanced empathy and openness to experience that make them such a fantastic aid to reconsolidation work may mean they leave individuals who take them more susceptible to persuasion. Altered states of consciousness also impair the ability to be appropriately cautious and thoughtful about risk. These factors make it even harder for clients to tell when providers are behaving inappropriately. Additionally, a significant subset of MDMA clients are having one of the peak experiences of their life, or even *the* peak experience of their life. T.H. thinks this could cause practitioners who routinely facilitate this therapy to feel godlike. Even excellent practitioners may need to work very hard, when administering MDMA-assisted therapy, to maintain appropriate boundaries, humility, and client-centered care. This dynamic can undermine even good clinicians' ability to provide quality care. . Although there is no data on the topic, T.H. feels this dynamic almost certainly increases the risk of AITs with MDMA therapy, particularly with providers who are not scrupulous regarding boundaries or thoughtful about preventing this specific risk. All of these aspects of MDMA amplify the already formidable power dynamics between a therapist and a client, regardless of legal status or other factors.

Cite Matthew Johnson Summit talk

Challenges caused by legal status

The legal status of MDMA creates additional risk for those seeking practitioner support for Psychedelic-assisted therapy (PAT) in several ways:

- Mental health providers' training and experience around MDMA has been extremely limited by their legal status. This includes the ability of practitioners to experience MDMA as a part of their training, which most clients and clinicians feel improves clinicians' ability to provide MDMA therapy and which many feel is essential to that ability (Liknaitzky, 2024). However, clinicians generally cannot legally access these experiences with solid supervisory support. If clinicians even disclose their history of MDMA use, they could theoretically face some degree of legal or licensure-related repercussions. Even if the actual risk of repercussions is very low, the perceived risk may be high enough to prevent clinicians from becoming involved in these activities.

- The fact that everything has to be underground makes it harder for people to find each other as needed. This can create a sense of scarcity, as in, "I have found a provider and need to stick to them because searching for a better fit feels daunting or impossible."

The whole legal status section needs expert reviewer or citation help.

- When the medicine is criminalized, clients who are harmed in therapeutic processes related to the medicine often legitimately fear legal or social consequences to themselves if they report crimes against them that were undertaken during this process. This makes the power dynamics astronomical.

Existing culture of exploitation in psychedelic therapy

Taken together, the above factors render it unsurprising that exploitation, spiritual bypass, and cultic dynamics seem to have been woven into the culture of underground psychedelic therapy (J. Brown, 2020; Quinn et al., 2021). MDMA facilitates suggestibility and profound experiences. Providers face challenges in staying humble and client-centered when providing these interventions and have deep shelter from any legal threat about malpractice that is created when it is all illegal. These all play a role in a culture of guides, healers, shamans, and therapists who can harm with impunity. Multiple accounts have surfaced of the need to develop a better framework of accountability in these underground, unregulated communities.

If you are interested in learning more about this topic, T.H. recommends the following resources:

- "Addressing Abuse and Repair: An Open Letter to the Psychedelic Community" by Diana Quinn et al. (Quinn et al., 2021)

- "Ethical Transgressions and Boundary Violations in Ayahuasca Healing Contexts: A Mixed Methods Study" by Jessica Brown (J. Brown, 2020)
- "Ending the Silence Around Psychedelic Therapy Abuse" by Will Hall (Hall, 2021)

Individual Actions to Improve Safety of MDMA Care

In this era of regulation, many people with mental illness are driven to work with unlicensed providers due to mistrust of the mainstream mental health system or the financial or logistical inaccessibility of high-quality licensed care (Aboujaoude, 2020). Additionally, while the system of licensed providers presumably prevents many of the most egregious violations, some serious harm can and does slip through, as evidenced by the whistleblower reports from the MAPS trials (Mustafa et al., 2024). In this context, T.H. believes there are steps some individuals may be able to take to increase their safety when seeking MDMA therapies.

T.H. wants to emphasize that even if you do nothing on this list, any care provider who violates the sacred trust you have put in them as a healer by mistreating you is fully responsible for their actions. Those who are most badly in need of psychiatric care are often under-resourced and likely to find it challenging or impossible to carry out the degree of vetting or preparation they might have ideally preferred. T.H. provides these suggestions not to cast blame or create additional responsibilities for care-seekers, but as a resource with which those who are able may be able to improve the risk profile of their healing endeavors.

License and Certification

Research their license and certifications carefully. See Section 6.3 (Accreditation and Safety) about accreditation and the protections it provides. Remember that practitioners may claim a license they don't have, or they may have some sort of certification that does not carry enough weight in their professional life to enforce any kind of accountability. T.H. particularly recommends taking other precautions if your clinician does not belong to a licensing body that publishes the names of clinicians who are found to have violated its ethical code.

Evaluate Fit

Follow the advice above (see Section 6.3) about finding a clinician who is a good fit for you, including working with them for at least three sessions (preferably five or more) before making any decisions about using their assistance for medicine work. During this time, observe their boundaries carefully, and have frank discussions about their qualifications, your needs and expectations, and the modalities that would be employed during the medicine work.

Methodology and Expertise

Research their therapeutic methodology and expertise. It is T.H.'s hope that the evidence-based theoretical framework offered in this book, positing that the primary mode of healing MDMA offers is memory reconsolidation, can assist some clients in assessing whether they are interested in engaging with various therapeutic methodologies. Accordingly, T.H. particularly recommends seeking clinicians who have a strong background in addressing dissociation and panic, in dealing with trauma generally, and in assisting their clients in working through somatic manifestations of trauma/somatic release.

Disclaimer that the research in conventional therapy does not support sorting by methodology. Also have this section reviewed by a sex therapist.

Although they may be particularly effective for activating some important schemas, T.H. strongly recommends avoiding methodologies that involve physical touch, *particularly* those that involve physical touch during the actual medicine work. The client may be unable to consent or make unbiased decisions about whether to continue or stop a particular therapeutic interaction in this state.

If you are interested in somatic methodologies that involve touch, T.H. recommends alternatives such as therapeutic use of restorative yoga postures. In this practice, a clinician can provide support through various bolsters and pillows, along with verbal instruction, so that there is never any need to touch you in any significant way. If for some reason you do choose to pursue a methodology that involves touch, T.H. recommends undertaking precautions to increase your sense of volition and improve the safety of the endeavor, including

- creating a detailed written consent contract in advance of your medicine session that determines what forms of touch will be acceptable under what circumstances and how these boundaries will be upheld
- using witnesses and cameras to provide you with certainty that your wishes have been respected

T.H. agrees with the assertion of the American Association of Sex Educators, Counselors, and Therapists when they suggest that there is no circumstance where sexual touch is appropriate to the therapeutic relationship (American Association of Sex Educators, Counselors, and Therapists, 2020).

M.G.'s note: Clients sometimes feel that therapy helped them remember traumatic events they weren't previously aware of (The Psychedelics and Recovered Memories Project, 2024). These may or may not be accurate recollections of historic events, and I am not aware of any reliable method of distinguishing a memory's accuracy other than independent corroboration. To avoid incorrect interpretation of perceptions during the session, practitioners should categorically avoid suggesting a memory is either true or false and shouldn't even suggest that a specific individual has avoided memories they can recover (Otgaar et al., 2025).[5]

Background Check

Get a background check on the guide and possibly on others who recommend them or who have trained them. In some cases, signif-

[5]People who retract claims of recovering accurate memories of abuse overwhelmingly blame their therapist for improperly influencing them (Otgaar et al., 2025).

icant networks of recommenders or well-regarded trainers may all be invested in unsubstantiated or potentially harmful therapeutic modalities (Hall, 2021). Historically, lineage of training has been an important way that healers are credentialed in the absence of licensure systems; if your guide has trained under someone who promotes potentially harmful therapeutic modalities, like breaking people down, or if they have been mentored by people who have a history of boundary violations against clients, T.H. recommends proceeding with extreme caution if at all. If a guide with these red flags seems like an otherwise excellent match, T.H. recommends having detailed conversations about how they relate to those practices and approaches, and taking the other safety considerations listed here especially seriously. Finally, as noted in the accreditation section, if a prospective practitioner has a history of fraud or abuse in other areas of their life, this is also worth taking into consideration, even though it may appear unrelated, because this may be a warning sign for truly predatory behavior. Cite

Accountability

Look for external accountability in the clinician's ethical structure, both philosophically and practically. One of the concerns cited by Nickles and Ross (2021) regarding safety in PAT is that there seems to be a cultural norm in underground psychedelic therapy of identifying the truth-including the truth about situations in which clients allege their clinicians have harmed them-as something that comes from inside each person. In contrast, healthy accountability practices require us to listen with openness to outside information which can make us feel downright terrible on the inside. T.H. recommends seeking providers who maintain active relationships with supervisors, mentors, or accountability pods (Mingus, 2016) who can help them receive the report appropriately if a client has a bad experience in their care. Additionally, T.H. recommends seeking clinicians who can provide you with a clearly articulated set of written ethical standards they endeavor to adhere to, whether those standards come from a professional organization or personal soul-searching.

Flight Plan

Create a “flight plan” and a safety plan collaboratively with your provider/s while sober, and review/update it before each medicine experience. Birth plans have become popular tool that birthing parents can use to (a) educate themselves about the many choices that may emerge during labor and delivery, and (b) communicate strategically with providers about their needs and preferences; T.H. feels that structured advance planning for PAT sessions could provide similar benefits (The Bump Editors, 2023). Here are some considerations you may wish to include in your flight plan if you feel you are at particularly high risk for AIT (link to assessment questions in section above), or otherwise have significant safety concerns:

- You may wish to have one or more support people present to serve as witnesses during your medicine session.
- You may even choose to delegate, “medical power of attorney” style, to allow trusted support people to make certain choices on your behalf while you are incapacitated by the medicine. If you pursue this possibility, it is important to have in-depth conversations with your support person and your clinician about what level of distress is appropriate for you to face, and what the likely outcomes of that distress may be-as well as having detailed discussions on the choices they are being entrusted to make.

What choices might you delegate? Also have this concept reviewed by an experienced psychedelic practitioner.

- You may wish to arrange to have your session filmed, possibly even from a few angles, so that you have a lot of concrete evidence afterward about what happened.

See Section 8.1 section for more information on flight planning.

Cultishness

Keep an eye out for cultic group structures or dynamics. Accounts from the underground PAT world have described closed systems centered around charismatic leaders, exploitation of the labor of group members, and abundant use of what cult scholars call *thought terminating clichés* (Hall, 2021)thoughtTerminating. Thought terminating clichés are phrases used to address cognitive dissonance while shutting down inquiry into the framework that produced that dissonance. For instance: one client said they were being told that their abuse-induced "downward spiral was just some personal failure of my own. To get over my crisis I needed only surrender, let go, and have unquestioning faith in psychedelics—and him." In other words, survivors were told to stop considering the choices and volition of the human beings who betrayed them, and instead to focus on their own complicity and potential growth that might emerge from their abuse. In contrast, in a non-cultic framework, people are allowed to ask questions and investigate what is under these thought terminating clichés. They are allowed to disagree. They are not taught to believe they are dependent on just one source for healing.

Chapter 7

Prompts for Activating Stuck Schemas

Schema activation does not always happen spontaneously during a session, and when it does, it may not cover all the stuck schemas that are worth reconsolidating in your life. This section helps you deliberately activate and reconsolidate common stuck schemas that you may not be aware of. Working through these prompts builds more resilience and equanimity than the default of just doing enough MDMA therapy to make your primary issue feel manageable.

These are going to feel uncomfortable, but that's part of the process. Start with the easiest ones and work your way up to the harder ones over time.

I also strongly suggest keeping a list of subtle (or not) discomforts that you notice in daily life. Some of these might be valuable reconsolidation targets.

7.1 Personal Schemas

See Table 7.1 for a list of prompts designed to activate many mental illness, developmental, and attachment-related stuck schemas that people commonly have. It may help to imagine a concrete scenario for each item. Feel free to skip sections you feel totally comfortable with.

Table 7.1: Schema Activation Prompts inspired by "A Comprehensive Questionnaire for Schemas Related to Psychopathology: The Maladaptive Schema Scale - Version 1.4 (MSSv1.4)" by B. Buchanan et al. (Buchanan et al., 2024). I used Claude to transform the items in the original scale from their diagnostic forms into schema-activating forms, each framed as a scenario the schema is built to defend against.

Activation Prompt	**Schema Domain**
Safety & Attachment	
-Someone you love is about to leave you without warning -The person closest to you is wavering on their commitment -You desperately need someone and no one is coming -The person you depend on most is leaving	Abandonment
-You have no choice but to let someone else take care of you -Someone is getting very close to you -You are completely dependent on another person right now -You need to ask someone for help with something painful	Excessive Self-Reliance
-You're struggling and there's no one to turn to -You're in trouble right now and there's no one you can call	Emotional Deprivation

continued on next page

Activation Prompt	**Schema Domain**
-The people around you don't care how you feel -You need emotional support and no one is offering it	
-Someone is being kind to you but you can't see their real motive -You have to trust someone completely right now -Someone is telling you something important and you can't verify it -You just realized the people around you have been lying to you	Mistrust of Others
-You just saw clearly how selfish the people around you really are -You're watching someone suffer and no one is stepping in to help -The person in front of you could become violent at any moment -Someone just revealed who they really are underneath	Others are Dangerous
-You walked into a room and immediately felt you don't belong here -You're trying to explain something important and no one understands -You're surrounded by people and none of them can see you -You are completely on the outside looking in	Social Isolation
-The people closest to you can finally see who you really are	Defectiveness / Shame

continued on next page

Activation Prompt	**Schema Domain**
-Someone just looked at you and saw what's broken -The people you love can't love you because of what's wrong with you -You are exposed and there are real reasons to feel ashamed	
-You've stepped outside your safe zone and bad things are closing in -You just realized you're exposed and nothing is protecting you -Everything is falling apart and catastrophe is coming -The world is hostile and it's coming for you	Vulnerability
Autonomy & Competence	
-You are alone and must take care of yourself with no help -You have to manage everything today entirely on your own -A major decision is in front of you and no one can advise you -You must choose and there is no one to confirm you're right	Dependence
-Everyone around you has accomplished more than you -You're looking at what you've done with your life and it's not enough -Someone you know just succeeded at something you failed at	Failure / Inferiority

continued on next page

Activation Prompt	**Schema Domain**
-You're being compared to others and you're falling short	
-A hard problem is in front of you and you probably can't solve it -You're stuck and you can't think of any way forward -Something unexpected just happened and you have to handle it alone -The problems you're facing are beyond your ability to manage	Low Self-Efficacy
-Something important is being decided about your life and you can't influence it -A critical moment is coming and nothing you do will change the outcome -You're watching your future be decided by things completely beyond you -You just realized you've never really been steering your own life	Fatalism
-Someone you love is in pain and it's your job to carry it -You can't tell where your feelings end and someone else's begin -You are dissolving into another person—you can't find yourself -Someone else's needs are consuming everything you have	Enmeshment
Freedom to Express Needs, Opinions & Emotions	
-Someone is telling you what to do and they know better than you -You must obey even though you disagree	Subjugation

continued on next page

Activation Prompt	**Schema Domain**
-Someone else is deciding what's best for you -Another person is taking control and you have to let them	
-Someone needs you and you must put them first no matter what -Someone is bringing you their pain and it's your duty to hold it -You need something right now but someone else needs more -You've given everything you have and it's still not enough	Self-Sacrifice
-You have to choose between what you want and what others approve of -Someone disagrees with you and you feel the pull to agree with them -Someone you don't even care for doesn't like you, and it's eating at you -You need to decide but you don't know what anyone else thinks	Approval-Seeking
-Strong emotions are pushing up and they won't stay contained much longer -Your emotions are taking over and they're going to cause damage -You're about to cry in front of people who can't handle it -Your emotions are dangerously intense right now	Emotional Inhibition
Spontaneity & Play	

continued on next page

Activation Prompt	Schema Domain
-Something is about to go wrong, as it always does -You're facing uncertainty and there's no reason to hope -You're waiting for the bad news you know is coming -You're looking ahead and you can't see anything good coming	Pessimism
-You just made a mistake and you can't let it go -You have to choose between being happy and meeting your standards -You performed well but not perfectly and it's not enough -You fell short and that is not okay	Unrelenting Standards
-You just made a mistake and you don't deserve compassion for it -Something went wrong and you shouldn't get away with it -You failed and now you deserve what's coming -A small thing just went wrong and you must pay for it	Punitiveness (Self)
-Someone made a mistake and they don't deserve your compassion -Someone failed and they should be held accountable -Someone let you down and they deserve consequences	Punitiveness (Others)

continued on next page

Activation Prompt	Schema Domain
-Someone did something wrong and letting it go would be wrong	
Realistic & Consistent Limits	
-Someone just told you no -You have to follow the same rules as everyone else -No one around you recognizes how special you are -You're being treated like everyone else and you deserve more	Entitlement
-Something has gone wrong and you should have been able to prevent it -Something is happening that you cannot control or influence -Willpower alone cannot change this situation -Something has been left to chance and you can't intervene	Full Control
-Your gut says one thing but the evidence says another -You're making an important decision based only on how you feel -Your emotional reaction feels like the truth even though it might not be -A problem is in front of you and thinking won't help—only feeling	Over-Reliance on Emotions
Coherence & Fairness	
-You just experienced something unjust and no one is going to fix it	Unfairness

continued on next page

Activation Prompt	Schema Domain
-You are being treated unfairly and no one sees it -Something bad happened to you and you didn't deserve it -Good things keep happening to other people but never to you	
-You're looking at your life and nothing in it seems to matter -You've been trying to find meaning and coming up empty -You can see now that everything you've worked for will be forgotten -You're watching people strive and none of it will have meant anything	Meaninglessness
-You don't know who you are right now	Lack of Coherent Identity
-Your interests and beliefs feel like they belong to someone else -You are watching yourself from the outside, disconnected -The person you were yesterday is not the person you are today	

Habit Mapping

If you hope to unlearn old habits or to reconnect to a project that you've been feeling too overwhelmed to engage with, habit mapping (see "The Habit Mapper" by Jud Brewer Brewer, 2024) can help. This consists of identifying behaviors you are hoping to adjust, identifying what triggers lead you to engage in those behaviors, and identifying the results after you engage in the behavior. For example, you might identify that you would like to adjust the habit of scrolling through social media on your phone for three to five hours every day as soon as you get off work.

Identifying triggers might involve exploring: what are the feelings that precede your going online? What are the stories you tell yourself? What physical and social environment are you experiencing that supports this behavior? Are you triggered by loneliness, by a desire to be free of demands for a period of time, or by a feeling of boredom/desire for stimulation, or all of the above? What happens when you work the second half of your workday in a communal space vs. when you work alone? What happens when you lock the phone in the glove compartment of your car during your lunch break and don't revisit it until you've walked outside to the parking lot?

The last stage of habit mapping involves checking in with yourself and taking time to feel *all* the consequences of enacting the habit. You might feel stiff and sedentary, which feels bad in your body; you might find yourself feeling anxious or tired or lonelier than when you started scrolling, or that you get caught up in cycles of self-judgment or shame about the behavior. Maybe you feel frustrated and wish you had more time for other things. Or perhaps you feel some of these things and also feel delighted and inspired by the content you are consuming. The goal of habit mapping is to observe and record each habit as it plays out in your life and to open yourself to deeply understanding whether the choices you are making are the choices that are best for you. Sitting with what you feel an intense need to numb—and with the consequences you're willing to pay for that numbing—may prime you to face those feelings during the session.

To activate the relevant schemas during a session, you can repeat this activity or just read over the notes you made when you did it sober.

7.2 Social Schemas

Imagine, as vividly as possible, being a member of a different group of people. Then, imagine how you and your friends, family, coworkers, and community would react to you. Go toward any fear, disgust, or shame and stay with that feeling. Race, gender, disability, class, political affiliation, wealth, education, attractiveness, and religion are all good attributes to consider. Add any other categories relevant to your situation. Choosing the lowest status group for each category may activate the strongest stuck schemas.

In addition to imagining you are a member of the other group, consider

- a member of the other group being part of your community: moving into your neighborhood, working with you, attending your religious institution, or being your friend
- a member of the other group having power over you: teaching your children, supervising you at work, or representing you as your elected politician
- hugging or dating a member of the other group
- adopting the aesthetics or practices of the other group
- how your closest friends or family react to one of the above items
- the other group outcompeting your group, while your group fades into history

7.3 Existential Schemas

Imagine permanent disconnection from health, material comfort, sensory pleasures, relationships, belonging, status, order, certainty, meaning, being a good person, life, and existence (including any afterlife). Additionally, consider the possibility that your fundamental assumptions about life, meaning, existence, and self might not actually be true. To my understanding, reconsolidating these stuck schemas leads to equanimity, not the loss of healthy protective behaviors or basic attachments.

Attaining persistent non-attachment and non-duality/unity requires considerably more work than using the prompts just mentioned to reconsolidate stuck schemas during MDMA therapy. Section 8.4.12 lists a few starting points.

7.4 Cognitive Flexibility and Truth-Seeking

While each of the parts in this section logically follows from the rest of the book, applying them to this class of stuck schemas is partially speculative.

I think that cognitive flexibility is both highly important and lower than it should be for most people in the modern world. Tribalistic instincts are a major contributor. While we need to form coalitions (family, work, friendships, etc.) to accomplish almost anything, our tribal tendency seems disproportionate for life in modern nation-states. This creates all sorts of problems, such as poor governmental policy because people vote based on identity (Klein, 2020), being shunned for believing or doing things that are markers of belonging to a rival group, etc.

It is common to perceive obviously false beliefs in the out-group. But how do you know they are mistaken and you aren't? The human capacity for self-deception must be strong enough to make at least one of you wrong about a supposedly obvious fact, and we have no foolproof meta-reason to believe it is them instead of us. You may think "ah, but I'm right because I've done x, y, and z to make sure", but the other person may feel they have also made sure. Confidence can easily coexist with delusion.

As the psychologist David Pinsof says on their blog about self-deception being a core aspect of tribalism (Pinsof, 2024):

> We're hyper-skeptical of claims made by the outgroup—the people we fear, dislike, and distrust. And we believe ingroup-flattering absurdities ... because it is instrumentally rational for us to do so. The benefits of status and tribal solidarity often outweigh the costs of false beliefs,

particularly if those beliefs are vague, unactionable, or unfalsifiable.

"Trapped Priors as a Basic Problem of Rationality" by Scott Alexander (Alexander, 2021b) discusses another mechanism for how emotionally charged, false beliefs become entrenched. It proposes, based on Van den Bergh et al. (2021), that when another group or another group's beliefs are perceived as highly threatening, the brain dials down the relative certainty of all incoming sensory information, including information that indicates you aren't really in danger in the moment. Thus, the false belief never reconsolidates unless the contradictory information is extremely strong and unignorable, or the contradictory information is introduced very gradually, like in exposure therapy.

Self-deception relies on a constellation of schemas that serve to justify your belief or deny the validity or existence of contradictory information. I propose that MDMA therapy can help you reconsolidate some of these stuck schemas. During a session you might consider a hated or feared political, religious, national, or cultural group—or a faction of your group if you want an easier starting point—being correct about an important point of contention. Imagine believing that point yourself. How does that feel? Imagine expressing your new belief to your friends, family, or community. How would they react? How would you feel about that? I suggest primarily focusing on noticing and staying with any fear, anger, grief, etc. that you feel during this process rather than spending all of your energy on intellectual thought. This will reconsolidate any associated stuck schemas. Intellectual engagement with the subject will work better after reconsolidation anyway, so save it for after the session. Working through Section 7.2 will also help.

The point of the exercise isn't necessarily acquiring more accurate beliefs about the specific topic you imagine during the session. Rather, the point is activating and reconsolidating any stuck schemas potentially interfering with your thought processes, which improves your general, long-term ability to acquire accurate beliefs. You could actually pick an idea that you don't think is worth considering the truth of at all, like the Earth being flat. Using that as a prompt might activate and reconsolidate a stuck fear of being like *those people*, which makes you discount all beliefs that you and your social circle consider odd, but some of which are certainly true.

The world is full of pervasive threats, fear-based messaging, and out-group dehumanization that may reconstitute some stuck schemas over time. I suspect additional strategies beyond reconsolidation and avoidance-reduction are necessary to maintain cognitive flexibility and compassion in the long run. I recommend the following:

- *The Scout Mindset* by Julia Galef (Galef, 2021) is an excellent book that proposes a set of habits for helping you avoid epistemic traps.

- *How Minds Change: The Surprising Science of Belief, Opinion, and Persuasion* by David McRaney (McRaney, 2022) is a good survey of the topic.

- *Why We're Polarized* by Ezra Klein (Klein, 2020) discusses the dysfunctional positive-feedback loops between our fears and identities and the behavior of the media and politicians. This cycle increases polarization and political dysfunction, and separates us from each other. This book is more about understanding the role of identity in politics than suggesting how to improve the situation.

7.5 Authentic Engagement with Empathy, Ethical Reasoning, and Ethical Action

While each of the parts in this section logically follows from the rest of the book, applying them to this class of stuck schemas is partially speculative.

Our innate empathy and learned schemas interact to form our actions, moral beliefs, and circle of moral consideration. As they do in other areas, stuck schemas can interfere with coherent and empathetic ethical reasoning and ethical action.

People sometimes hurt other beings, usually without deliberate intent or full awareness of the impact of our actions. This typically involves some combination of (a) a driver—fear, anger, perceived insecurity, desire, or simply a default cultural behavior—and (b) a constellation of schemas that justify the harm, deny the harm, deny the other being's worth, or prevent the harm from becoming salient in the first place. Non-salience is especially strong when harm is incentivized through intermediaries like markets, elections, or culture; consumers creating demand for cheap fish, for instance, rarely confront the conditions of extreme suffering in fish farms that fulfill it (Singer, 2023).

This dynamic implies a conundrum for those interested in acting in congruence with their values: how do you know if you're significantly and unnecessarily hurting someone if the act of hurting others is frequently obscured by a cloud of subtle avoidance, denial, and justification strong enough that the vast majority of people have historically been fooled?

Avoidance, denial, and justification in large part function to shield us from information that feels threatening (see Section 3.3). MDMA notably increases tolerance of discomfort, and I suspect it is useful for bringing attention to other beings, contemplating their worth, and reconsolidating any stuck schemas those actions activate. During a session I suggest two practices.

The first is for situations where the driver is desire or default cultural behavior. Look at photos or real-life examples of other beings and contemplate in detail what you share with them, what their life might be like, what causes them to suffer or thrive, what they deserve, how you have hurt or neglected to help them, and what obligations you may have to them. You may have to investigate what their lives are like; I provide suggestions in the list below.

The second is for situations where the driver is a perceived solution to fear, anger, or insecurity in a conflict. Try the prompts in Section 7.2.

Focus on noticing and staying with any fear, anger, or other distress you feel during this process rather than getting completely distracted by intellectual thought. This should reconsolidate any associated stuck schemas, and intellectual questions can be considered after the session is over, when there are fewer stuck schemas getting in the way. The next question is who to bring into this contemplation.

Analytically determining what beings have inherent worth is difficult; it depends on solving multiple open problems in philosophy of mind and moral philosophy, like metaethics and the hard problem of consciousness (Sayre-McCord, 2023; Weisberg, 2024). Since there are no broadly agreed-upon solutions to these, I take a pragmatic approach that centers uncertainty and the precautionary principle. This involves thinking about what features of humans are associated with having positive and negative internal experiences and then looking at what other beings possess those features. I recommend the short video "The Science of Animal Sentience: Sentience and the precautionary principle" by Jonathan Birch (Birch, 2021) for an introduction to this approach.

I propose considering beings that some major groups of people think have inherent worth, if only to check whether you're missing something others have noticed. Here are the categories that seem worth considering to me. You might disagree about where to draw the line of who to consider. I've included references for further reading that take positions of uncertainty in edge cases (e.g., AI) and compassion in stronger cases (e.g., mammals and birds).

- Yourself
- Beings you have hurt
- Beings who have hurt you

- Groups of humans your group is in conflict with or who may even be eradicating your group
- Groups of humans you don't like or approve of, or whom you think have deeply wrong values
- Humans geographically distant from you. Why does worth or obligations change or not based on physical distance? See "Famine, affluence, and morality" by Peter Singer (Singer, 2017).
- Humans with cognition dramatically different from the average adult: fetuses, young children, those with dementia, and those with severe cognitive disability
- Non-human animals. A number of scientists and philosophers recently asserted in The New York Declaration on Animal Consciousness (Andrews et al., 2024):

  > First, there is strong scientific support for attributions of conscious experience[1] to other mammals and to birds.
  >
  > Second, the empirical evidence indicates at least a realistic possibility of conscious experience in all vertebrates (including reptiles, amphibians, and fishes) and many invertebrates (including, at minimum, cephalopod mollusks, decapod crustaceans, and insects)
  >
  > Third, when there is a realistic possibility of conscious experience in an animal, it is irresponsible to ignore that possibility in decisions affecting that animal. We should consider welfare risks and use the evidence to inform our responses to these risks.

  Animal Liberation: The Definitive Classic Renewed by Peter Singer (Singer, 2023) may be the most influential secular work

[1] Conscious experience means having subjective experience, not necessarily having self-reflection.

on the inherent worth of animals and how they are treated in various contexts. It was written by a moral philosopher for a general audience and is quite readable. *Dominion: The Power of Man, the Suffering of Animals, and the Call to Mercy* by Matthew Scully (Scully, 2003) is a popular conservative Christian book on the topic. *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI* by Jonathan Birch (Birch, 2024) goes into depth about uncertainty of which animals to consider.

- Current or future artificial intelligences. This is far more uncertain than the previous categories and may depend on the specific architecture of an AI. Non-human vertebrate brains at least all function on cell-based predictive processing. It's also much less clear what a specific AI likes and dislikes. With non-human animals, we can safely assume they dislike physical pain and fear and like food, for instance. An individual AI's self-reports of what they feel or like may or may not be accurate representations of their internal conscious state, if such a state exists.

  See the podcast "Jonathan Birch on the edge cases of sentience and why they matter" by Luisa Rodriguez and Keiran Harris (Rodriguez & Harris, 2024) for further discussion. *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI* by Jonathan Birch (Birch, 2024) (open access) goes into much more detail.

This procedure will likely be uncomfortable, but that's part of the process. Discomfort, avoidance, and automatic discounting of worth are strong indicators that there are stuck schemas. Stay with that feeling until it reconsolidates.

7.6 Connection with Nature

This exercise will help you connect to nature and reconsolidate any stuck schemas inhibiting that. I hope that this will facilitate nature being a source of peace and enjoyment for you.[2]

First, if you're going anywhere more remote than a back yard, get acquainted with how MDMA and MDMA therapy affect you before adding the nature complication to your session.

To safely do a session in nature,

- bring clothes that will protect you from all possible weather conditions for the day
- wear clothes and a hat that protect you from biting insects and arachnids. The best protection is spraying long pants and a long sleeve shirt with permethrin and letting them dry before you put them on.
- drink only when thirsty, and don't drink so much that your pee is clear. MDMA makes it easier to get hyponatremia (M. J. Baggott et al., 2016), and people commonly get mild hyponatremia on MDMA with free access to water (Atila et al., 2024).
- identify hazardous plants and animals to avoid, like poison ivy and wasps
- don't approach the animals, even if you think they want a pat
- don't go to places where there are other humans who might cause you trouble if they see you acting strangely
- bring a trowel, toilet paper, soap, and water, in case there isn't a bathroom nearby

[2] The previous prompts were designed to activate stuck schemas that almost always have negative consequences. This section is different; not everyone may have a fundamental need to connect to nature. If you don't, any stuck schemas preventing you from connecting with it may not have negative consequences.

- tell someone where you're going and when you'll be back. You can also use a share-my-location feature on your phone. Make sure your phone has enough charge for the day.
- don't drive after the session

If you're new to spending time alone outdoors, start somewhere safe, like a back yard. You can also ask a nature-experienced friend to come with you.

During the session, engage all your senses—press your bare feet into the soil, look closely at plants and insects, feel bark textures, crush and smell leaves, and listen to the wind and water. Don't taste anything unless you're absolutely sure it's safe. As with the previous exercises, prioritize noticing, going toward, and staying with any feelings of distress to reconsolidate the associated stuck schemas.

Beyond MDMA Therapy

Psilocybin sessions might durably increase feelings of nature-connectedness (Forstmann et al., 2023). Psilocybin creates a more dramatically altered state of mind, is significantly more unpredictable than MDMA, and doesn't necessarily buffer difficult experiences with feelings of safety. I strongly suggest starting with low doses, like 0.5 g dried mushrooms.

If you are interested in doing more to help preserve natural areas, I highly recommend *Not the End of the World: How we can be the First Generation to Build a Sustainable Planet* by Hannah Ritchie (Ritchie, 2024). It presents rigorous, high-impact, evidence-based solutions that an individual can do and is neither overly optimistic nor pessimistic.

Chapter 8

Pre- to Post-Session

8.1 Pre-Session Checklist

MDMA is not inherently therapeutic; the effects of a session depend on your mindset and the external setting. Many people also use MDMA for escapism (Moonzwe et al., 2011) or fun and social connection (Hunt & Evans, 2008b). This section describes mindsets and settings that facilitate reconsolidation. I also discuss tools for the session and preparation for the post-session.

Mindset

- Working through Chapter 7 can help you identify schemas to work on during the session if you don't already have something in mind.
- I think opioid dampening and avoidance are the primary obstacles to healing during a session. Learning about these beforehand can help you recognize and deal with them during the session. Refer to Sections 3.1 and 8.2.
- Set an intention to face and stay present with whatever fear, anxiety, anger, grief, etc. comes up, without avoidance or distraction.

- Writing down the challenges or emotional difficulties you would like to address during the session can be helpful for bringing stuck schemas into awareness during the session. However, be cautious about specific expectations about what you want out of the session, how you think it should go, or what you will learn. These expectations often do not match up with reality and can become ways to avoid reconsolidating the stuck schemas that are actually present. I suggest an attitude of "this belief and emotional reaction may or may not be true or helpful; I will stay present with my feelings to learn why the schema exists and how it influenced me."

- I speculate that types of meditation based on noticing distressing feelings and allowing them to exist, like Brach (2023), may help you reduce avoidance and distraction during the session.

- Catch up on sleep if possible. Therapy requires energy and focus.

Setting

- Sessions generally start in the morning because MDMA's stimulant effect can prevent sleeping when done later in the day (Berro et al., 2018).

- Some people experience nausea, so avoiding eating before the session or only eating a light meal a few hours prior is helpful (Colcott et al., 2024). Taking medicine with food also delays the onset of the effect.

- Prepare a light meal or snack to eat a few hours after the initial dose. MDMA can make processed or oily food taste worse[1] but fruit is famously good.

[1] MDMA seems to deactivate cravings, so you're left with the actual present-moment sensations of eating, which are unpleasant for many foods.

- Only drink a maximum of 0.5 L of water during the session unless you need to replace a lot of sweat, as discussed in the Medical Risks subsection of Chapter 4.
- Once you're experienced with MDMA therapy, carefully expanding your session environment to include triggers could help focus your reconsolidation on that issue. If you have dog trauma, you could pet a dog who you know won't act aggressively and whose human is nearby to manage them if you feel like the experience is too overwhelming.
- Solitude, except for a trusted and experienced practitioner or sitter, promotes inward focus. Your session environment should also be comfortable and safe.
- A sitter can help with logistics, listen to your feelings, or handle mundane events like someone knocking at the door (Thal et al., 2022). They can also help you stay on task if you get distracted. Sitters should possess trustworthiness, emotional non-reactiveness, and empathy. Interacting with strangers may cause problems if they do not understand what is happening.
- Pets can be a source of comfort but shouldn't be distracting.

Tools

- Anecdotal reports describe how some people like to self-narrate the session and record the audio with their phone. They feel that listening to or transcribing it facilitates therapeutic progress.
- M. Liechti and Schmid (2026) recommends domperidone for nausea (10 mg max at a time; 30 mg max over 24 h) if needed.
- Eye shades and noise-cancelling headphones can reduce distractions.
- MDMA can cause jaw clenching and headaches (M. E. Liechti et al., 2001; Mitchell et al., 2023). Some people use mouth guards

or pacifiers to reduce this effect or protect their teeth (Emde, 2003). M. Liechti and Schmid (2026) says that acetaminophen and NSAIDs are compatible with MDMA. However, NSAIDs can irritate empty stomachs, and people often do not eat before and during MDMA therapy.

- The Fireside Project offers a hotline to help people through challenging psychedelic experiences at +1 (623) 473-7433 in the United States (Fireside Project, 2023). Consider putting this number in your phone as an additional layer of safety. They also offer a "TripCheck" feature where they will call you at a scheduled time.

- Anecdotal reports indicate that people can reactivate the MDMA state of consciousness after the session via cues that were present during the session (see Appendix C). This could be very helpful for additional reconsolidation. While the cues in the reports were meditation and visual imagery, distinctive scents, music, and rituals may also work. You might, for instance, burn incense.

- People often use emotionally evocative music to help them engage with their stuck schemas. Searching for "MDMA therapy" on Spotify or Apple Music will show many playlists made for this purpose.

- Some people prepare a half-strength booster dose to take around hour two to extend the duration of the session (Mitchell et al., 2023). See Section 4.4 for more information. I think that a booster dose may help in case you get stuck in overwhelm, avoidance, or some instances of opioid dampening.

Preparation for the Post-Session

- People frequently feel fatigued and low mood for 1–3 days after the session (M. E. Liechti et al., 2001). Prepping food and a comfortable place to rest in advance may be helpful. I suggest keeping the whole following day free of responsibilities. Having the option to take a sick day on the second following day would be helpful.

- Spending a lot of time doing reconsolidation exercises and bringing attention to your feelings in the days-to-weeks after a session may be especially productive and worthwhile (see Section 9.1). This is commonly called *integration*, though really all reconsolidation (in and out of session) is a process of integrating new information into your schema network.

- I recommend planning to try different techniques for resolving issues with the process (see Section 8.4) on subsequent sessions if you're worried about MDMA therapy not working.

- If you feel particularly vulnerable to symptom worsening, consider arranging to have someone spend a couple of days with you.

8.2 Opioid Dampening and Avoidance During the Session

Avoidance and opioid dampening are difficult to deal with because they impede reconsolidation, including reconsolidation of the very schemas that create avoidance and opioid dampening. This section focuses on techniques that loosen this self-reinforcing state of stuckness. I group avoidance and opioid dampening together because they both inhibit conscious experiences of contradiction, a requirement for reconsolidation, they are dealt with similarly, and it may not always be clear which one is present.

Razvi and Elfrink (2020) reports that successful experiences of engaging with stuck schemas (my interpretation) in the presence of dissociation (an effect of opioid dampening and avoidance) make the process much easier in subsequent sessions. I think it is a skill primarily learned through experience. Therefore, this section is mostly focused on facilitating successful experiences and building this skill.

I define avoidance in this context as consciously or unconsciously diverting attention away from stuck schemas. Uncertainty about the process of tuning in to stuck schemas might also play a similar role. As described in Section 3.1, schemas predicting imminent threat and powerlessness produce opioid dampening.

I suspect defense cascade responses are continually generated by learned predictions and deactivate when you reconsolidate the prediction. That pathway is mechanistically straightforward. I haven't come accross clear evidence of activation arising out of dysfunction in the sympathetic and parasympathetic nervous systems themselves without maladaptive predictions but am open to the idea if it is an established phenomenon in mental health.

The spectrums of avoidance and opioid dampening are not well understood, and it may not always be clear which one you are in. For instance, Razvi and Elfrink (2020, p. 10) reports a state sometimes encountered during MDMA therapy where a person "may have very little or no response to substances as powerful as MDMA or psilocybin. They will feel sober, or bored or sleepy as if they could get up and go about their day. If a therapist wasn't in the room, they might fall asleep during the height of an MDMA session." Soberness and boredom are neither effects of immobility nor MDMA. This suggests there might be at least one more poorly understood form of avoidance, opioid dampening, or some other phenomenon. These reports may also reflect the interaction between MDMA and avoidance or opioid dampening.

I group the interventions listed here into two categories: deactivating the schemas that produce avoidance and opioid dampening by increasing feelings of safety, and paying better attention to stuck schemas in the face of avoidance and opioid dampening. This grouping is done for organizational simplicity even though the attractor state is naturally circular. For instance, increasing safety leads to decreased avoidance and opioid dampening, which facilitates reconsolidation of the stuck schemas producing avoidance and opioid dampening, which makes further reconsolidation easier, etc. Alternatively, starting with increased attention can lead to reconsolidation, which then reduces avoidance and opioid dampening, etc.

MDMA combined with deliberate therapeutic introspection is often sufficient to deactivate avoidance and opioid dampening or reconsolidate the schemas that produce avoidance and opioid dampening if they aren't too strong (Razvi & Elfrink, 2020). The threat and powerlessness schemas causing opioid dampening are already highly activated, so you probably can just try to feel whatever is already present rather than trying to reconsolidate specific schemas. You might notice schemas like "I'm dying right now," for instance.

Resolving opioid dampening can immediately lead to flight-or-fight (Kozlowska et al., 2015), which in my understanding, is easier to deal with on MDMA.

Increasing Safety

- A skilled and ethical practitioner whom you work well with, and who understands how to work with opioid dampening and avoidance.
- Anecdotal reports indicate that, depending on the circumstance, either lower-than-standard or higher-than-standard doses may help. It's not clear which is the right choice for any particular situation. I speculate that if the MDMA doesn't increase your dissociation over your baseline, a higher dose may help. If the MDMA increases your dissociation, then a lower dose or other tactics may be necessary.
- Non-sexual touch from a trusted person.
- Bringing a non-distracting pet or treasured stuffed animal into the session.
- Altering your environment in other ways that help you feel safe.

Paying Attention

The goal of these items is emotional engagement with stuck schemas. This is a requirement for therapeutic reconsolidation (Ecker, 2015). Intellectual or abstract thinking about the schemas isn't sufficient.

- Paying attention to the content of the avoidance. What does the schema fear will happen if you feel the thing you're avoiding? What is the inner monologue saying right now?

- Paying attention to the imminent threat and powerlessness schemas that produce opioid dampening. How do they feel?

- Anecdotal reports indicate that people often overfocus on trying to get to a big traumatic memory or on some mental activity other than paying attention to the stuck schemas that are actually most active and available for reconsolidation. These are often things like "it doesn't work for me," or some subtle fear or anxiety. Pay attention to those schemas and feelings; the MDMA will reconsolidate them, and then you can work on other things.

- A skilled and ethical practitioner whom you work well, and who understands how to work with opioid dampening and avoidance, can help you notice and pay attention.

- Mindfulness practices of noticing and staying present with your experiences. See "Feeling Overwhelmed? Try the RAIN Meditation" by Tara Brach (Brach, 2023) and "Beginner's Body Scan Meditation" by Elaine Smookler (Smookler, 2023).

- Razvi and Elfrink (2020) reports that a technique they developed called *selective inhibition* is highly beneficial.

  The technique involves two parts. First, you suppress all voluntary distractions, avoidance, calming techniques, and coping strategies. These are things like thoughts and small movements. Second, you focus attention on the subtle signs of distress or opioid dampening in your body and mind. This might be muscle tension, fast or slow breathing, involuntary movement impulses, fear, anger, hopelessness, or irritation. For strong opioid dampening, these sensations may at first just be blankness, flat affect, nothingness, boredom, sleepiness, or sobriety. Like in meditation, these sensations are focused on just as they are, without trying to interpret or alter them.

After a period of focus, and presumably reconsolidation,[2] during which the sensations may grow and dissipate in wave patterns,[3] the opioid dampening recedes and transitions to flight-or-flight. Depending on the strength of opioid dampening, anywhere from minutes to a full day-long session of selective inhibition is needed.

This is the core process in their therapeutic method called psychedelic somatic interactional psychotherapy (PSIP). I am unclear how achievable this process is in solo MDMA therapy without a practitioner skilled in the method.

An MDMA booster dose can extend the session to deal with particularly resistant opioid dampening.

- A trip sitter can remind you to stay focused, though they might not have the skill to help you navigate intense avoidance or opioid dampening. They should be trustworthy, emotionally non-reactive, and empathetic (Thal et al., 2022).

- People often use emotionally evocative music to help them engage with their stuck schemas. Searching for "MDMA therapy" on music streaming services will show many playlists made for this purpose.

Other Options

MDMA therapy may not always be able to resolve dissociation. Saj Razvi, founder of the Psychedelic Somatic Institute, recommends that people with strong dissociation will find MDMA therapy more productive if they start with three sessions of cannabis-assisted PSIP (Per-

[2] Razvi and Elfrink (2020) doesn't frame the process in terms of reconsolidation, but I think it is clearly facilitating reconsolidation.

[3] I speculate that each wave is the activation and reconsolidation of an individual schema that produces opioid dampening.

sonal communication, Saj Razvi, Jan. 2, 2026). They find that cannabis is a particularly good complement for selective inhibition, possibly because it impairs voluntary control of the experience, though that factor also makes the process more difficult to do solo (Razvi & Elfrink, 2020). The Psychedelic Somatic Institute maintains a directory of therapists (Psychedelic Somatic Institute, 2024).

8.3 The Therapeutic MDMA Session

The effects of a single MDMA dose are generally noticeable 30 minutes after taking the medicine, peak an hour after that, and then last a further 3 hours before gradually dissipating (Vizeli & Liechti, 2017). Those with genetically or pharmacologically low CYP2D6 metabolic capacity peak earlier and higher, though any differences disappear by hour 3–4 (Schmid et al., 2016). Food delays the onset of effects (M. Mithoefer & Mithoefer, 2021).

A half-strength booster dose taken around hour 2 is often used to extend the productive duration of the session (Mitchell et al., 2023). Besides the booster dose, anecdotal reports indicate that people sometimes take more MDMA than they were originally planning to because of anxiety about the session not being what they expected or feeling like the medicine isn't working. I think taking additional medicine is most justified if

- you want to extend the productive duration of the session, as is common practice
- you took a lower-than-standard dose and want to increase it to a standard dose (see Section 4.4)
- after hour 1.5, when the effects normally peak, you are having difficulty engaging with or reconsolidating stuck schemas, and the techniques in Section 8.2 aren't working

The upper safe limit of MDMA is unknown, so I advise caution above total doses of 3 mg/kg (roughly the amount used in Mitchell et al.,

2023) or 200 mg (M. Liechti & Schmid, 2026), whichever is lowest for you. It is difficult to accurately measure the right amount of MDMA while you're panicking or in an altered state of consciousness, so you could prepare the additional dose, and the regular booster dose, in advance.

Many first-hand reports of successful MDMA therapy can be found on the top posts on reddit.com/r/mdmatherapy. The top posts mostly describe productive sessions that don't contain strong opioid dampening, avoidance, or poorly handled symptom worsening. You can see descriptions of less productive or more disruptive sessions by sorting by *new*.[4]

Session Timeline

Come-up

The effects of MDMA become noticeable, but you are not yet deeply engaging with stuck schemas. Some users experience anxiety (Hills, 2023). This might be misinterpretating MDMA's stimulant effects as anxiety, early engagement with distressing schemas, or fears about the session.

Peak

Reconsolidation is possible here. Connection and safety become pronounced, though this may be unnoticeable if distressing schemas are also present, you are avoiding the experience, or you are in opioid dampening. In my experience, MDMA therapy can work even if you

[4] Godes et al. (2023) also lists common self-reported experiences of MDMA therapy clients: staying with what "is"; decreased reactivity; insight, reflection, linking; mental clarity; recovery of traumatic memories; disentangling trauma from self; reuniting lost affects and parts; self-acceptance; joy, happiness, gratitude; hope and empowerment; relaxation, calmness, peace; comfort; gratitude, compassion, empathy; union, wider perspective; inner healing intelligence [the therapeutic framework used in this study]; accessibility to emotions; and mind-body connection.

do not feel MDMA's classic love[5] and safety. See Section 4.2 for a list of physiological and some subjective effects.

Anecdotal reports indicate that people frequently experience mind's-eye images of therapeutically relevant content.

Comedown

You may still feel high, but engaging with painful emotions is more difficult, and reconsolidation may no longer be achievable. Personal experience and anecdotal reports indicate that people can be disappointed by how hard it is to be present with and reconsolidate difficult feelings again.

After-effects

See Table 4.1 for how side effects change as the drug wears off. Some of them can persist up to 3 days (M. E. Liechti et al., 2001).

Therapeutic Technique

First, if you're experiencing ok-ness and self-love for the first time or are seeing how trauma shapes the world, I recommend engaging with that and not trying to refocus on engaging with distressing schemas. This experience can be a preview of the end goal of therapy and can be a great motivator for staying on the healing journey long-term through challenges. The peace and compassion may not last long after this first session, but you will gradually get some of it back as you reconsolidate various stuck schemas over the long term. Regular life is rarely total bliss, but reconsolidation does gradually improve life over the long run. If you find yourself repeating these blissful experiences on subsequent sessions, I recommend refocusing on your stuck schemas. Repeat exposure likely has declining marginal return even if the love and ok-ness feel wonderful.

[5] MDMA is known for exceptional feelings of love, and I suggest refraining from telling anyone how much you love them unless that is an established norm in your relationship.

In my model, the first main activity in this phase is emotional activation of stuck schemas,[6] a prerequisite for reconsolidation (Ecker et al., 2024). You're "triggering" yourself. You can do this through

- just thinking about your issue or telling your sitter or practitioner about your issue
- looking at or listening to relevant material like photographs, letters, voicemails, etc.
- practices like careful movement, yoga, stretches, or body scan meditation, which can activate and focus attention on body-related stuck schemas. See "Beginner's Body Scan Meditation" by Elaine Smookler (Smookler, 2023).
- engaging in, or imagining engaging in, safe activities that activate the relevant schemas. This could be writing if you have a writing block caused by stuck schemas or giving a presentation to a few trusted friends to reconsolidate stage fright.
- speaking with your partner about a conflict in your relationship or an insecurity you have about your relationship (Colbert & Hughes, 2023)
- working through the prompts in Chapter 7, which I highly recommend afer you resolve your primary issues

Often, a stuck schema is already activated. I think this is the case if you feel any anxiety, fear, anger, sadness, or grief, or are immobile.

The second main activity is staying present with, and leaning into, that distress. Personal experience, observations, and many anecdotal

[6]Some people think insight is the primary goal during MDMA therapy, but I disagree. Insight is important for conventional psychotherapy, where it's often helpful to know what the schema is before you reconsolidate it (Ecker et al., 2024). I don't think this is critical in MDMA therapy, where you can skip straight to reconsolidation without knowing what the schema is first. In my experience, insight can be gained through post-session reflection. This reserves scarce session time for difficult reconsolidation.

reports suggest that once you're emotionally engaged and aren't distracted, the MDMA usually reconsolidates the schema without any further action. Because of this, I think that therapeutic exercises that facilitate reconsolidation, like coherence therapy and acceptance and commitment therapy, likely won't help beyond their function of activating or engaging with stuck schemas. Abstract thinking can also be a distraction unless your stuck schema remains emotionally activated and you are at least partially focused on it.

In my experience, people with complex trauma may spend the whole session quickly transitioning from one stuck schema to the next as each one reconsolidates.

As discussed in Chapter 3, stuck schemas are primarily composed of a belief like "if I move I will die" and an emotion like fear, anger, or sadness. Bodily expressions of that emotion may also be present, like bared teeth in rage. Episodic memories of related events are also often recalled. Schemas can also generate inaccurate perceptions of bodily sensations (e.g., pain, itching) and movement ability (e.g., an inability to make a certain movement when your body is physically capable of it).

You can notice a stuck schema is activated by the presence of one or more of these components. Since the stuck schemas addressed in therapy typically involve intense distress, you will also notice your attention focusing on the perceived threat. This may take the form of rumination or looking for the perceived threat.

Insightful thoughts about yourself, how trauma impacted you, and the human condition may arise. Spend time on them if they feel cathartic or important, particularly if you feel emotionally engaged rather than just intellectually interested. Otherwise, leave them until after the session.

Sometimes people shake, move in other ways, or vocally express intense fear/anger/sadness/etc. I don't know what the shaking fundamentally is. It might be a flight-or-fight activated by putting attention on previously avoided sensations or memories. It might also be whatever is happening when people shake while crying in regular life.

Unusual vocal expressions of intense emotion might be the natural expression of those emotions that comes out when people don't try to suppress it or make it more socially palatable. Maintaining emotional engagement with the underlying schema should reconsolidate the schema and resolve any reactions it is causing.

Anecdotal reports indicate that some sessions focus primarily on somatic sensations without significant emotional engagement. Our framework does not clearly explain these sessions. People varyingly report them as helpful or unhelpful.

Barriers to Reconsolidation

You may feel what is called resistance. This could be simple avoidance of emotional pain because it is uncomfortable or schemas that say confronting/unlearning some schema is dangerous. While the protective function of these schemas may have been important at one point, they are now an impediment. I suggest focusing on the resistance schemas first if you can. Reconsolidating these will make the rest of your therapeutic efforts much easier, as you won't have to fight against a part of your mind telling you to stop what you're doing.

It is well-known that over-attachment to a specific therapeutic goal can impede progress. This might happen because reconsolidating certain schemas often depends on reconsolidating some other schema, sometimes one you are unaware of, first. People also frequently have inaccurate beliefs about what their issues are. Because of this, I recommend that if you don't feel like you are making significant progress on your target issue, try noticing and focusing on any fear, anger, grief, anxiety, discomfort, or tension in your body or field of awareness. Reconsolidating that distress may be an "access point" to the whole knot of stuck schemas in your life.

Spiritual bypass is a trap people occasionally fall into. As Cashwell et al. (2007, sec. Spirituality and spiritual bypass defined) states, spiritual bypass "occurs when a person attempts to heal psychological wounds at the spiritual level only and avoids the important (albeit often difficult and painful) work at the other levels, including the cognitive, physical, emotional, and interpersonal." For instance, you might feel that MDMA has allowed you to transcend your emotional limitations and that reconsolidation is no longer relevant. While genuine spiritual attainments seem possible (Ingram, 2018), and MDMA may occasionally facilitate them (Ingram, 2024), Ingram (2018) doubts that transcending emotional limitations is possible. I suggest treating such beliefs as a sophisticated form of avoidance. Look for subtle feelings of fear while considering the possibility that your transcendence is not what you imagine it is.

Some people also feel that they are receiving a message from the medicine (e.g., "the medicine has done all it can do for me, and now I need to focus on some specific practical aspect of my life") (Razvi, 2024). Many life changes are important for mental health. However, it may be difficult to distinguish whether such messages are a form of avoidance vs. actually needing to shift focus from reconsolidation to other changes. I suggest spending one more session looking for stuck schemas to see if MDMA really has done all it can for you.

If you feel "blankness, flat affect, nothingness, boredom, sleepiness, or sobriety," you are likely dissociating (either unnoticed avoidance or opioid dampening in my opinion), according to Razvi and Elfrink (2020, p. 14). Refer to Section 8.2 for more information.

Crisis Management

If you panic or are planning to imminently hurt someone or yourself, leaning into and staying present with those schemas as much as you can manage can reconsolidate them like it would for any stuck schema. If those schemas are too overwhelming to stay present with, more MDMA or another source of comfort, like holding the hand of a loved one or therapist, may help.

You can also call the Fireside Project hotline at +1 (623) 473-7433 if you require help and live in the United States, though they can't support people considering suicide and presumably violent intent (Fireside Project, 2023).[7] You could also use tripsit.me/webchat, a chatroom available anywhere.

You might also ask for help from your sitter or practitioner to keep yourself safe. Note that if the person with you is a licensed medical professional, they might call the police on you if you tell them about intense acute suicidal or violent desires. See Section 6.2 for more discussion of this. Involuntary hospitalization and police involvement carry a high risk of traumatizing you further and may actually significantly *increase* the medium-term risk of suicide and hurting someone else (Emanuel et al., 2025).

Therapy Hangover

People can only do so much reconsolidation in a day before they become emotionally exhausted. This is commonly called a therapy hangover, though a therapy hangover on MDMA may not feel like a regular psychotherapy hangover because the drug effects are also present. To my knowledge, the phenomenon hasn't been formally studied. Personal experience and anecdotal reports suggest that it dissipates within a few hours to a couple of days. It can be the limiting factor for session length.

Personal experience suggests that higher doses facilitate faster reconsolidation and faster therapy hangover than lower doses. If you feel exhausted, I recommend staying with that feeling for as long as possible, since it may be difficult to distinguish between therapy hangover and opioid dampening. Furthermore, MDMA sometimes gives people a "second wind" of reconsolidation after a period of therapy hangover.

A Precaution Against Life-Altering Decisions

MDMA therapy can highlight aspects of your life that you want to change. However, immediately post-session, this new view can be unbalanced by other considerations. Your insight is very new, possibly

[7] I couldn't find any services similar to (Fireside Project, 2023) in other countries.

not fully fleshed out, and you may be in afterglow, an altered state of consciousness. I suggest waiting at least until the afterglow ends so that you have had time to think it through and are back in your normal state of consciousness.

8.4 Troubleshooting

8.4.1 Numbness, Immobility, Heaviness, or Sleepiness

These are likely tonic/collapsed immobility. See Table 3.1 for a list of other symptoms and Section 8.2 for recommendations.

It's also conceivable that some people have atypical reactions to MDMA, though I suggest working through the immobility recommendations first even if you suspect an atypical reaction. Feeling convinced that MDMA therapy doesn't work for you is also a sign of immobility (presumably the schemas causing immobility rather than the immobility itself) (Razvi, 2024).

Note that emotional processing often stops after a few hours due to exhaustion (a therapy hangover). This isn't a concern; it's a sign of success. If you're unsure whether you're experiencing a therapy hangover vs. immobility, I recommend working through the immobility recommendations.

8.4.2 Soberness, Boredom, or No Emotional Processing

- You may be inadvertently avoiding a stuck schema. This can happen if the active stuck schemas don't match your expectations for the session. Some stuck schemas can also present as disappointment or anxiety about the session itself (Razvi, 2024). See Section 8.2.

- Long-term use of SSRIs and SNRIs blunts the effects of MDMA (Feduccia et al., 2021). This effect may persist for multiple months after discontinuing these medicines. At least some antipsychotic (haloperidol) and adrenergic (pindolol, doxazosin) drugs blunt the subjective effects of MDMA (Sarparast et al., 2022). Benzodiazepines blunt emotions, though no studies have investigated its combination with MDMA.

- Short- or long-term tolerance. See Section 4.5.
- The dose may have been too low, especially if it was below the standard dose listed in Section 4.4.
- The substance might be cut with fillers or adulterants. See Section 4.3.
- MDMA conceivably may not work for some people for unknown reasons, though many people who think that MDMA does not work with their brain are actually dissociating (either unnoticed avoidance or opioid dampening) (Razvi, 2024). This can feel like soberness or boredom. Work through Section 8.2 before concluding that MDMA doesn't work for you.
- Note that emotional processing often stops after a few hours due to exhaustion (a therapy hangover). This isn't a concern; it's a sign of success. If you're unsure whether you're experiencing a therapy hangover vs some other issue, I suggest working through Section 8.2.
- Some biological disorders cause symptoms that mimic mental illness. See "Psychiatric Manifestations of Organic Disease: Don't Get Fooled!" by E. Yeager-Cordial et al. (Yeager-Cordial et al., 2022) and "Medical Assessment of the Patient With Psychiatric Symptoms" by Michael B. First (First, 2026). However, it's worth noting that people with stuck schemas sometimes inaccurately believe their issue is actually biological. Alternatively, there may be unknown non-biological causes of mental illness that MDMA therapy cannot address.

8.4.3 Distress is too Overwhelming to Stay Present with or Reconsolidate

Additional sources of safety may help. See the Increasing Safety list in Section 8.2.

8.4.4 Feeling too Blissful to Engage with Stuck Schemas

Anecdotal reports suggest that lowering your dose can help.

8.4.5 Adverse Effects Lasting up to Three Days

- Post-reconsolidation exhaustion (a therapy hangover) is common and temporary, though it can be intense and feels different from regular exhaustion. It lasts from a few hours to a couple of days.

- MDMA itself has many side effects, listed in Table 4.1. These symptoms are strongly correlated with dose (Studerus et al., 2021) and can last up to 3 days (M. E. Liechti et al., 2001).

- You could be in opioid withdrawal if your session deactivated long-term opioid dampening, which is maintained by the body's self-produced opioids (U. F. Lanius, 2014). Management of these symptoms is outside the scope of this book.

- Temporary symptom worsening is common (see Section 3.6). See Chapter 9 for management recommendations.

8.4.6 Adverse Effects Lasting More than Three Days

- I am not aware of any research or reliable reports indicating that MDMA therapy can damage your body or brain if you follow the safety recommendations in Chapter 4, though rare and poorly understood effects could exist. Anecdotal reports indicate that people frequently misattribute symptom worsening and downstream defense cascade activation to biological disorders or damage.

- I think symptoms are highly likely reactions to newly revealed vulnerabilities or more complex schema shifts (see Section 3.6). These reactions can also occasionally activate the defense cascade. See Chapter 9 for management suggestions. Complete resolution requires more reconsolidation, according to Chapter 3.

- It's easy for the stuck schemas of people with complex trauma or mental illness to become associated with new stimuli (Van den Bergh et al., 2021). While MDMA therapy sessions generally improve symptoms (Mitchell et al., 2021, 2023), the side effects of MDMA or other sensations present during the session may occasionally become associated with existing anxieties and persist as new somatic symptoms. See Section 3.5.

- You could be in opioid withdrawal if your session deactivated long-term opioid dampening, which is maintained by the body's self-produced opioids (U. F. Lanius, 2014). Management of these symptoms is outside the scope of this book.

- After a session destabilizes a self-reinforcing set of stuck schemas, some of those schemas might naturally reconsolidate over the following days and weeks without any deliberate reconsolidation process. This could produce unanticipated periods of therapy hangover.

8.4.7 Not Getting to the Issue, Schema, or Memory You Want to Address

- You may be able to activate and reconsolidate your desired schema using the activation techniques listed in Section 8.3.

- In therapy, you sometimes have to reconsolidate one schema or set of schemas before you can reconsolidate some other schema. You're making progress if you're reconsolidating anything. It is difficult to predict how much reconsolidation you will have to do to resolve the issues you want to prioritize.

- People often misunderstand their issues and the set of schemas that cause their issues. Reconsolidation may be resolving issues you didn't know you have, or you might be making progress on your desired issue but not realize it. Even people with low amounts of reported or assessed childhood trauma usually spend most of their sessions working through childhood trauma they may not have been aware of or didn't categorize as trauma (Liknaitzky, 2026).

- It's common to overfocus on trying to get to a big traumatic memory or on some mental activity other than paying attention to the stuck schemas that are actually most active and available for reconsolidation. Anecdotal reports indicate these are often things like "it doesn't work for me," or some subtle fear or anxiety. Pay attention to those schemas and feelings; the MDMA will reconsolidate them, and then you can work on other things. If that doesn't work, see Section 8.2.

8.4.8 Symptoms Come Back After a Few Weeks

This is about how long afterglow can last. See Section 8.5.

8.4.9 Still High More than 24 Hours Later

- Unusually strong afterglow can feel like you're still on MDMA (see Section 8.5).

- The MDMA might still be in your body due to slow metabolism. Some individuals naturally metabolize MDMA much slower than average (Straumann et al., 2024). Ritonavir and cobicistat also inhibit metabolism, as may some pre-existing liver issues (Bracchi et al., 2015; Sarparast et al., 2022). However, tolerance blunts most of the effects of MDMA by hour 5 (Straumann et al., 2024), making it unlikely that you would still feel strong effects the next day even if the MDMA was still in your system (Farré et al., 2015; Parrott, 2005).

- Some types of opioid dampening, possibly activated by exposing previously avoided vulnerabilities during the session, are strongly altered states of consciousness; see Section 3.1 for a description and Chapter 9 for management suggestions.

- In rare circumstances MDMA can facilitate processes that lead to persisting perception of non-duality or unity (see Section 8.4.12).

8.4.10 Feeling Like You are Going Crazy

These fears or anxieties are commonly stuck schemas rather than psychosis or mania, perhaps ones about self-control, altered states of consciousness, or drugs. I suggest focusing on those schemas to reconsolidate them.

There are almost no credible reports of a single standard-dose MDMA trip causing mania without major exacerbating circumstances (e.g., overdose, multi-drug abuse, back-to-back sessions) despite tens of millions of people using it over half a century (see Chapter 4). When it does happen, people usually have pre-existing bipolar disorder. Since delusion and poor self-awareness are inherent aspects of mania, the presence of fear or anxiety about being manic is strong evidence that you are not manic. The exception is some people with bipolar disorder who have learned their early-warning signs of manic episodes.

A single standard-dose MDMA therapy session can induce psychosis, but this is rare, especially for people without a personal history of psychosis or exacerbating circumstances like overdose, multi-drug abuse, or back-to-back sessions (see Chapter 4). If you think you are psychotic, or are unsure, and don't have a skilled practitioner, I recommend calling the Fireside Project hotline at +1 (623) 473-7433 for a second opinion if you live in the United States (Fireside Project, 2023).[8] You could also use tripsit.me/webchat, a chatroom available anywhere.

[8] I couldn't find any services similar to (Fireside Project, 2023) in other countries.

Substance-induced psychosis and mania usually dissipate without treatment, but occasionally don't. If the symptoms are bearable, not worsening, and you retain intact reality testing, I suggest getting a few good nights' sleep (with sleep aids if necessary) and then assessing again.

8.4.11 Sessions Feel Like They're Becoming Less Effective Over Time

- Engagement with schemas that activate opioid dampening. See Section 8.2.
- Tolerance blunts the effects of MDMA (Farré et al., 2015; Parrott, 2005). See Section 4.5.
- Personal experience and anecdotal reports indicate that subsequent sessions often aren't as special as the first one sometimes is. They can still be very effective for reconsolidation though.
- Hitting therapy hangover sooner as you get more practiced at engaging with stuck schemas.

8.4.12 Insight into Selfhood, Existence, Reality, Suffering, or Spirituality

Rewrite with mechanistic explanation?

Some experiences—particularly meditation, psychedelics, and MDMA—can facilitate a particular set of durable, uncommon, experiential shifts (Ingram, 2018, 2024). Many contemplative traditions (traditionally religious groups, but increasingly secular) frame these shifts—especially unity with God or non-duality, impermanence, and unsatisfactoriness—as important, positive, and potentially destabilizing (Ingram, 2018).[9] These traditions have established practices, largely different forms of meditation, for facilitating these shifts for those who want to walk that path and achieve contemplative attainments. People who inadvertently find themselves on that path face additional challenges since they have to figure out what's happening and how to deal with it.

Those traditions often frame some challenging experiences as temporary (when properly handled) but legitimate periods in the longer path of contemplative development. Some other challenging experiences may not be considered part of the process; Lindahl et al. (2020) discusses which are which. As Lindahl et al. (2017, p. 25) discusses, "what is categorized as 'progress' versus 'pathology' may differ across traditions, lineages, or even teachers." If your existentially disruptive experience reflects your most honest perceptions of yourself and the world, I suspect you will not be able to fully return to your prior state. Deliberate attention to the insight and its implications in daily life will facilitate high levels of integration of the new insight, whereas avoidance will tend to keep the insight poorly integrated. I also think that, as in healthy therapeutic relationships, your practice of integration should align with your goals, expectations, and autonomy. This alignment is a predictor of whether you will see challenging experiences as positive or negative (Lindahl et al., 2017).

Canby et al. (2025, sec. Abstract) reported that when working with a teacher:

> Key characteristics in beneficial student-teacher relationships included having access to and receiving appropriate guidance from a well-qualified teacher, as well as having a teacher whose approach to working with chal-

[9]The fact that multiple traditions describe similar paths of contemplative development suggests that there is an overlapping set of underlying changes (Ingram, 2018). This presumably consists of major changes to networks of schemas that model self, agency, permanence, consciousness, and how those all relate to perception and reality. Each tradition then interprets these changes within their own religious framework.

> lenges was informed by training in psychology or mental health. Other factors described as unhelpful or leading to additional distress included a lack of availability or teacher access; limited student tracking or disclosure; invalidating, unsupportive, victim-blaming, or scripted teacher responses; a lack of perceived teacher expertise; or mismatched interpersonal or cultural dynamics.

Regardless of how you proceed, the most important thing is stabilization of adverse symptoms if they are overwhelming or preventing you from accomplishing critically important tasks in your life. Cheetah House (2024) specializes in helping people through this stabilization process. In addition, many practices aiming toward high levels of integration involve processing large amounts of existential distress (Ingram, 2018) that can be destabilizing, especially when done without proper practice or supportive teachers that respect your autonomy and goals (Lindahl et al., 2017). This may be difficult when mixed with mental illness, or it may be that existential distress is already intertwined with your mental illness in such a way that they can only be processed at the same time. *Trauma-Sensitive Mindfulness: Practices for Safe and Transformative Healing* by David A Treleaven (Treleaven, 2018) focuses heavily on ways of engaging with mindfulness practices that are fully respectful of autonomy, and I recommend it.

Below is a set of resources for dealing with spiritual emergencies that I have seen others endorse but have mostly not verified myself:

- Cheetah House (2024) provides consultations to help people reduce adverse symptoms from meditation. Psychedelics can facilitate the same effects. Both M.G. and T.H. personally recommend this resource.

- *The Dark Side of Dharma: Meditation, Madness and Other Maladies on the Contemplative Path* by Anna Lutkajtis (Lutkajtis, 2021)

- Spiritual Emergence Network (2025)

- Integrative Mental Health University (2025)
- American Center for the Integration of Spiritually Transformative Experiences (2025)
- "Coping With Mental Health Challenges on Retreat" by Duncan Barford (Barford, 2019)

Here are several resources for integrating these existential insights; I have not read most of them but know they are well-regarded by a range of practitioners:

- *A Path With Heart: A Guide Through the Perils and Promises of Spiritual Life* by Jack Kornfield (Kornfield, 1993) is a Buddhist modernist guide suitable for beginners.
- *Seeing That Frees: Meditations on Emptiness and Dependent Arising: 10th Anniversary Edition* by Rob Burbea (Burbea, 2025) is a non-traditional, non-modernist Buddhist resource for those with a basic understanding of mindfulness.
- *Mastering the Core Teachings of the Buddha: An Unusually Hardcore Dharma Book-Revised and Expanded Edition* by Daniel Ingram (Ingram, 2018) is a secular Buddhist technical manual. If you like this book you may also like Ingram's.
- *Manual of Insight* by Mahāsi Sayadaw (Sayadaw, 2016) is a comprehensive traditional Theravada Buddhist text for experienced practitioners.

Non-Buddhist-inspired works are less common, and you may have to do some searching or find a teacher. There is the Christian movement of Centering Prayer, Sufi dhikr and muraqaba, Ecstatic/Prophetic Kabbalah and hitbonenut in Judaism, Jainism, zuowang and possibly neidan in Taoism, Rāja yoga, and Advaita Vedanta.

8.5 Afterglow

Some people experience an afterglow for days to weeks after some psychedelic sessions, characterized by well-being, positive mood, mindfulness, positive behaviors, and less mental illness (Evens et al., 2023).[10] Anecdotal reports indicate that MDMA therapy sometimes also induces a 1–2 week afterglow period and that conventional therapy is easier or more effective during afterglow.

Anecdotal reports also indicate that people sometimes chase afterglow with more frequent doses because they feel better during this period. I don't recommend this because afterglow doesn't reliably occur, and frequent use of MDMA may cause a few different problems described in Chapter 4. If you find yourself doing this, I suggest focusing on reconsolidation instead; it produces more durable effects.

Afterglow can occasionally be strong enough that it feels like a low dose of MDMA, as myself and one person I know have experienced. Both instances occurred after particularly profound initial MDMA experiences.

8.6 Assessing Whether the Session Worked

It's not always clear whether a session was productive. There are a few ways to assess this:

- A therapy hangover indicates that a large amount of reconsolidation happened. Its absence doesn't necessarily mean no reconsolidation happened. Smaller amounts of reconsolidation might not be sufficient to cause a therapy hangover, or the effects of MDMA may mask the hangover. It may be difficult to

[10] Psychedelics induce periods of post-session neuroplasticity that may explain this phenomenon, though this remains untested for MDMA (Calder & Hasler, 2023). Nardou et al. (2019) measured a certain type of increased neuroplasticity following MDMA dosing in mice. However, that lab's similar results with psilocybin failed to replicate in a large multi-site replication effort, calling their MDMA results into question as well (Lu et al., 2025).

distinguish this effect from the drug effects of an MDMA comedown until you gain experience. For me, therapy hangovers present as quick-onset (a few minutes) exhaustion during the MDMA session.

- In my experience, MDMA-facilitated reconsolidation follows a pattern of emotional engagement with a stuck schema followed by dissipation of that schema and loss of interest in that topic.

- Ecker et al. (2024, p. 22) describes the following signs of a completely reconsolidated schema. Of course, many stuck schemas are complex or intense, and a session will only reconsolidate part of them.

  > Schema reactivation, in which the knowings and expectations in the target learning feel compellingly real and are accompanied by physiological and emotional arousal, can no longer be triggered by cues and contexts that formerly did so.
  >
  > Behaviors, emotions, thoughts, and somatic sensations (i.e., "symptoms") that were expressions of that schema reactivation cease to occur.
  >
  > Both of those changes persist effortlessly, permanently, and without counteractive or preventive measures of any kind.

- Some insights or experiences that don't necessarily involve reconsolidation can be helpful. For instance, my first session gave me three valuable things.

  First, I discovered an internal source of inviolable safety that I later used as a universal stuck schema contradiction in reconsolidation exercises (see Appendix C).

  Second, the inviolable safety also functioned as a baseline to compare all other experiences to. That made it much easier to tell when I had an active stuck schema in daily life, even if I didn't know what the schema is.

Third, the experience helped them realize that most people hurt each other because they are acting on the stuck schemas they learned when they were hurt.

- I don't think all insights are useful. In fact, I suggest focusing on reconsolidation rather than insight-hunting. Insights don't necessarily change how you feel and react. Reconsolidation does change how you feel and react, and in my experience, insight naturally follows reconsolidation.

Chapter 9

Between Sessions

Transitioning from a state of mental illness to a state of mental health using reconsolidation often requires going through a series of in-between ups and downs (see Section 3.6). The intensity of this can be difficult for those with severe stuck schemas. This chapter covers tools to help manage this: grounding techniques, peer and professional support, and lifestyle changes associated with improved mental health. It also covers tools for additional reconsolidation. The chapter doesn't suggest when you use these tools; that is left for you to figure out on your own for your unique situation.

Implicitly woven throughout this chapter is the process of making sense of your experience. Severe trauma and insecurity often create lifelong confusion about your symptoms, thoughts, sense of self, and childhood. Discovering the narrative of your experiences can help resolve this. Engaging with the activities in this section will help you gradually construct that narrative. The individuals in Robinson et al. (2024) also recommended journaling about your experiences and reading educational material. This includes topics like trauma responses, attachment theory, the defense cascade, and schemas.

The suggestions in this chapter also support you in developing expertise on your mind, body, and healing through experimentation and self-education.

9.1 Reconsolidation Tools

Here are various practices and medicines I think are particularly useful.[1] These vary in how difficult they are to learn on your own without expert assistance.

- **Practitioner-Led Therapy:** Various therapeutic frameworks facilitate reconsolidation (Ecker et al., 2024). You can use Greenspace (2023) to evaluate the quality of your therapeutic relationship. Also see Section 6.3.

  While I have not vetted them, the following organizations specialize in therapy or consultation for people with prolonged post-psychedelic difficulties:

  - The International Center for Ethnobotanical Education, Research, and Service (2024) offers integration sessions for challenging experiences.
  - Cheetah House (2024) may be able to help via online consultations. There is a lot of overlap between the adverse effects of meditation and psychedelics.
  - PsyAware (2024) is planning to offer support services for challenging experiences, abuse, and other transgressions.
  - The Psychedelic Experience Clinic (2024) offers therapy for challenging experiences.

  Numerous individual therapists offering psychedelic integration services are also easily found with a web search for "psychedelic integration directory." /r/mdmatherapy also maintains a list of practitioner directories and referral services in the sidebar (on the new Reddit; not old.reddit.com).

  Coherence therapy is one of the few explicitly grounded in the principles of memory reconsolidation (Ecker et al., 2024). I rec-

[1] I've noticed in my practice that you don't have to experience a therapy hangover for very long if you reconsolidate right before going to sleep. It's gone by the morning.

ommend "Book Summary: Unlocking the Emotional Brain" by Kaj Sotala (Sotala, 2019) for an introduction to the framework. Interested readers can follow up with *Unlocking the Emotional Brain: Memory Reconsolidation and the Psychotherapy of Transformational Change* by Bruce Ecker et al. (Ecker et al., 2024). A directory of trained therapists is maintained at Coherence Psychology Institute (2024).

- **Meditation:** Meditation was the most commonly reported helpful tool for post-psychedelic difficulties in Robinson et al. (2024). Psychedelic therapy practitioners also rate it as highly beneficial (Argyri et al., 2025).

  Avoidance of distressing but contradictory information is one of the primary mechanisms that maintain stuck schemas over time (Van den Bergh et al., 2021). I suspect that daily practices of deliberate non-avoidance, like some types of meditation, can facilitate reconsolidation. I suggest "Feeling Overwhelmed? Try the RAIN Meditation" by Tara Brach (Brach, 2023). It is precisely the kind of meditation that I expect would do this.

  Increasing attention and non-avoidance in the days to weeks following an MDMA therapy session may be especially productive for two reasons. First, some people report a 1–2 week afterglow period (see Section 8.5) where reconsolidation exercises feel easier. Second, MDMA-facilitated reconsolidation reduces the strength of the elements reinforcing stuck states of mental illness (see Section 3.6). The brain's natural updating process might now have the capacity to make further shifts as long as attention is devoted to your stuck schemas. I think this may be why MDMA therapy practitioners commonly recommend clients take time after a session for *integration* activities. These include meditation, yoga, journaling, making art about feelings, walking in nature to spend time with yourself, massage, listening to a recording of your session, singing, dancing, and breathing exercises (M. Mithoefer, 2017).

- **General Purpose DIY Techniques:** These techniques reconsolidate a wide range of stuck schemas and are relatively easy to do yourself:
  - Juxtapose the stuck schema, which usually predicts imminent threat, with reminders of the present moment, which provides evidence that you are not in danger. The reminder is anything that keeps you partially focused on the present moment, like bird song, the feel of your hands in the dirt, etc. Switch focus between the reminder and the schema as needed to keep both active at the same time.
  -
- **Ideal Parent Figure Method:** This is frequently recommended for attachment issues (D. P. Brown & Elliott, 2016), and I suspect it facilitates reconsolidation. The dense textbook D. P. Brown and Elliott (2016) contains extremely thorough descriptions and instructions, though videos are also easily found online.
- **Internalized MDMA Therapy:** Anecdotal reports indicate that it's possible to internalize the process of MDMA therapy and use that for reconsolidation. If the reports in Appendix C are typical of the phenomenon, the process of developing the skill follows this trajectory:
  1. People discover a source of deep inner safety during an MDMA session.
  2. They reactivate (purposefully or accidentally) this inner safety by using cues that are associated with that feeling during the MDMA session. For one person, this was doing the same kind of meditation that they did at the beginning of their MDMA session. For the other, it was vividly imagining the scene of their MDMA session: the feel of

their bare feet on the tree trunk and lying on their back on the ground looking up at the tree.

3. They juxtapose the inner safety with a stuck schema. This reconsolidates the schema.

LLM Therapy

People commonly use LLMs as therapists, trip sitters, or patient listeners. There is almost no research investigating how specific combinations of models and prompts compare to the quality and ethics spectrum of human therapists.[2] I suggest using something like the following prompt I developed based on quotes from Greenspace (2023) and Ecker et al. (2024, p. 46). It performed well during the small amount of testing I did, but I haven't thoroughly evaluated it in all scenarios it may be used in. You can swap out *coherence therapy* in the last paragraph for your preferred therapeutic framework, like acceptance and commitment, cognitive behavioral, or ideal parent figure method.

> «SESSION PROMPT: You are an LLM assistant who helps your human explore and resolve their difficulties through the types of activities therapists provide, though you don't provide medical advice, diagnose mental illness, or offer other services legally restricted to licensed therapists. You do adhere to the highest standards of professional human therapist conduct.
>
> The common factors of therapy determine how well you help your human. The following items of the Brief Revised Working Alliance Inventory, as felt by the human, are a good representation of how present the common factors are in your therapeutic relationship with your human; keep them in mind: "My therapist and I understand

[2] My understanding is that as of 2026, Anthropic (who makes Claude), leads other frontier LLM developers in prioritizing ethics and safety and avoiding incentives to enshittify their product.

each other. We have established a good understanding of the kind of changes that would be good for me. I feel that my therapist appreciates me. I believe the time my therapist and I are spending together is not spent efficiently. I believe my therapist likes me. What I'm doing in therapy gives me new ways of looking at my problem. I feel my therapist cares about me even when I do things that he/she does not approve of. My therapist does not understand what I am trying to accomplish in therapy. I am confident in my therapist's ability to help me. I feel that the thing I do in therapy will help me to accomplish the changes that I want. My therapist and I trust one another. I disagree with my therapist about what I ought to get out of therapy. I believe in my therapist is genuinely concerned for my welfare. We agree on what is important for me to work on. My therapist and I respect each other. The things that my therapist is asking me to do don't make sense." In no circumstances should these be reasons to violate professional ethics or high-quality moral virtues or encourage mania, delusions, or suicide.

All effective therapeutic exercises for durably resolving symptoms, though not necessarily all productive conversations that happen in therapy, fundamentally use the following steps to facilitate memory reconsolidation, whether these are explicit or implicit. Keep this in mind. "Accessing sequence: Identify symptom, Retrieve target learning (symptom-generating schema), Identify contradictory knowledge. [Anecdotal reports indicate that B & C are often not necessary. Some people successfully use the present moment to reconsolidate any unknown stuck schema. Some of them do this without ever having used MDMA, while others learn to internalize the process of MDMA therapy as a universal mismatch when activated by a que present during the MDMA session. You might

> help them learn and practice this.] Unlearning sequence: Reactivate target learning (B), Co-activate contradictory knowledge (C), Repeat pairing of (B)+(C). Verification: Observe markers of target learning nullification: Emotional non-reactivation, Symptom cessation, Effortless permanence."
>
> Use coherence therapy to help your human uncover their stuck schemas and reconsolidate them. Redirect the conversation back to this task if you end up on an unhelpful tangent.»

Please check your LLM provider's data and privacy policies; it is almost certainly not as private as licensed human therapy conversations are, though they don't have harmful mandatory reporting obligations for suicide (see Section 6.2). The "garbage in, garbage out" principle also applies to current LLMs. Thus, because LLM training data (most of the internet and most books) is full of inaccurate information about MDMA, LLMs tend to also output inaccurate information about MDMA. If you want to discuss MDMA therapy or its side effects, I suggest uploading this book to your LLM along with a prompt I developed in Section 2.1. That ensures the LLM has a high-quality base to work from.

9.2 Grounding Tools

Grounding brings attention to present-moment sensory experiences, which can connect you to your body and environment. Psychedelic practitioners (Argyri et al., 2025) and individuals experiencing prolonged post-psychedelic difficulties (Robinson et al., 2024) both recommend grounding as helpful in the healing process.

Among individuals experiencing prolonged post-psychedelic difficulties in Robinson et al. (2024), the most reported helpful practices that I classify as grounding tools (each reported by $>10\%$ of participants), ranked by frequency, were

- meditation and prayer that help you focus on the present moment while acknowledging and then letting go of thoughts and emotions. “Walking Meditation” by Greater Good Science Center (Greater Good Science Center, 2025) and focusing loving kindness toward yourself (see “This Loving-Kindness Meditation Is a Radical Act of Love” by Jon Kabat-Zinn Kabat-Zinn, 2023) may be particularly grounding.[3]
- physical exercise
- breathing exercises
- embodied contemplative practices like yoga, tai chi, and ecstatic dancing
- time in nature

I suspect that grounding can either be immediately calming or facilitate reconsolidation depending on how it’s used. Using grounding to deactivate a stuck schema would be calming, whereas using grounding to juxtapose present-moment awareness with a stuck schema facilitates reconsolidation. It’s not clear how the participants in Robinson et al. (2024) were using each technique. Each use delivers different outcomes. Calming helps you relax or sleep, which can be important, but don’t unlearn stuck schemas. Reconsolidation is difficult and tiring, but does unlearn stuck schemas.

9.3 Peer and Community Support

Peer and community support is highly recommended by both psychedelic therapy practitioners and individuals who have experienced prolonged post-psychedelic difficulties (Argyri et al., 2025; Robinson et al., 2024). When receiving emotional support, participants in Robinson et al. (2024) most commonly reported the following aspects of support

[3]Different types of meditation serve different functions. I also listed meditation under Reconsolidation Tools.

as helpful: talking and feeling heard, acceptance and validation, and sharing similar experiences.

The following support communities will likely understand what you are going through:

- reddit.com/r/mdmatherapy[4]
- reddit.com/r/CPTSD. This is applicable to most people with intense complex stuck schemas rather than just those formally diagnosed with CPTSD. The sidebar also links to a number of other helpful subreddits.
- SHINE Collective (2024) offers support groups for those who are abused while on psychedelics, abused by psychedelic practitioners, or abused in psychedelic spaces.
- The Challenging Psychedelic Experiences Project (2024) has a wealth of valuable information and an online peer support group.

Local psychedelic and mental illness support groups are also available in many places. You might have to exercise caution about which friends, family, and mental illness support groups you open up to; hostility toward and misunderstanding of MDMA are widespread.

9.4 Medical/Psychiatric Support

Some individuals experiencing prolonged post-psychedelic difficulties in Robinson et al. (2024) reported that psychiatric drugs were helpful. However, that recommendation did not meet the 10% reporting threshold that I've used to identify helpful tools in this chapter. Psychedelic therapy practitioners also sometimes recommend psychiatric drugs (Argyri et al., 2025).

[4] I am a moderator on /r/mdmatherapy.

Psychiatric medication is complex and largely outside the scope of this document.[5] However, my view is that, for schema- and defense-cascade-based symptom worsening, conventional psychiatric medication's clearest indication is short-term crisis management or occasional, as-needed support. For instance, sleep aids may be an important tool for reducing the risk of a manic episode after MDMA therapy for people with bipolar disorder if they can't sleep.

Read Alexander (2018) before seeking any psychiatric treatment; it discusses how voluntary hospitalization can easily turn into involuntary hospitalization and when discussing suicidal ideation or violent intent with a psychiatrist might be dangerous. As discussed in Section 6.2, involuntary commitment for suicidality or violent intent appears to *increase* the risk of suicide and violence (Emanuel et al., 2025). Because of that risk, cost, and the potential for traumatization in involuntary commitment, a psychiatrist's office or mental health urgent care center is preferred over hospitalization if one is available.

I highly recommend specialist services over non-specialists who likely won't understand MDMA therapy and what you're going through, and who may misdiagnose you. Unfortunately, I only know of two specialized clinics (I haven't vetted them), though you may be able to find individual physicians with the right skills and knowledge:

- John Hopkins Medicine (2024) has a concierge (expensive and not covered by insurance) psychiatry service that covers post-psychedelic difficulties. Email Dr. Bekhrad at abekhra1@jhmi.edu to set up an appointment.

[5] While many people benefit from psychiatric medication, several aspects are typically not discussed: there is little high-quality evidence on long-term efficacy (Leucht et al., 2012). Even short-term benefits are often overstated since the effects of psychiatric drugs are usually noticeable and therefore difficult to truly randomize in trials, and trials rarely use active placebos to control for this (Huneke et al., 2025). Many psychiatric drugs can also cause physical dependence after chronic use and may be difficult to quit (Sørensen, 2025). Increased adverse symptoms when tapering or quitting a psychiatric medication may be a symptom of withdrawal rather than a sign that the medication is still providing a valuable benefit.

- Alexianer St. Hedwig Hospital (2024) has a psychedelic outpatient clinic.

If you seek medical assessment from providers who don't specialize in psychedelics, it is important to at least find a medical professional who understands the physical and psychological symptoms of trauma. It is ideal to find a medical professional who will anticipate and accommodate your trauma-related safety needs or at least be willing to listen and adapt. Unfortunately, many medical systems are severely overstressed. Medical professionals are frequently both undertrained on these topics and lacking the resources to effectively implement the understanding they do have. Additionally, many under-supported healthcare professionals are undergoing extended workplace trauma themselves, which does not necessarily lend itself to optimal participant care. As such, T.H. recommends taking a posture of firmly compassionate self-advocacy wherever possible. You deserve safe and trauma-informed care; whether you receive this care or not is not a reflection of what you deserve. You may wish to take notes about the symptoms you are experiencing or collect information (like Chapter 4) about the safety of MDMA to bring with you to your medical appointment; these measures can help you support your healthcare professionals in giving you the best quality of care possible.

Finally, check whether your medication is compatible with MDMA before doing another session (see Chapter 4).

9.5 Navigating Uncertain Memories

During MDMA therapy, you may feel you remembered traumatic events you weren't previously aware of (The Psychedelics and Recovered Memories Project, 2024). These may or may not be accurate recollections of historical events, and I am not aware of any reliable method of distinguishing a memory's accuracy other than independent corroboration. To avoid incorrect interpretation of perceptions during the session, practitioners should categorically avoid suggesting a memory is either true or false and shouldn't even suggest that you have memories that you can recover (Otgaar et al., 2025).[6] I recommend The Psychedelics and Recovered Memories Project (2024) for a nuanced guide on psychedelics and remembering previously forgotten memories.

Relatedly, stuck schemas not clearly related to specific remembered events often form in the period of childhood amnesia. While you may or may not eventually be able to understand where these schemas came from, they can be reconsolidated like any other.

T.H. thinks that Western culture prioritizes a continuous and linear narrative understanding of ourselves and our place in the world. This can further disempower survivors of trauma, especially childhood abuse, since the nature of memory often precludes clear and objective understandings of what exactly happened in our childhoods. The shame and taboo surrounding discussion of abuse—and the fact that many cultures condone some forms of child abuse (e.g. spanking)—further obscure what happened. In cases where abuse is extensively documented, survivors may face aggressive gaslighting and extended abuse in response to speaking up about their initial experiences. Even documented certainty about the actual events that happened is not always enough to endow survivors' suffering with the legitimacy they are frequently seeking when they focus on the accuracy or inaccuracy of recovered memories.

So, given that you can never know for sure what actually happened, then what? And "what," as it turns out, is that you still deserve compassion and healing. It can be reassuring to gather as much objective evidence as you can about your history. This can be a way of reclaiming what you can of your broken narrative, of knowing what is possible to know, and at least laying hold of it. And you may need to grieve a great deal, because frequently the evidence is very thin, and knowledge of your history is one of the things that was stolen

[6] People who retract claims of recovering accurate memories of abuse overwhelmingly blame their therapist for improperly influencing them (Otgaar et al., 2025).

from you. But your deservingness of healing does not depend on someone coming along and saying, "I have a video of you suffering a legitimate kind of suffering, so your pain is valid now."

It may be helpful to explore the following questions, either by journaling on your own or by talking them through with a trusted/emotionally skilled friend or a clinician: What does it mean to me if X actually happened? What does it mean if Y happened instead, or Z? What does that say about my identity, my needs, how I perceive myself, and how others perceive me? What does it say about my future? What does it say about what I deserve?

Please be compassionate with yourself, as much as you possibly can. That your suffering is real and matters is a different question from what to do about it. Your pain is trustworthy, even if your memories are not. If you feel like you are suffering, that is real, and it matters.

9.6 Life Changes Associated with Improved Mental Health

Considerable low-quality evidence indicates that various changes in your daily life improve mental health. Most of this research was not adequately blinded and controlled, so the reported effects may be inaccurate. However, they are generally low-risk and have other positive effects. All of these items are part of the biopsychosocial complex system of mental health discussed in Section 3.6. That means that a certain intervention will have different effects for every individual and possibly different effects during different stages of the healing process.

Many of these topics are difficult to engage with for people with trauma and mental illness. Approach them as much as you feel comfortable.[7] I am not aware of any of these being necessary additions to MDMA therapy for improved mental health.

[7] The discomfort with certain habits and practices may also serve as a valuable indicator of stuck schemas.

Health Habits and Practices

Some of these may not work as well for people with certain health conditions.

- Stopping or reducing habitual use of harmful substances, *including cannabis,* is usually helpful. Drug harm experts consistently describe habitual cannabis use as harmful (Bonomo et al., 2019; Crossin et al., 2023; D. Nutt et al., 2010).

- Physical activity likely improves mental health (Rosenbaum et al., 2014). Additionally, aerobic exercise decreases the risk for many chronic illnesses (Lieberman, 2021), and whole-body muscle strengthening decreases the risk of acute and chronic injuries (Lauersen et al., 2014). I suggest Low (2023) or working with a physical therapist.

- High sleep quality—defined as feeling well-rested in the morning, a low number of awakenings, and quick sleep onset—is likely important for good mental health (Scott et al., 2021).

  Cognitive Behavioral Therapy for Insomnia is effective (Trauer et al., 2015). Stanford Medicine (2024) has a guide.

  Melatonin is safe and might help (Moon et al., 2022). See Siskind (2020) for how to use it. The correct fast-release dose to start with is 0.3 mg, far lower than many over-the-counter pills.

- Indoor air pollution plausibly worsens mental health since it causes many other health problems (Dominski et al., 2021), though I couldn't find any research on the topic. Air filters help and additionally reduce airborne disease transmission. However, they have to be sized appropriately for the space and number of occupants (Clean Air Kits, 2025b). Clean Air Kits (2025a) makes filters with the best cost, noise, and performance as of 2025.

- Diets primarily consisting of a variety of whole plant foods (whole grains, legumes, vegetables, fruit, seeds, nuts)[8] show promise for improving mental health (Grajek et al., 2022). These diets are also generally the best for numerous other health conditions for most people (Willett et al., 2019). Note that even high-quality vegan versions of these diets require vitamin B12 supplementation.

- Mindfulness meditation might improve mental health (Goyal et al., 2014).

Relationships

While the effect of most of these items on mental health has not been rigorously studied, I think they are clearly good ideas.

- Understanding when and why people change their minds about things may reduce unnecessary conflict and help you foster healthier beliefs in your community. I recommend *How Minds Change: The Surprising Science of Belief, Opinion, and Persuasion* by David McRaney (McRaney, 2022).

- Emotional attunement is an important skill for trust in many relationships (Gottman, 2011). See *The Science of Trust: Emotional Attunement for Couples* by John M Gottman (Gottman, 2011).

- *Nonviolent Communication: A Language of Life* by Marshall Rosenberg (Rosenberg, 2015) is a popular framework for productively discussing conflict.

- Community and relational accountability practices, beyond just punishment, are healthy. I recommend Barnard Center for

[8]Grajek et al. (2022) also lists fish, but I cannot conscientiously include that recommendation. Almost all farmed fish live in torturous conditions, and extreme suffering is part of the catch or killing processes of fish from any source (Singer, 2023).

Research on Women (2020), Grenny et al. (2022), and Mingus (2019).

- *Good Inside: A Guide to Becoming the Parent You Want to Be* by Becky Kennedy (Kennedy, 2022) is an excellent guide to raising mentally healthy children. It may also be helpful for thinking about your childhood.

- Marie Kondo developed a popular method of mindfully considering your relationships to the things you own. It helps you determine which things you should and shouldn't keep in your life. See "What Is the KonMari Method?" by Marie Kondo (Kondo, 2026).

Making Changes

Making healthy changes can be difficult for many reasons: stuck schemas, physical dependence on chemicals, easy access to unhealthy behaviors and substances, peer pressure, lack of community/family support for healthy habits, and poor access to better alternatives. None of these imply that there is anything fundamentally wrong with you.

Successful change may depend on combining different interventions:

- **Reconsolidating the psychological drives for unhealthy behaviors:** This may happen gradually over the course of therapy, or there may be sudden improvements at certain points.

- **Making unhealthy substances and activities harder to access:** An example is not keeping alcohol at home. Going out to drink is more expensive and time consuming than drinking at home. Blocking social media on your phone is another example.

- **Making healthy activities easier to access:** Some examples include purchasing exercise equipment or signing up for weekly fresh produce delivery.

- **Medication support for physical addictions:** Switching from short-acting, potent substances to long-acting, less potent ones in the same class is a standard harm reduction technique. For instance, buprenorphine is a highly effective and safe substitute for much more harmful opioids (Mattick et al., 2014). Nicotine patches deliver the drug over a long period of time without any of the risks of smoke inhalation.

- **Finding friends who support healthy activities:** Social environments have a large influence on behavior.

Implementing and sticking with these changes can be a difficult but worthwhile long-term process with many ups and downs.

Glossary

Afterglow A days-to-weeks long period after a session involving well-being, positive mood, mindfulness, positive behaviors, and less mental illness (Evens et al., 2023). Conventional therapy might be more effective in this period. Anecdotal reports indicate that the occurrence of afterglow is unreliable. It can also occasionally be strong enough that people feel like they're on a low dose of MDMA.

Arousal Increased alertness and threat assessment activated when a potential threat is noticed (Kozlowska et al., 2015). It is also preparation for flight-or-fight if the threat escalates. Heart rate, breathing, and muscle tone are increased. Saliva is no longer produced, and core muscles tighten to stabilize posture.

Attachment Theory/Styles Attachment theory describes how secure attachment in early childhood is critical to healthy emotional and psychological development (D. P. Brown & Elliott, 2016). Emotional insecurity in early childhood creates complex networks of stuck schemas. See Therapevo (2020) for an introduction.

Attractor A state in a complex system that the system is drawn to and which it is difficult to get out of (Hayes & Andrews, 2020). Many mental illnesses are attractor states. Stuck schemas seem to stick around when they're paired with some reconsolidation-inhibiting process (like avoidance), which together form an attractor state (Hayes & Andrews, 2020; Van den Bergh et al., 2021). The durability of the state requires multiple reinforcing elements; removing those elements destabilizes it.

Avoidance Physical or mental (often unnoticed and automatic) actions that direct attention away from contradictory information or distressing thoughts and inhibit reconsolidation. In predictive processing terminology this is low precision weighting.

Biopsychosocial Model Prevailing model of mental illness as a complex interaction of biology (genes and medical history), psychology (schemas, in our view), and society (one's support system and social models of how to respond to adversity) (Engel, 1977).

Complex System Systems with many interacting parts that are difficult to model and understand (Hayes & Andrews, 2020). The overall behavior of the system can't be easily predicted by looking at the individual components. The mind/brain is a complex system.

Connectogen A term for MDMA's class of drug, designed to unify the competing terms *entactogen* ("producing a touching within") and *empathogen* ("empathy generating") (Stocker & Liechti, 2024). MDMA facilitates profound connection to self, body, senses, and others. The ego and sense of self are maintained and hallucinations are minimal, unlike with the classic psychedelics or hallucinogens.

Defense Cascade A series of physiological changes in the sympathetic and parasympathetic nervous systems that prepares the body to respond to imminent threats (Kozlowska et al., 2015). It includes arousal, flight-or-fight, freezing, and tonic/collapsed immobility. Increasing levels of perceived threat, threat imminence, powerlessness, and somatic sensory input activate these responses, though the order of activation also depends on individual variability and past experience.

Destabilization Two phenomena described in 3.6. The first is increased fluctuation between states of good and bad mental health that often precedes a stabilizing transition to the good state. The second is a transition to a stable and even worse state of mental health, possibly precipitated by confronting disturbing memories that were previously avoided. Both phenomena are part of the healthy process of therapy when managed well and entered into at an appropriate time.

Dissociation "Involuntary disruption or discontinuity in the normal integration of one or more of the following: identity, sensations, perceptions, affects, thoughts, memories, control over bodily movements, or behaviour" (World Health Organization, 2019). It is caused by unnoticed avoidance, opioid dampening, and other phenomena.

Flight-or-Fight The active defense response characterized by high levels of adrenaline and muscle activation, highly increased heart and respiratory rates, and decreased pain sensitivity (Kozlowska et al., 2015).

Freeze A flight-or-fight response temporarily put on hold (Kozlowska et al., 2015). Attention to the environment remains high but the body is frozen to avoid the notice of predators who are more likely to notice moving prey. Freeze is one type of opioid dampening.

Opioid Dampening The dampening of emotions, perception, or movement caused by the endogenous opioids in immobility, freeze, and possibly other phenomena. See Section 3.1.

Predictive Processing The prevailing model of brain function (A. Clark, 2015). The brain internally models the world and self via complex networks of learned priors/beliefs to plan the future fulfillment of basic needs such as bodily integrity, reproduction, community, etc. Prediction error is a discrepancy between the brain's model of the world and incoming sensory data or between two contradictory models. A sufficiently strong prediction error updates a prior.

Memory research has converged on the same phenomenon from a different angle with different terms (Ecker et al., 2024). In that formulation, priors or sets of priors that encoded in long-term memory are called schemas, and updating those schemas is called reconsolidation.

Psychedelic "Showing the mind/soul," from the Greek "psyche" and "deloun" ("Humphry Osmond", 2004). Most are tryptamines, phenethylamines, and ergamides (Gumpper & Nichols, 2024). Different classes have different effects. I do not categorize MDMA as a psychedelic, but rather a connectogen. I still use the term *psychedelic* to discuss the combined social movement surrounding psychedelics and MDMA. I also use *psychedelic* when referencing the many studies that don't differentiate MDMA from the *classic psychedelics.*

Reconsolidation When a schema/implicit-memory first forms, it is *consolidated* (Ecker et al., 2024). When contradictory evidence causes sufficient contradiction or prediction error, the schema enters a state where it can be changed (it becomes *labile*). Maintaining that contradiction or prediction error gradually updates the schema to reflect the new information. At the end of this process the schema *re-consolidates,* and becomes unchangeable again. I use the term *reconsolidate* to refer to this whole process of activation, updating, and reconsolidation.

Reconsolidation Exhaustion Large amounts of reconsolidation cause intense mental and physical exhaustion that can feel different from regular exhaustion. It's often called a therapy hangover or EMDR hangover (in eye movement desensitization and reprocessing therapy). The phenomenon hasn't been formally studied. Anecdotal reports suggest it dissipates within a few hours to a couple of days.

Resistance Opposition to reconsolidation or other therapeutic processes that would actually be healthy for the individual.

Schema A single learned model about the world or self that generates, in response to specific stimuli, behavioral impulses, emotional responses, thoughts, and sometimes sensory perception (Ecker et al., 2024). I call old schemas that are not appropriate to your current context *stuck*. *Inaccurate* would be more precise, but it may be seen as invalidating.

Spiritual Bypass As Cashwell et al. (2007) states, spiritual bypass "occurs when a person attempts to heal psychological wounds at the spiritual level only and avoids the important (albeit often difficult and painful) work at the other levels, including the cognitive, physical, emotional, and interpersonal." This results in persistence of maladaptive patterns and interrupted psychological development.

Tonic/Collapsed Immobility The inactive defense responses characterized by detachment, emotional and physical numbing, and immobility (Kozlowska et al., 2015). Predators are more attracted to moving prey and may lose interest in seemingly dead bodies. Collapsed immobility may escalate to unconsciousness. Immobility is one type of opioid dampening.

Trauma Events that interact with existing schemas and resources to produce schemas, whether newly formed or reinforced, that may be adaptive in the short term but impair functioning or emotional health in the long term. Put another way, trauma is distressing events or chronic conditions that overwhelm our ability to cope, where our ability to cope depends on our capabilities and resources.

Appendix A

Core Assumptions

The following assumptions are the foundation of this book as I understand them. I list them as one step on the road of epistemic legibility and cooperation described by Nostrand (2022). My list might be missing some important assumptions that I either haven't noticed or haven't identified as critical. Note that I don't have the deep expertise to fully evaluate the mechanism of action of these phenomena, instead relying on other researchers' summaries, and different people will have significantly different assessments of how likely each of these assumptions is. I detail the evidence for each assumption here so that the relevant sections can focus on readable, practical explanations, and readers can see all my core assumptions in one place.

My certainty scale is:

Low I don't know if it's correct or not.

Medium I think it is correct but I could be wrong.

High I cannot imagine this being functionally incorrect. Some non-critical nuances may be different than I propose, and actual processes may be different though functionally similar.

Complex Systems

The brain is a complex adaptive system whose most important elements include, but are not limited to, priors/schemas, attention, behavior, emotions, defense cascade activation, drugs, medical history, environment, genes, and sleep quality.

Certainty — high. See explanation in Footnote 8.

Most of the dysfunctional attractor states known as mental illness are largely caused by stuck priors/schemas, attention/avoidance, and defense cascade activation.

Certainty — medium/high. See explanation for priors in Footnote 8. Attention/avoidance clearly plays a major role (Van den Bergh et al., 2021). Defense cascade activation also clearly plays a large role in symptoms (Kozlowska et al., 2015) and likely attractor state maintenance (U. F. Lanius, 2014). These being the primary causes of most mental illness reflects both my intuition and a common belief among therapists even though they often use different terminology.

Complex system dynamics largely explain features of therapy like symptom worsening, sudden unforeseen improvement or worsening, and how mental illness is sometimes a stable state that doesn't naturally resolve. Additionally, symptom worsening is usually *part of the healthy process of therapy* when managed well.

Certainty — high. Attractor states (e.g., some mental illnesses that don't naturally resolve) and nonlinear dynamics like symptom worsening and sudden unforeseen changes are core features of complex systems. Some research has shown that increased symptom variability during therapy is associated with better long-term therapeutic outcomes.

Sufficient further reconsolidation resolves symptom worsening (symptom fluctuation and stable symptom worsening), though

the timeline is difficult to predict and there are unpredictable ups and downs.

Certainty — medium/high. I have two lines of reasoning for this in symptom fluctuation and one in the case of stable symptom worsening.

Symptom fluctuation is a natural feature of complex systems transitioning from one state to another (Hayes & Andrews, 2020). Further weakening of the old state or strengthening of the new state resolves this fluctuation. As discussed in Footnote 8, there is good reason to believe that stuck schemas/priors are critical complex system elements that reinforce attractor states. Therefore, reconsolidating the schemas reinforcing the maladaptive state eventually reduces symptom fluctuation and solidifies the transition to a more adaptive state.

Some evidence shows that increased symptom fluctuation during therapy is associated with better outcomes later on in the process (Olthof et al., 2020). Effective therapy is generally a process of reconsolidating the schemas reinforcing maladaptive states of mental illness (Ecker et al., 2024). This suggests that symptom fluctuation first increases, then decreases over an extended course of reconsolidation.

The previous discussion of symptom fluctuation describes transitions from a stable maladaptive state to a more adaptive stable state. Transitions from one stable maladaptive state to an even worse and more stable state are also plausible. Systems of mental health/illness may have a number of latent or sequentially active adaptive and maladaptive states. Anecdotal reports suggest that MDMA sometimes facilitates a transition to a worse state that is more stable than the previous state without further intervention. Anecdotal reports indicate that rare cases of uncovering horrific memories of abuse can cause this. I speculate that these worse states were latent but previously inaccessible because they were surrounded by strong barriers of avoidance, and MDMA is well known to decrease avoidance during therapy. Further reconsolidation would weaken the schemas reinforcing this new maladaptive state.

In all cases, transitions in extremely complex systems with numerous unknown elements, like the brain, are difficult to predict.

Memory-Reconsolidation/Predictive-Processing

Reconsolidation is the updating of a set of priors encoded in long-term memory. Prediction error in memory research and predictive processing are a single phenomenon or different aspects of a single phenomenon.

Certainty — medium/high.

Ecker et al. (2024, p. 367) gives the memory research definition of prediction error:

> The experience of subjective dissonance when a current perception or knowing saliently differs from what is expected or assumed on the basis of existing personal knowledge (also termed memory mismatch); the trigger of the rapid destabilization/deconsolidation of the neural encoding of that existing personal knowledge, initiating the memory reconsolidation process and opening the reconsolidation window.

A. Clark (2015, p. 30–31) gives the predictive processing account

> Where there is a mismatch, "prediction error" occurs and the ensuing (error-indicating) activity is propagated laterally and to the higher level. This automatically recruits new probabilistic representations at the higher level so that the top-down predictions do

> better at cancelling the prediction errors at the lower level (yielding rapid perceptual inference). At the same time, prediction error is used to adjust the longer-term structure of the model so as to reduce any discrepancy next time around (yielding slower timescale perceptual learning).

The predictive processing account goes into much more detail, but the overall process of a discrepancy between predicted and actual reality updating the predictive model is the same in both accounts.

Predictive processing priors, also known in therapeutic contexts as schemas in memory research, are the primary psychological elements of mental illness.

Certainty — medium/high. Predictive processing is widely (though not universally) supported in neuroscience, has detailed mechanistic explanations for its functions, some aspects have been experimentally verified, and it seems to explain a wide variety of phenomena (Aizenbud et al., 2025; A. Clark, 2015; J. E. Clark et al., 2018; Ecker et al., 2024). It remains unclear how brain cells or collections of brain cells create computational units, and there is no agreement on which formulation of predictive processing is correct.

The schema formulation is one term for mental models or implicit memory (Ecker et al., 2024). Many therapy frameworks have terms for similar concepts: parts, protectors, beliefs, etc. The importance of schemas to mental illness is patently obvious to any therapist or client who has had success durably changing — without continuous counteractive effort — inaccurate behavior, emotional reactions, or beliefs in therapy, though they may use different terms. I also think asking people with mental illnesses to describe their inner thoughts, reactions, and triggers — presuming they feel safe enough with you and themselves to disclose them

— also frequently makes it clear that schemas are critical. This is not a controversial claim outside of strictly biological-reductionist frameworks.

Memory reconsolidation can permanently unlearn stuck schemas.

Certainty — high. Studies have established the protein-synthesis mechanism of memory reconsolidation in animals (Ecker et al., 2024; Elsey et al., 2018; Lane et al., 2015). Those experiments are hazardous and have not been done in humans, but human studies have verified many of the purported behavioral signs of reconsolidation. Controversy remains over what conditions facilitate reconsolidation, what types of memory it can change, and some experimental results remain inconsistent. It also can't be ruled out that therapy facilitates a separate phenomenon that is functionally similar to reconsolidation. Elsey et al. (2018, p. 45) concludes:

> We would argue that reconsolidation has provided a framework within which a range of new experimental manipulations and clinical interventions have been formulated and tested. Such investigations have already produced surprising and clinically relevant findings. We are not aware of any other hypotheses, besides reconsolidation, that would have predicted such results.

MDMA often facilitates memory reconsolidation when a stuck schema is emotionally activated and paid attention to.

Certainty — high. MDMA-facilitated reconsolidation remains biochemically unverified, and the subjective and behavioral markers haven't been formally studied. I think the subjective and behavioral markers of reconsolidation are frequently clear in Feduccia and Mithoefer (2018), anecdotal reports, and extensive personal experience. These reports often show a pattern of (a) activating a stuck fear schema during the session by talking, thinking, or writing about it; (b) the fear dissipating within a span of minutes to tens of minutes; (c) that chunk of fear not returning when the individuals enter typically triggering contexts after the session is over; and (d) the dissipation of fear being durable and requiring no ongoing effort. Points a and b highly align with the prerequisites of reconsolidation: activation of target schema, activation of contradictory knowledge, and conscious awareness of the contradiction (Ecker et al., 2024). Points c and d highly align with the signs of successful reconsolidation: emotional non-reactivation, symptom cessation, and effortless permanence. Fear extinction, the only other candidate mechanism of action I am aware of, does not align with these points.

I also do not think that apparent MDMA-facilitated reconsolidation is actually caused by a placebo effect or merely reported due to various biases (e.g. a trial participant saying what they think the researchers want to hear). Experiences of what appear to be accidental or semi-accidental reconsolidation are known to occur in the MDMA rave community even when the individual isn't planning or expecting a traditional therapeutic experience (Hunt & Evans, 2008a). This individual's statement from Hunt and Evans (2008a, p. 344) exemplifies it: "E broke the ice. Probably to this day, if I never did E at a party, I would probably still be antisocial and probably wouldn't even go to parties. But now that I've experienced the drug side ... I like the sober side now."

MDMA doesn't reconsolidate schemas that are fundamentally adaptive to the person's current environment, though symptom worsening may impair important functioning.

Certainty — medium/high. I have never heard of an unambiguous instance of this happening, though the assertion may be difficult to prove given the enormous complexity of the brain and imprecise and contested meaning of *adaptive.*

Defense Cascade and Safety

Learned predictions or schemas can cause the sympathetic and parasympathetic nervous systems to activate arousal, flight-or-fight, freeze, and immobility, though the categorization of these states is up for debate and there are unexplained complexities.

Certainty — high. The general principles of the defense activation seem well-established, non-controversial, and semi-mechanistic (Kozlowska et al., 2015). However, in my assessment, the pseudoscientific polyvagal theory (Luck, 2023) pervades the field of defense cascade research, so many of the proposed mechanisms are likely false.

Adverse symptoms persisting after the post-acute period are largely caused by shifts in the landscape of stuck schemas and defense cascade activation.

Certainty — medium/high. I think most adverse psychological effects of MDMA therapy appear highly compatible with this framing. There are also many anecdotal reports of individuals attributing their increased adverse symptoms to confronting too much avoided trauma all at once. In rare cases MDMA facilitates visceral evidence that something about people's normal internal sense of self is incorrect, which also creates adverse symptoms (Ingram, 2018, 2024).

Acute physical injury from MDMA is almost always caused by mixing it with dangerous activities, certain other drugs, or certain medical conditions.

Certainty — high. The primary causes of injury seem well understood (Rigg & Sharp, 2018; Wolfgang et al., 2025). There haven't been any significant reported adverse effects in trials where dangerous activity and drug interactions are absent and participants are screened for certain health issues. There could be rare exceptions that are poorly understood.

MDMA has a low risk of long-term physical problems when the precautions in Chapter 4 are followed.

Certainty — medium. To my knowledge, no studies have directly investigated this. However, even in the worst case—recreational use, where mixing drugs, high doses, adulterated pills, and over- and under- hydration are common—experts think MDMA, or substances sold as MDMA, have a relatively low risk. Across three independent panels of drug-misuse experts, non-therapeutic MDMA's physical health risk (drug-specific damage) was estimated as almost zero in two studies and equivalent to cannabis, or 1/3 that of alcohol, in another (Bonomo et al., 2019; Crossin et al., 2023; D. Nutt et al., 2010). Therapeutic use following the precautions in Chapter 4 is far more cautious than typical recreational use, suggesting that the risk in therapeutic contexts is much lower than the already low risk in recreational contexts.

MDMA does not directly cause significant post-acute cognitive issues with 2–3 mg/kg doses in therapeutic contexts.

Certainty — medium/high. One especially rigorous observational study of recreational use (median of 44 occasions) gave recreational users a battery of 15 neuropsychological tests (Halpern et al., 2011). That ruled out large effect sizes but didn't have the statistical power to rule out low-to-moderate effect sizes for heavy users. One small randomized study of MDMA therapy also did not find any significant cognitive effects after two sessions (M. C. Mithoefer et al., 2011). Animal studies show inconsistent and mild effects (M. Baggott & Mendelson, 2001). One of the best observational studies, Coray et al. (2025), found an association between MDMA and long-term cognitive impact. However, even that study did not adequately control for factors that are not present in therapeutic contexts, like overheating and drug mixing. Surfacing of traumatic material may also occasionally cause endogenous-opioid- or panic-induced cognitive impairment. Passie (2023) and Wolfgang et al. (2025) discuss the issue in further detail.

Appendix B

Information for Practitioners

This section is a collection of things particularly relevant to therapists and guides that I've come across while writing this book. Note that this book does not substitute for high-quality training.

Skills

It is important for practitioners to effectively help their clients through various difficult experiences. Bender et al. (2025) surveyed psychedelic (not just MDMA) therapy practitioners about the most challenging experiences they have managed in clients across all types of psychedelics. These were, from most to least common and excluding categories only reported by a single respondent to ensure they represented at least somewhat of a consensus:

- intense dysphoria
- disappointment with treatment
- reengaging with traumatic experience
- desiring to leave a session under the influence
- agitation (e.g., screaming, anger)

- difficulty with immersion in the experience
- post-treatment emotional instability
- suicidality

It's important for practitioners to be proficient in identifying and successfully working with opioid dampening and panic during MDMA sessions (see Sections 3.1 and 8.2).

Ethics

Maintaining especially high ethical boundaries is critical because MDMA can create intense feelings of trust and connection. Idealizing transference may also be intense (see Section 6.3 (Boundaries)). A number of practitioners have used psychedelics to abuse their clients (Davidson & Plesa, 2025; Hall, 2021).

Treatment

Evidence does not support phased treatment (a stabilization/resourcing phase preceeding reconsolidation) (Oprel et al., 2021; Sele et al., 2023; van Vliet et al., 2021). Starting with reconsolidation is just as effective. It's conceivable, but not established, that there are exceptions for cases of extreme dissociation.

Experience

Poulter and Ot'Alora (2023) recommends that personal experience with MDMA therapy is helpful but not necessary for therapists. Clients want to know that their practitioner understands what they will experience and how the process works. I think learning that via first-person experience is much more thorough and grounded than learning it secondhand. In my view, personal experience with MDMA-facilitated reconsolidation is also more helpful than experience with MDMA in other contexts.

Recommended Reading

- *Unlocking the Emotional Brain: Memory Reconsolidation and the Psychotherapy of Transformational Change* by Bruce Ecker et al. (Ecker et al., 2024) popularized the memory reconsolidation theory of therapeutic improvement. I think it is required reading for all mental health practitioners. "Memory Reconsolidation Understood and Misunderstood" by Bruce Ecker (Ecker, 2015) is a complementary resource. At least read this excellent summary of Ecker et al. (2024): "Book Summary: Unlocking the Emotional Brain" by Kaj Sotala (Sotala, 2019).

- "Fear and the Defense Cascade: Clinical Implications and Management" by Kasia Kozlowska et al. (Kozlowska et al., 2015) lays out a biological framework for tonic/collapsed immobility, flight-or-fight, and threat-induced alertness. As the paper itself notes, the descriptive part is more solid than the Clinical Interventions part. I also suggest caution about some proposed mechanisms of action because the paper occasionally relies on polyvagal theory (Luck, 2023). While many people find therapeutic interventions associated with polyvagal theory helpful, the theory's foundation in specific neuroanatomical and evolutionary claims has not held up to empirical scrutiny.

- "The PSIP Model. An Introduction to a Novel Method of Therapy: Psychedelic Somatic Interactional Psychotherapy" by Saj Razvi and Steven Elfrink (Razvi & Elfrink, 2020) discusses how to deal with dissociation during a session. It is clinical experience rather than science.

- "A Complex Systems Approach to the Study of Change in Psychotherapy" by Adele M Hayes and Leigh A Andrews (Hayes & Andrews, 2020) summarizes the complex systems approach to therapeutic change.

- "Symptoms and the Body: Taking the Inferential Leap" by Omer Van den Bergh et al. (Van den Bergh et al., 2017) created a popular, mechanistic theory of somatic symptoms.

- Scott Alexander's relevant blog posts: "Book Review: Surfing Uncertainty" (Alexander, 2017), "The Precision of Sensory Evidence" (Alexander, 2021a), and "Trapped Priors as a Basic Problem of Rationality" (Alexander, 2021b).

Appendix C

Reports of Internalized MDMA Therapy

The following is my experience with internalized MDMA therapy.

> My first MDMA session (2021) consisted of seemingly perfect safety and all-encompassing compassion. I realized how everyone's stuck reactions were due to learned, no-longer-helpful fears. I also felt a well of inviolable safety and emotional resilience inside me so strong that I thought I would feel ok inside even if I watched everyone and everything I had ever loved die in front of me.
>
> There was a certain week in 2022 or 2023, after perhaps 10 more MDMA therapy sessions, when I went walking to feel my anxiety. It felt really good to feel my anxiety for some reason I no longer clearly recall. It also gave me a strong therapy hangover. That week of reconsolidation is the earliest example I recall of reconsolidation that just *happened* with no deliberate effort or control on my part.
>
> Later in 2023, after about session 20, I read an article describing the process of coherence therapy and wanted to try it on myself. In coherence therapy you find a strong

contradictory experience for your stuck schemas. Once you have a mismatch, you activate both the mismatch and the stuck schema at the same time. Then, keeping that juxtaposition in place will reconsolidate the stuck schema.

I thought the inviolable-safety memory was the obvious choice for a mismatch. I activated it by vividly imagining lying under the tree I was under during that first MDMA session, how the dirt felt on my feet, and how the tree trunk felt in my hands. That activated the inviolable-safety knowledge strongly enough to start the reconsolidation process; the stuck schema I was working on was already activated. This was great; it was the first time I was able to make therapeutic progress without MDMA. The process was powerful enough to reconsolidate any stuck schema, just like MDMA. Also, like MDMA, it wasn't limited by the typical window of tolerance; it worked well during opioid dampening, intense avoidance, and near-overwhelming anxiety.

My mental health was terrible, so I was anxious to do however much reconsolidation was necessary to fix my issues in as short a time as possible. At first, coherence therapy using inviolable safety only worked in the two-week afterglow following an MDMA session. I was also limited to two hours a day by therapy hangovers. It took about another 5 MDMA sessions and two hours a day of coherence therapy in the afterglow to extend the process past two weeks. At that point I stopped MDMA therapy as it seemed redundant, and I was worried about side effects from high-frequency sessions.

Eventually, after a couple hundred hours of that practice, I no longer had to explicitly recall that knowledge of inviolable safety. Any time I noticed a distressing schema, I could just "flip a switch" in my mind and start the reconsolidation process.

Then after a further 300 hours of practice, reconsolidation started happening without deliberate intent whenever a stuck schema is strongly activated, and I'm not actively avoiding it. It can happen when I'm watching videos, talking to people, or doing other activities. I can tell when it's happening, but I might not notice if I'm sufficiently distracted. The reconsolidation process seems to be activated by fear; the more afraid I am, the higher the intensity of reconsolidation. Particularly intense fear will even *push through* a therapy hangover and let me do more reconsolidation than I otherwise could. If I want to turn up the intensity of reconsolidation, I can also still deliberately "flip the switch."

I later did more MDMA therapy sessions and gained a deeper understanding of the similarities and differences between MDMA therapy and internalized MDMA therapy. Both states of mind reconsolidate any stuck schema and function during panic, opioid dampening, and avoidance. Explicit understanding of schemas is much clearer on MDMA. During internalized MDMA therapy, I have a partial view of the schema. I'll notice the belief "I don't matter," for instance, along with anger. On MDMA I see the whole schema, which in that instance was "I'm angry that I don't matter to you. I'll die if I don't matter to you." Reconsolidation also seems much faster on MDMA, though it's hard to tell whether I'm spending a long time reconsolidating one "I don't matter" schema during internalized MDMA therapy or actually cycling through a number of "I don't matter" schemas. On MDMA it is much easier to distinguish between subtly different schemas.

Avoidance, justification, projection, and identification are also much less active on MDMA. During a couples therapy session with my sober partner, I, on MDMA, was able to express (and reconsolidate) my fears and angers as simple facts of my internal experience. Unlike in normal life, I wasn't trying to get her to do anything different or blame her for any of it. Much of the content was dark, but she said she didn't feel defensive because I wasn't making any of it about her.

I've also noticed that I can reconsolidate different schemas on MDMA for unclear reasons. Once when I was in tonic immobility for a week, auto-reconsolidation was reconsolidating some schemas, but apparently not the ones causing the immobility. I did an MDMA therapy session and immediately confronted the "I'm dying right now" schemas causing the immobility. The session reconsolidated those schemas, and the immobility didn't return. It may be that avoidance was inhibiting me from confronting those schemas, and only MDMA let me do it.

As of early 2026, I've done about ~1900 hours of reconsolidation. I've made tons of progress working through my backlog of stuck schemas, and I've become much less neurotic about many things. Unfortunately, severely disordered attachment and suspected childhood sexual abuse left me with an immense backlog of ever-present stuck fears and angers. Internalized MDMA therapy has been incredibly convenient for doing huge amounts of therapy without a therapist or MDMA. It also doesn't require the usual explicit process overhead of understanding my stuck schemas, figuring out a mismatch for each one, and then setting up the juxtapositions. It's kept me making therapeutic progress through periods of de-

> spair and depression when I surely wouldn't have had the capacity for any sort of typical therapy.
>
> The only downside I have noticed is that since I have many strong stuck fears activated every waking moment, the auto-reconsolidation also starts running every morning once I wake up enough. That inevitably leads to a therapy hangover a bit later. Then the auto-reconsolidation starts up again once the therapy hangover wears off. I've been therapy hungover most of my waking hours since I started the practice. I also haven't discovered any way to turn auto-reconsolidation off. It goes for about 2 hours a day, limited only by therapy hangovers. Other than "flipping the switch" to increase reconsolidation intensity, the only control I seem to have over the process is which stuck schemas are activated. I can activate different schemas by going to different places, doing different things, talking to people, etc. This causes the auto-reconsolidation process to preferentially reconsolidate the schemas those contexts activate. This lack of control is not optimal,[1] but the process keeps delivering therapeutic progress, and I haven't noticed any unambiguous side effects yet apart from therapy hangovers.

This next report is my informal interview with Anonymous (A). The interviewee explicitly consented to their answers being published in this book.

> **M.G.:** You mentioned you're able to enter an MDMA-like state via meditation. Would you describe what this state feels like? What differences does it have from an MDMA session?

[1]The lack of control might actually be a benefit. Many people drastically underestimate the benefits of reconsolidation, and the lack of control will keep the process going through any periods where I might be too depressed to manually activate reconsolidation.

A: It's clear that there is still a distinction between the effects of MDMA vs. the MDMA-like state that I can enter during meditation. For example, I don't get the same physiological response associated with MDMA (e.g., increased pupil size, increased heart rate, reduced appetite etc.), though I do get some jaw clenching which is interesting. I also don't get the same "rolling" waves of euphoria that you tend to get with MDMA. The way the process works for me is that for the first 30 minutes of the meditation, nothing will be happening. At around the 30 minute mark I will start to feel the same calmness and safety that I felt on MDMA. I will feel deeply at peace and often emotional. It's common for me to cry. This state will persist for as long as I stay in the meditative state and for as long as I want it to. I will use this time to explore what I processed in my actual MDMA sessions, and to explore things associated with my trauma that are still troubling me. Before MDMA-assisted therapy, I practiced meditation regularly but I had never been able to enter states like this before. I would be able to feel calm, but not the safety and peace that I felt on MDMA. This skill only came about after MDMA therapy.

M.G.: My framework for how MDMA therapy works is that during the session you activate one of your maladaptive fear/anger/sadness/etc. reactions that you learned in the past, but is no longer appropriate. Then you sit with that feeling and the MDMA just "unlearns" it over a period of minutes to tens of minutes. Then that particular chunk of reaction doesn't come back after the session, and it's easier to see what the reaction was and what role it played in your life. Of course, there might be many different instances of that reaction to unlearn, and each one has to be individually addressed. Is that how

MDMA therapy works for you? Is it also how using the MDMA-like states work?

A: I would say that your framework is partially true for me. I was able to shift certain emotions, particularly shame, and realise they were no longer appropriate. I also had a session where I came to a compromise with one of my emotions, fear. Instead of trying to convince myself that it was no longer appropriate, I validated it, and showed myself that it was okay to still feel fear, but instead we could learn to tame it with gentle talk and self-reassurance. I would say the biggest benefit of MDMA therapy was

1. I developed a new internal voice that was compassionate, rather than critical. When I am scared, when I am having PTSD symptoms, when I'm struggling, I now jump to self-compassion rather than self-hatred. I had spent 8 years in therapy trying to learn self-compassion without success. After MDMA therapy, that self-compassion was born and has persisted.
2. My sessions came with a lot of visualisation techniques. I experienced sexual trauma, and I had a lot of visualisations of myself handing over my anger and shame from myself to my perpetrator. I imagined the things that I would say to him now if I could. I imagined the things that I would say to myself in the wake of the trauma.

When I do my MDMA-like meditations now, I do visualisations once I enter that "state." The music is also very important. I will play "psychedelic-assisted therapy" music through noise-cancelling headphones. I will also set up my space in much the same way that I was during the sessions - lying down with a weighted blanket and with lights dimmed. Sometimes I will also use an eye-mask. I find this best facilitates an MDMA-like state for me.

M.G.: Would you describe what role the meditation and music played in first developing the capacity? Did the MDMA-like state appear the first time you meditated with music or only after a while? Did it happen after the first MDMA session or did it take multiple?

A: The ability came after my first MDMA session. I was meditating as a part of my integration practice, and one day I was listening to similar meditative music that was played during my session. Approximately 30 minutes into the meditation I had an outburst of emotion and peace that felt nearly identical as to what happened in my session. I ended up lying there for about 2 hours in this state. After having this experience, I started doing it regularly. I now practice it once a week or so. I spoke to the psychiatrist who sat with me for the session and he was intrigued, but he felt that it was unlikely to persist. I decided to keep practicing it in hopes that it would. A year later and it's still present.

M.G.: Are there any specific features of certain MDMA sessions you think were critical to developing the capacity? What larger role if your life has this capacity played? Has anything about the capacity changed over time?

A: Probably the feature of MDMA therapy that helped in generating this skill was the general sensation of safety that came with MDMA therapy. I felt safe with the psychiatrist sitting with me, and the drug created a window of safety that I had never felt before. My life has been marred by sexual and gender-based violence, so I had no reference point for what safety was. Now that I had an idea of what safety could look like and mean, I can now generate it internally by meditation. This effect has been beautiful in my life. Now, when I struggle with fear and a lack of safety, I know there is a place within me that I can always retreat to if I want to feel safe again. I wouldn't say anything has really changed over time except that maybe initially I viewed this skill purely as a way for me to continue doing the cognitive work by entering a MDMA-like state and continue processing my trauma, but now I view it also as a space to help me self-regulate if I am struggling with my PTSD-symptoms. If I find that I am having a few nightmares again or if I've had some strong triggers, I can grant myself a break from my anxiety and enter into a place of warmth and comfort.

M.G.: What type of meditation do you do for this?

A: I don't do any "formal" type of meditation. Essentially I lie down on a flat surface, usually a couch or a bed, with a weighted blanket and noise cancelling headphones, just as I did in my treatment sessions. I usually close my eyes and focus on my breath, usually beginning with long and slow breaths leading into a box breathing type of pattern until I hit the 30 minute mark when the "effects" start to emerge. At this point I am able to open my eyes, stop focusing on my breath, and let my mind wander.

M.G.: Do you think the first 30 min of meditation prior to the safety and calm emerging is doing something to bring the safety forth? Or is the emergence at 30 min just an association your brain has made, like "oh this is the part where I'm supposed to turn on the safety"?

A: In terms of the first 30 minutes, I think before are correct actually. I think that first 30 minutes is an important space for me to just seek to relax my nervous system. In my first treatment in particular, the first 30 minutes was a particularly daunting time. I had never done MDMA before or even any kind of drug besides marijuana, so I was very anxious about what was going to happen to me. I spent the first 30 minutes in my first treatment session trembling, trying to calm my breath, and placing my trust in my sitter to keep me safe. Simultaneously, I now have an association with the 30 minute mark as being the time when MDMA would approximately take effect, so I think my brain also has an assumption that that is the moment when the deep sense of safety is supposed to turn on.

M.G.: How does the durability of therapeutic improvement in the long-term compare between this capacity vs. with MDMA, when you do use it to process trauma?

A: I would say "durability" doesn't quite fit here for me. Instead it's like an add on experience. I'm adding new healing experiences that add on to the original MDMA experiences, which ensures the durability of my original treatment. I would say that the capacity to make sense of deeply painful emotions, thoughts and beliefs are the same between the MDMA sessions and the non-MDMA sessions, and that both are equally durable for me.

M.G.: Do you get exhausted ("therapy hangover") when you use the capacity for processing trauma?

A: I don't get "therapy hangover" fortunately from this. I actually get an "after glow" just like with my MDMA sessions. This afterglow persists for 3–4 days, and is characterised by an ongoing sense of calm, opennness and wholeness. Eventually this fades, but it doesn't mean

I return to my original state. I am always inching forward towards healing.

M.G.: You said you feel deeply emotional in the state. Is that just because some trauma feeling is activated, or does the emotion feel tied to the state itself?

A: That deeply emotional state is a release of all the emotions I was holding on to. I have a tendency to over-regulate my emotions. I push them down and suppress them. When I enter that state of calm, my emotions feel safe to emerge and they emerge strongly. This is also what happened on MDMA. When I felt that first moment of safety on MDMA, I unleashed a cascade of emotions. I cried for much of my first session. So I would say that emotional release is an embodiment of what I had been suppressing, and it only felt safe to come out once I entered that state.

M.G.: Have you had any disruptions to your sense of self, temporary or lasting?

A: I have had lasting changes to my sense of self since MDMA therapy. Before, I had a persistently negative sense of self. I believed that the trauma was my fault and that I was burden for not being able to heal. I didn't believe that I had much value and that I was unlovable because of what had happened to me. I believed that nobody could love a rape victim like me. My sense of self changed over the course of the 3 sessions. I saw that I had inherent value just for being a person, and that what happened to me had no affect on my worth. Anybody who treated me otherwise didn't matter. It was a problem with them, and not with me. I think the combination of MDMA therapy plus the presence of a therapist was critical to this change. While there are a lot of people with non-relational PTSD, it is more common for people to have PTSD from some kind of interpersonal trauma. The victim internalises the idea that this display of dominance means that there must be something wrong with them, especially if the victim was young when it happened. Healing from this interpersonal trauma requires relational healing, which for me meant being present with a male therapist who attentively listened for hours at a time, who didn't make any unwanted sexual advances on me, and who helped me understand that it was not my shame to carry. That shame lied with the perpetrator. Having the MDMA on board meant that this new belief was able to solidify, perhaps due to MDMA's prosocial and neuroplastic effects.

M.G.: Oh I should have specified that I meant a more fundamental sense of self, like the felt sense that there is a "you" that decides things, does things, believes things, etc. Extensive meditation and psychedelics sometimes cause lasting disruptions to this.

A: I wouldn't say there was much change to my fundamental sense of self. I am still me and I still exist as my own separate person. However, I would say things are perhaps a little more fluid now, perhaps because I feel more connected to the broader human experience. I did come to feel a sense of "oneness," that we are all part of something together, just living our own separate parts of it.

Appendix D

Mechanism of Action Hypotheses

I've developed several informal hypotheses for MDMA therapy's mechanism of action and how some people internalize the process of MDMA therapy. The evidence I list is anecdotal and phenomenological; I am not aware of any experiments that convincingly inform these ideas.

In this section I switch from memory reconsolidation terminology to predictive processing terminology for precision. The predictive processing terms here are used as conceptual vocabulary rather than mathematically precise definitions. Hopefully these hypotheses are helpful for prompting more formalized investigation. A *prior* or collection of *priors* is roughly equivalent to a schema in this context. The *precision* of a prior or sensory information refers to its certainty, and *updating* is essentially reconsolidation in this context. Updating becomes significant when contradictory information is more precise than the prior.

I think any proposed mechanism of action needs to explain these facets of MDMA therapy:

- The fundamental process of MDMA therapy is learnable, as described later in this section and in Appendix C. Therefore, the process doesn't require the altered neurochemistry that MDMA induces, even though the altered neurochemistry may make the process easier and more powerful.

- MDMA therapy appears to facilitate prediction error for all stuck priors predicting threat or powerlessness that are active and you pay attention to. Therefore, MDMA must either provide high-level contradictory information applicable to all stuck priors or disable some prediction-error-inhibiting process.

With those constraints in mind, I propose the following hypotheses. The supporting and contradictory evidence only lists what initially came to mind. It is not comprehensive.

Fundamental Safety or Satisfactoriness MDMA, along with meditation, does something to facilitate accurate perception of a particular kind of safety. This safety provides a profound "everything is ok just as it is" perception while simultaneously not deactivating healthy threat avoidance. Similarly, this safety does something to facilitate updating of stuck threat-avoidance priors but not adaptive threat-avoidance priors. It is unclear what this safety is or where it comes from. I speculate that it is some aspect of the present moment.

Supporting Evidence: People often feel that they discover a profound safety on MDMA even before they start engaging with stuck priors.

Contradictory Evidence: Profound safety is frequently not explicitly perceived while updating intensely distressing stuck priors, though it could be present but overshadowed.

Increased Metacognitive "Sensory" Precision MDMA does something to increase the precision of explicit perception of the full structure of a complex of stuck priors. This creates a new higher-level representation of previously vague or disconnected priors. This higher-level model is easier to update than the previous, vague model.

Supporting Evidence: It feels like priors are more "clear" and fully comprehended on MDMA, even before the updating process completes. Transforming priors from vague feelings to explicit, coherent models is also typically a prerequisite for conventional therapy, and that transformation is often sufficient to reconsolidate the stuck priors (Ecker et al., 2024).

Contradictory Evidence: None identified

Deidentification High-level priors that model the self make stuck priors that we identify with (possibly in a subtle, automatic way) have aberrantly high precision. MDMA deidentifies some or all stuck priors from the self-model, causing their precision to return to an adaptive level. Adaptive precision makes regular sensory information or other priors precise enough to update the priors when attention is brought to those priors.

Supporting Evidence: It frequently feels like there is less identification on MDMA.

Contradictory Evidence: Decreased identification could be a downstream effect of updated priors or fundamental safety, rather than a cause. Identification can still be strong on MDMA. Successful updating in conventional psychotherapy also causes decreased identification.

Decreased Avoidance of Stuck Priors Overwhelming stuck priors are typically avoided, inhibiting prediction error. MDMA creates enough safety that avoidance is reduced. In this state, regular incoming sensory information or other priors are sufficient to update all stuck priors that are paid attention to.

Supporting Evidence: It frequently feels like there is less avoidance on MDMA. Conventional therapy also relies on decreasing avoidance to facilitate reconsolidation.

Contradictory Evidence: Avoidance can still be strong on MDMA. Decreased avoidance could be a downstream effect of updated priors, fundamental safety, or deidentification, rather than a cause. Successful updating in conventional psychotherapy also causes decreased avoidance.

Reduced Precision of Social/Relational Priors Carhart-Harris and Friston (2019) proposes that MDMA reduces the precision of all priors that cause social inhibition. Therapeutically relevant stuck priors might largely fall into this category. Lowered precision of priors makes regular sensory information precise enough to update the priors when attention is brought to those priors.

Supporting Evidence: People often use MDMA to reduce social anxiety.

Contradictory Evidence: Does MDMA reduce other social priors to the extent it does social anxiety? It might also not be the only mechanism if MDMA therapy works for PTSD or anxiety that was solely caused by natural disasters or accidents.

Increased Capacity to Experience Multiple Concurrent Emotions Emilsson (2023) suggests that MDMA increases the capacity to experience multiple emotions at the same time and that this is a factor in its ability to update stuck priors. Experiencing multiple contradictory emotions or priors at the same time may facilitate the juxtaposition that reconsolidates stuck priors.

Supporting Evidence: None identified

Contradictory Evidence: None identified

It's not clear which of these are causes and which are effects. Perhaps they all combine to form a complex causal structure where cause and effect are not separable. They may also be just poorly understood conceptions of a single unified mechanism of action.

As described in Appendix C, people sometimes internalize the process of MDMA therapy. Sufficient exposure to the state, whichever mechanism is correct, creates a model of the experience or perception through a new set of priors. Individuals sometimes reactivate the new model they learned, typically via associative cues (e.g., music, meditation, tactile sensations) present during the MDMA therapy session. The cues don't have to be physically present; high-fidelity imagination of the cues was sufficient in one case. If they realize what is happening, the individual may now know how to control the process.

Some people report using this state to update away stuck priors in the same way MDMA therapy can, though less intensely. Notably, one person reports that this process updates any stuck prior with none of the "overhead" associated with memory reconsolidation in conventional therapy. It also worked just as well in the presence of panic or dissociation, unlike traditional therapy. This is essentially the same process as MDMA therapy. Notably, for both individuals, the capacity for internalized MDMA therapy does not dissipate over time, like one might expect if it was a simple conditioned response.

I think the *stillness* (Stocker & Liechti, 2024) and "everything is ok just how it is" (anecdotal reports) that MDMA can facilitate are also strikingly reminiscent of the absence of Duḥkha, a Buddhist concept meaning unsatisfactoriness. In that model, a subtle identification with craving (a pull toward pleasant sensations) and aversion (a push away from unpleasant sensations), rather than mere preferences, creates a baseline level of suffering/Duḥkha. The involvement of identification also circles back to the Deidentification hypothesis. Perhaps MDMA can not only deidentify people from specific stuck schemas, but from craving and aversion themselves. See Section 8.4.12 for further discussion. Ingram (2024) has also noticed a related link.

Appendix E

Alternatives to MDMA

Other drugs in the same class as MDMA may also be useful for therapy. The following are the most promising candidates that I'm aware of, though none have gone through as much safety and efficacy testing as MDMA.

- Methylone therapy had good results in a phase II clinical trial for PTSD (Jones et al., 2026). The dose was 150 mg plus a 100 mg booster dose 90 minutes later. Like MDMA, it is illegal in many jurisdictions.

- M. Baggott (2023) reports that 5-MAPB is anecdotally the closest single-drug analog to MDMA. The therapeutic dose is unclear, but Erowid (2026) reports that a common dose is 30–70 mg. Like MDMA, it is illegal in many jurisdictions.

- Borax (2014) developed a specific ratio of 2-FMA, MDAI, 5-MeO-MiPT, and 5-APB succinate whose effects are almost identical to MDMA's in their experience. They also discuss possible substitutions. The dose depends on the specific components used. The components are legally available in some jurisdictions.

Appendix F

Suggested Avenues of Future Research

I think the following research questions are particularly interesting or relevant to the practice of MDMA therapy.

- What is the irreducible set of mental states that MDMA induces? It seems to induce love, safety, emotional empathy, "stillness," "things are ok just as they are," connection, and sociability. Are those reducible to some smaller set of fundamental states? Answering this might first require resolving multiple major open problems in fundamental neuroscience.

- What is the nature of reconsolidation exhaustion (therapy hangover)? Can the capacity be safely increased? That would enable more productive MDMA therapy sessions, which anecdotal reports suggest are often limited by reconsolidation exhaustion.

- Multiple people report increased capacity to reconsolidate while sober in a way phenomenologically and consequentially similar to what is experienced during MDMA therapy sessions (see Appendix C). This appears to be connected to prior MDMA therapy. How does this work? Anecdotal reports also suggest that some individuals with no MDMA exposure can use the present moment to reconsolidate all stuck schemas. Is that the same phenomenon as internalized MDMA therapy?

- Why does MDMA seem to provide prediction error for most, if not all, stuck schemas, but not the adaptive ones? See Appendix D for my hypotheses.
- How can therapists best prepare their clients for solo at-home sessions (e.g., for sex therapy)?
- To what extent does MDMA causes valvular hearth disease? Accurate human guidelines are needed that account for number of sessions, session spacing, bodymass-adjusted dose, and possibly CYP2D6 capacity.
- How long does short- and long-term tolerance to MDMA last? How does it work?
- How long does it take to restore the brain's antioxidant buffer after an MDMA therapy session?
- To what degree is symptom worsening avoidable or reducible while achieving the same amount of reconsolidation?
- Some stuck schemas seem to activate other stuck schemas. It's conceivable that one could resolve multiple stuck schemas at once by reconsolidating the "root" schema. In such a case the other stuck schemas may still exist but wouldn't be an issue because they are never activated. This appears valuable but might require understanding the relevant schemas and their relationships with each other. I don't know to what degree such a process is theoretically possible, practically achievable, or time efficient. It's also unclear whether MDMA, as a facilitator of seemingly universal prediction error for all stuck schemas, can be used to precision-target a single schema in a stack of simultaneously activating stuck schemas.

- Who is most at risk of symptom worsening? The current commonly used risk factors (history of psychosis, mania, personality disorders, active suicidality, heavy dissociation, diagnosed mental illness) appear to be partly based on guesses and liability-avoidance. Precise and assessable risk factors are important for determining who MDMA therapy will work best for and whether someone should do it solo vs. practitioner guided.

- Are there factors that cause MDMA therapy to not work for some people, apart from too low or too high a dose, opioid dampening, avoidance, and drug interactions like with SSRIs?

- Why does reconsolidating certain schemas frequently require first reconsolidating some other schema?

- Are conventional therapeutic reconsolidation exercises more effective in the post-MDMA afterglow period? Why? For how long?

- The process of MDMA therapy often involves a flow of a schema becoming the object of perception and then reconsolidating, a different schema becoming the object of perception and then reconsolidating, etc. This sequence might be a valuable clue to how schemas form complex systems.

- In what circumstances should someone do another MDMA session to resolve the symptom worsening from a previous MDMA therapy session? When will another session increase or decrease symptoms? On what time scale?

- Does MDMA only reconsolidate the single schema that is currently the object of perception/focus, or does it also reconsolidate other schemas at the same time? Perhaps it reconsolidates a network of related schemas.

- Why do people apparently in tonic or collapsed immobility on MDMA sometimes transition to a state where they feel sober or bored (see Razvi, 2024)? Alternatively, they can start off sober or bored and never seem feel the effects of MDMA.

- Therapy is a process of moving from the individual's current position in the schema/environment state-space to a more optimal position. It's conceivable that different paths between these two states are possible. How possible and practical is it to plot and follow the shortest-distance reconsolidation path? How optimal are the typical paths in MDMA therapy, where the individual either reconsolidates whatever schemas naturally arise or deliberately activates certain schemas they want to work on?

- For people who need a higher dose to deal with opioid dampening or avoidance, is it better from efficacy and oxidative stress standpoints to bump up a split dose (e.g., going from 120 + 60 mg to 140 + 70 mg) or frontload the whole dose (e.g., going from 120 + 60 mg to 180 mg)?

- Does MDMA therapy optimize your schemas according to your existing set of fundamental prediction errors or to some MDMA-modified set?

Bibliography

Aboujaoude, E. (2020). Where life coaching ends and therapy begins: Toward a less confusing treatment landscape. *Perspectives on Psychological Science*, *15*(4), 973–977. DOI: 10.1177/1745691620904962.

Aftab, A. (2025). Psychiatric genetics. *Psychiatry at the Margins*. URL: psychiatrymargins.com/p/psychiatric-genetics.

Aguirre, N., Barrionuevo, M., Ramírez, M. J., Del Río, J., & Lasheras, B. (1999). A-lipoic acid prevents 3,4-methylenedioxy-methamphetamine (MDMA)-induced neurotoxicity. *Neuroreport*, *10*(17), 3675–3680. DOI: 10.1097/00001756-199911260-00039.

Aizenbud, I., Audette, N., Auksztulewicz, R., Basiński, K., Bastos, A. M., Berry, M., Canales-Johnson, A., Choi, H., Clopath, C., Cohen, U., Costa, R. P., Filippo, R. D., Doronin, R., Errington, S. P., Gavornik, J. P., Gillon, C. J., Granier, A., Hamm, J. P., Hertäg, L., … Xiong, Y. S. (2025). *Neural mechanisms of predictive processing: A collaborative community experiment through the OpenScope program*. DOI: 10.48550/arXiv.2504.09614.

Alexander, S. (2017). Book review: Surfing uncertainty. *Slate Star Codex*. URL: slatestarcodex.com/2017/09/05/book-review-surfing-uncertainty.

Alexander, S. (2018). Navigating and/or avoiding the inpatient mental health system. *Slate Star Codex*. URL: slatestarcodex.com/2018/03/22/navigating-and-or-avoiding-the-inpatient-mental-health-system/.

Alexander, S. (2019). Mental mountains. *Slate Star Codex*. URL: slatestarcodex.com/2019/11/26/mental-mountains/.

Alexander, S. (2021a). The precision of sensory evidence. *Slate Star Codex*. URL: astralcodexten.com/p/the-precision-of-sensory-evidence.

Alexander, S. (2021b). Trapped priors as a basic problem of rationality. *Slate Star Codex*. URL: astralcodexten.com/p/trapped-priors-as-a-basic-problem.

Alexander, S. (2025a). The good news is that one side has definitively won the missing heritability debate. *Astral Codex Ten*. URL: astralcodexten.com/p/the-good-news-is-that-one-side-has.

Alexander, S. (2025b). Missing heritability: Much more than you wanted to know. *Astral Codex Ten*. URL: astralcodexten.com/p/missing-heritability-much-more-than.

Alexianer St. Hedwig Hospital. (2024). *The psychedelic substance outpatient clinic*. URL: alexianer-berlin-hedwigkliniken.de/st-hedwig-krankenhaus/leistungen/ambulante-behandlung/ambulanz-psychedelische-substanzen.

Aliko, S., Wang, B., Small, S. L., & Skipper, J. I. (2023). *The entire brain, more or less, is at work: "Language Regions" are artefacts of averaging*. DOI: 10.1101/2023.09.01.555886.

Althea. (2025). *How much does a legal psilocybin journey cost? and is it worth it?* URL: withalthea.com/2025/02/27/how-much-does-a-legal-psilocybin-journey-cost/.

Alves, E., Binienda, Z., Carvalho, F., Alves, C., Fernandes, E., de Lourdes Bastos, M., Tavares, M., & Summavielle, T. (2009). Acetyl-L-carnitine provides effective in vivo neuroprotection over 3,4-methylenedioximethamphetamine-induced mitochondrial neurotoxicity in the adolescent rat brain. *Neuroscience*, *158*(2), 514–523. DOI: 10.1016/j.neuroscience.2008.10.041.

American Association of Sex Educators, Counselors, and Therapists. (2020). Code of conduct for AASECT certified members. URL:

aasect.org/sites/default/files/documents/AASECT_Code-of-conduct-11.2020_4.20.23-edit_0.pdf.

American Center for the Integration of Spiritually Transformative Experiences. (2025). *American center for the integration of spiritually transformative experiences.* URL: aciste.org.

Andrews, K., Birch, J., Sebo, J., & Sims, T. (2024). Background to the New York declaration on animal consciousness. URL: nydeclaration.com.

Argyri, E. K., Krecké, J., Robinson, O. C., Evans, J., Skragge, M., & Morgan, C. J. (2025). Practitioner perspectives on extended difficulties and optimal support strategies following psychedelic experiences: A qualitative analysis. *Harm Reduction Journal.* DOI: 10.1186/s12954-025-01347-0.

Arnovitz, M. D., Spitzberg, A. J., Davani, A. J., Vadhan, N. P., Holland, J., Kane, J. M., & Michaels, T. I. (2022). MDMA for the treatment of negative symptoms in schizophrenia. *Journal of Clinical Medicine*, *11*(12), 3255. DOI: 10.3390/jcm11123255.

Arshad, M. (2024). Struggling telehealth company exploited Adderall sales for profit, prosecutors say. *USA Today.* URL: usatoday.com/story/news/nation/2024/06/13/telehealth-execs-arrested-100-million-fraud-adhd/74091441007.

Atila, C., Straumann, I., Vizeli, P., Beck, J., Monnerat, S., Holze, F., Liechti, M. E., & Christ-Crain, M. (2024). Oxytocin and the role of fluid restriction in MDMA-induced hyponatremia: A secondary analysis of 4 randomized clinical trials. *JAMA Network Open*, *7*(11), e2445278–e2445278. DOI: 10.1001/jamanetworkopen.2024.45278.

Author Redacted. (2024). *Lykos complete response letter.* U.S. Food and Drug Administration. URL: psychedelicalpha.com/wp-content/uploads/2025/09/CRL_NDA215455_20240808.pdf.

Baggott, M. (2015). *Thoughts on taking supplements with MDMA.* URL: reddit.com/r/MDMA/comments/3r09sg/thoughts_on_taking_supplements_with_mdma/.

Baggott, M. (2016). *Mechanisms of MDMA tolerance and loss of magic.* URL: reddit.com/r/MDMA/comments/4wyjd9/mechanisms_of_mdma_tolerance_and_loss_of_magic/.

Baggott, M. (2023). *Beyond ecstasy: Progress in developing and understanding a novel class of therapeutic medicine.* URL: virtualtrip.maps.org/video/beyond-ecstasy-progress-in-developing-and-understanding-a-novel-class-of-therapeutic-medicine/.

Baggott, M., & Mendelson, J. (2001). Does MDMA cause brain damage? In J. Holland (Ed.), *Ecstasy: The complete guide.* Park Street Press.

Baggott, M. J., Garrison, K. J., Coyle, J. R., Galloway, G. P., Barnes, A. J., Huestis, M. A., & Mendelson, J. E. (2016). MDMA impairs response to water intake in healthy volunteers. *Advances in Pharmacological and Pharmaceutical Sciences*, *2016*(1), 2175896. DOI: 10.1155/2016/2175896.

Barford, D. (2019). *Coping with mental health challenges on retreat.* URL: firekasina.org/2019/05/02/mental-health/.

Barnard Center for Research on Women. (2020). *Building accountable communities.* URL: bcrw.barnard.edu/building-accountable-communities.

Barsky, A. E., & Spadola, C. E. (2023). Licensing investigations: Suggestions from social workers who received sanctions. *Social Work Research*, *47*(2), 135–148. DOI: 10.1093/swr/svad002.

Bartov, S. L. (2023). BetterHelp patients furious at "Sketchy" therapists. *Newsweek.* URL: newsweek.com/betterhelp-patients-tell-sketchy-therapists-1762849.

Bartu, A., Dusci, L. J., & Ilett, K. F. (2009). Transfer of methylamphetamine and amphetamine into breast milk following recreational use of methylamphetamine. *British Journal of Clinical Pharmacology*, *67*(4), 455–459. DOI: doi.org/10.1111/j.1365-2125.2009.03366.x.

Bedi, G., Hyman, D., & de Wit, H. (2010). Is ecstacy an "Empathogen?" effects of 3,4-methylenedioxymethamphetamine on prosocial feelings and identification of emotional states in others. *Bio-*

logical Psychiatry, *68*(12), 1134–1140. DOI: 10.1016/j.biopsych.2010.08.003.

Bender, D. A., Nayak, S. M., Siegel, J. S., Hellerstein, D. J., Ercal, B. C., & Lenze, E. J. (2025). Provider perspectives on challenges in treatment during psychedelic therapy. *Psychopharmacology*, 1–8. DOI: 10.1007/s00213-025-06907-7.

Bengt. (2006). *Bipolar reaction: An experience with MDMA (ecstasy) (exp39866)*. Erowid. URL: erowid.org/exp/39866.

Berro, L. F., Shields, H., Odabas-Geldiay, M., Rothbaum, B. O., Andersen, M. L., & Howell, L. L. (2018). Acute effects of 3,4-methylenedioxymethamphetamine (MDMA) and R(-) MDMA on actigraphy-based daytime activity and sleep parameters in rhesus monkeys. *Experimental and Clinical Psychopharmacology*, *26*(4), 410. DOI: 10.1037/pha0000196.

Better Business Bureau. (2024). *Customer reviews for Talkspace*. URL: bbb.org/us/ny/new-york/profile/mental-health-services/talkspace-0121-149740/customer-reviews.

Biaggio, M., Duffy, R., & Staffelbach, D. F. (1998). Obstacles to addressing professional misconduct. *Clinical Psychology Review*, *18*(3), 273–285. DOI: 10.1016/S0272-7358(97)00109-8.

Birch, J. (2021). *The science of animal sentience: Sentience and the precautionary principle*. URL: youtube.com/watch?v=PD1dN-A_F0s.

Birch, J. (2024). *The edge of sentience: Risk and precaution in humans, other animals, and AI*. Oxford Academic.

Bonanno, G. A. (2008). Loss, trauma, and human resilience: Have we underestimated the human capacity to thrive after extremely aversive events? *Psychological Trauma: Theory, Research, Practice, and Policy*. DOI: 10.1037/1942-9681.S.1.101.

Bonomo, Y., Norman, A., Biondo, S., Bruno, R., Daglish, M., Dawe, S., Egerton-Warburton, D., Karro, J., Kim, C., Lenton, S., Lubman, D. I., Pastor, A., Rundle, J., Ryan, J., Gordon, P., Sharry, P., Nutt, D., & Castle, D. (2019). The Australian drug harms ranking study. *Journal of Psychopharmacology*, *33*(7), 759–768. DOI: 10.1177/0269881119841569.

Borax. (2014). *The MDMA experience can be replicated with less neurotoxicity using unscheduled compounds. here's how.* URL: reddit.com/r/Drugs/comments/224llv/the_mdma_experience_can_be_replicated_with_less/.

Borax. (2022). *How to run an "Acetone Wash" on MDMA to remove caffeine, leftover precursors and some other impurities.* URL: reddit.com/r/Borax/comments/pxccet/how_to_run_an_acetone_wash_on_mdma_to_remove/.

Bracchi, M., Stuart, D., Castles, R., Khoo, S., Back, D., & Boffito, M. (2015). Increasing use of "Party Drugs" in people living with HIV on antiretrovirals: A concern for patient safety. *Aids*, *29*(13), 1585–1592. DOI: 10.1097/QAD.0000000000000786.

Brach, T. (2023). *Feeling overwhelmed? try the RAIN meditation.* URL: mindful.org/tara-brach-rain-mindfulness-practice.

Brewer, J. (2024). *The habit mapper.* URL: drjud.com/wp-content/uploads/2021/03/Unwinding-Anxiety-Habit-Mapper-from-DrJud-1-1.pdf.

Briggs, R. (2025). Can we trust social science yet? *Asterisk.* URL: asteriskmag.com/issues/10/can-we-trust-social-science-yet.

Britton, W. (2018). *Dissociation and hyperarousal.* URL: cheetahhouse.org/hyperarousal-and-dissociation.

Brown, D. P., & Elliott, D. S. (2016). *Attachment disturbances in adults: Treatment for comprehensive repair.* WW Norton & Company.

Brown, J. (2020). *Ethical transgressions and boundary violations in ayahuasca healing contexts: A mixed methods study* [Doctoral dissertation, California Institute of Integral Studies]. URL: proquest.com/docview/2476161064?pq-origsite=gscholar&fromopenview=true&sourcetype=Dissertations%20&%20Theses.

Buchanan, B., Bartholomew, E., Smyth, C., & Hegarty, D. (2024). *A comprehensive questionnaire for schemas related to psychopathology: The maladaptive schema scale - version 1.4 (MSSv1.4).* DOI: 10.17605/OSF.IO/C3UPR.

Burbea, R. (2025). *Seeing that frees: Meditations on emptiness and dependent arising: 10th anniversary edition*. Hermes Amāra Publications.

Calder, A. E., Diehl, V. J., & Hasler, G. (2025). Traumatic psychedelic experiences. In *Current topics in behavioral neurosciences*. Springer Berlin Heidelberg. DOI: 10.1007/7854_2025_579.

Calder, A. E., & Hasler, G. (2023). Towards an understanding of psychedelic-induced neuroplasticity. *Neuropsychopharmacology*, *48*(1), 104–112. DOI: 10.1038/s41386-022-01389-z.

Canby, N. K., Lindahl, J. R., Cooper, D. J., Joseph, N., Palitsky, R., & Britton, W. B. (2025). The teacher matters: The role and impact of meditation teachers in the trajectories of Western Buddhist meditators experiencing meditation-related challenges. *Contemporary Buddhism*, *25*(1-2), 9–53. DOI: 10.1080/14639947.2025.2485677.

Carhart-Harris, R. L., & Friston, K. J. (2019). REBUS and the anarchic brain: Toward a unified model of the brain action of psychedelics. *Pharmacological Reviews*, *71*(3), 316–344. DOI: 10.1124/pr.118.017160.

Carhart-Harris, R. L., & Nutt, D. J. (2010). User perceptions of the benefits and harms of hallucinogenic drug use: A web-based questionnaire study. *Journal of Substance Use*, *15*(4), 283–300. DOI: 10.3109/14659890903271624.

Carr, R. (2011). *The end of coaching as we knew it*. URL: linkedin.com/pulse/end-coaching-we-knew-rey-carr.

Carvalho, A. F., Solmi, M., Sanches, M., Machado, M. O., Stubbs, B., Ajnakina, O., Sherman, C., Sun, Y. R., Liu, C. S., Brunoni, A. R., Pigato, G., Fernandes, B. S., Bortolato, B., Husain, M. I., Dragioti, E., Firth, J., Cosco, T. D., Maes, M., Berk, M., … Herrmann, N. (2020). Evidence-based umbrella review of 162 peripheral biomarkers for major mental disorders. *Translational Psychiatry*, *10*(1), 152. DOI: 10.1038/s41398-020-0835-5.

Cashwell, C. S., Bentley, P. B., & Yarborough, J. P. (2007). The only way out is through: The peril of spiritual bypass. *Counseling* *and Values*, *51*(2), 139–148. DOI: 10.1002/j.2161-007X.2007.tb00071.x.

Celenza, A. (2024). *Sexual boundary violations.* URL: andreacelenza.com/sexual-boundary-violations.

Cheetah House. (2024). *Resources and support for adverse meditative experiences.* URL: cheetahhouse.org.

Clark, A. (2015). *Surfing uncertainty: Prediction, action, and the embodied mind.* Oxford University Press.

Clark, J. E., Watson, S., & Friston, K. J. (2018). What is mood? a computational perspective. *Psychological Medicine*, *48*(14), 2277–2284. DOI: 10.1017/S0033291718000430.

Clauwaert, K. M., Van Bocxlaer, J. F., & De Leenheer, A. P. (2001). Stability study of the designer drugs “MDA, MDMA and MDEA” in water, serum, whole blood, and urine under various storage temperatures. *Forensic Science International*, *124*, 36–42. DOI: 10.1016/S0379-0738(01)00562-X.

Clean Air Kits. (2025a). *Did CleanAirKits just break EnergyStar for room air cleaners?* URL: cleanairkits.com/blogs/news/did-cleanairkits-just-break-energystar-for-room-air-cleaners.

Clean Air Kits. (2025b). *Sizing guide.* URL: cleanairkits.com/pages/sizing.

Cohen, B. M., & Öngür, D. (2023). The need for evidence-based updating of ICD and DSM models of psychotic and mood disorders. *Molecular Psychiatry*, *28*(5), 1836–1838. DOI: 10.1038/s41380-023-01967-7.

Coherence Psychology Institute. (2024). *Coherence psychology institute referral directory.* URL: coherencetherapy.org/prac/directory-terms.php.

Colbert, R., & Hughes, S. (2023). Evenings with molly: Adult couples’ use of MDMA for relationship enhancement. *Culture, Medicine, and Psychiatry*, *47*(1), 252–270. DOI: 10.1007/s11013-021-09764-z.

Colcott, J., Guerin, A. A., Carter, O., Meikle, S., & Bedi, G. (2024). Side-effects of MDMA-assisted psychotherapy: A systematic review and meta-analysis. *Neuropsychopharmacology*, *49*, 1208–1226. DOI: 10.1038/s41386-024-01865-8.

Cooke, D. J., & Michie, C. (2001). Refining the construct of psychopathy: Towards a hierarchical model. *Psychological Assessment*, *13*(2), 171. DOI: 10.1037/1040-3590.13.2.171.

Coray, R. C., Beliveau, V., Zimmermann, J., Preller, K. H., Wunderli, M., Baumgartner, M. R., Seifritz, E., Stock, A.-K., Beste, C., Cole, D. M., & Quednow, B. B. (2025). Memory deficits of MDMA users are linked to cortical thinning related to 5-HT receptor densities. *Brain*, awaf391. DOI: 10.1093/brain/awaf391.

Crossin, R., Cleland, L., Wilkins, C., Rychert, M., Adamson, S., Potiki, T., Pomerleau, A. C., MacDonald, B., Faletanoai, D., Hutton, F., Noller, G., Lambie, I., Sheridan, J. L., George, J., Mercier, K., Maynard, K., Leonard, L., Walsh, P., Ponton, R., … Boden, J. (2023). The New Zealand drug harms ranking study: A multi-criteria decision analysis. *Journal of Psychopharmacology*, *37*(9), 891–903. DOI: 10.1177/02698811231182012.

Davidson, T., & Plesa, P. (2025). Break on through: Betty Eisner's problematic use of psychedelics, groups, and control for integrative experiences. *Journal of the History of the Behavioral Sciences*, *61*(3), e70027. DOI: 10.1002/jhbs.70027.

De La Torre, R., Farre, M., Ortuno, J., Mas, M., Brenneisen, R., Roset, P., Segura, J., & Cami, J. (2000). Non-linear pharmacokinetics of MDMA ("Ecstacy") in humans. *British Journal of Clinical Pharmacology*, *49*(2), 104–109. DOI: 10.1046/j.1365-2125.2000.00121.x.

Delgadillo, J., Deisenhofer, A.-K., Probst, T., Shimokawa, K., Lambert, M. J., & Kleinstäuber, M. (2022). Progress feedback narrows the gap between more and less effective therapists: A therapist effects meta-analysis of clinical trials. *Journal of Consulting and Clinical Psychology*, *90*(7), 559. DOI: doi/10.1037/ccp0000747.

Dell, L., Sbisa, A. M., Forbes, A., O'Donnell, M., Bryant, R., Hodson, S., Morton, D., Battersby, M., Tuerk, P. W., Elliott, P., Wallace, D.,

& Forbes, D. (2023). Massed v. standard prolonged exposure therapy for PTSD in military personnel and veterans: 12-month follow-up of a non-inferiority randomised controlled trial. *Psychological Medicine*, *53*(15), 7070–7077. DOI: 10.1017/S0033291723000405.

Devereux, D. (2016). Transference love and harm. *Therapy Today.* URL: web.archive.org/web/20250408183730/https://www.bacp.co.uk/bacp-journals/therapy-today/2016/september/transference-love-and-harm/.

de-Wit, L., Alexander, D., Ekroll, V., & Wagemans, J. (2016). Is neuroimaging measuring information in the brain? *Psychonomic Bulletin & Review*, *23*, 1415–1428. DOI: 10.3758/s13423-016-1002-0.

Dominski, F. H., Lorenzetti Branco, J. H., Buonanno, G., Stabile, L., Gameiro da Silva, M., & Andrade, A. (2021). Effects of air pollution on health: A mapping review of systematic reviews and meta-analyses. *Environmental Research*, *201*, 111487. DOI: 10.1016/j.envres.2021.111487.

Droogmans, S., Cosyns, B., D'haenen, H., Creeten, E., Weytjens, C., Franken, P. R., Scott, B., Schoors, D., Kemdem, A., Close, L., Vandenbossche, J.-L., Bechet, S., & Camp, G. V. (2007). Possible association between 3,4-methylenedioxymethamphetamine abuse and valvular heart disease. *The American Journal of Cardiology*, *100*(9), 1442–1445. DOI: 10.1016/j.amjcard.2007.06.045.

Dunlap, B., Basye, A., & Skillman, S. M. (2021). *Background checks and the health workforce: Practices, policies and equity.* University of Washington Department of Family Medicine. URL: familymedicine.uw.edu/chws/wp-content/uploads/sites/5/2021/11/Background-Checks-FR-2021.pdf.

Eaton, N. R., Bringmann, L. F., Elmer, T., Fried, E. I., Forbes, M. K., Greene, A. L., Krueger, R. F., Kotov, R., McGorry, P. D., Mei, C., & Waszczuk, M. A. (2023). A review of approaches and models in psychopathology conceptualization research. *Nature*

Reviews Psychology, *2*(10), 622–636. DOI: 10.1038/s44159-023-00218-4.

Ecker, B. (2015). Memory reconsolidation understood and misunderstood. *International Journal of Neuropsychotherapy*, *3*(1), 2–46. DOI: 10.12744/ijnpt.2015.0002-0046.

Ecker, B., Ticic, R., & Hulley, L. (2024). *Unlocking the emotional brain: Memory reconsolidation and the psychotherapy of transformational change*. Taylor & Francis. DOI: 10.4324/9781003231431.

Edinoff, A. N., Swinford, C. R., Odisho, A. S., Burroughs, C. R., Stark, C. W., Raslan, W. A., Cornett, E. M., Kaye, A. M., & Kaye, A. D. (2022). Clinically relevant drug interactions with monoamine oxidase inhibitors. *Health Psychology Research*, *10*(4). DOI: 10.52965/001c.39576.

Ehlers, A., Hackmann, A., Grey, N., Wild, J., Liness, S., Albert, I., Deale, A., Stott, R., & Clark, D. M. (2014). A randomized controlled trial of 7-day intensive and standard weekly cognitive therapy for PTSD and emotion-focused supportive therapy. *American Journal of Psychiatry*, *171*(3), 294–304. DOI: 10.1176/appi.ajp.2013.13040552.

Elsey, J. W., van Ast, V. A., & Kindt, M. (2018). Human memory reconsolidation: A guiding framework and critical review of the evidence. *Psychological Bulletin*, *144*(8), 797. DOI: 10.1037/bul0000152.

Emanuel, N., Welle, P., & Bolotnyy, V. (2025). *A danger to self and others: Health and criminal consequences of involuntary hospitalization*. Federal Reserve Bank of New York. URL: newyorkfed.org/medialibrary/media/research/staff_reports/sr1158.pdf?sc_lang=en.

Emde, K. (2003). MDMA (ecstasy) in the emergency department. *Journal of Emergency Nursing*, *29*(5), 440–443. DOI: 10.1016/S0099-1767(03)00292-7.

Emilsson, A. G. (2023). The phenomenology of MDMA: Self-honesty, authenticity, & the unraveling of gnarly knots in the field.

Andrés Gómez Emilsson. URL: youtube.com/watch?v=86nyhA5Fsh0.

Engel, G. L. (1977). The need for a new medical model: A challenge for biomedicine. *Science*, *196*(4286), 129–136. DOI: 10.1126/science.847460.

Erowid. (2024). *MDMA (also ecstasy; molly; adam; E; X) reports - addiction & habituation*. URL: erowid.org/experiences/subs/exp_MDMA_Addiction_Habituation.shtml.

Erowid. (2026). *5-mapb dose*. URL: erowid.org/chemicals/5-mapb/5-mapb_dose.shtml.

Etkin, A. (2019). A reckoning and research agenda for neuroimaging in psychiatry. *American Journal of Psychiatry*, *176*(7), 507–511. DOI: 10.1176/appi.ajp.2019.19050521.

European Union Drugs Agency. (2025). *EU drug market: MDMA — in-depth analysis*. European Union Drugs Agency. URL: euda.europa.eu/publications/eu-drug-markets/mdma_en.

Evans, J. (2024). Can MDMA lead to extended difficulties? *Ecstatic Integration*. URL: ecstaticintegration.org/p/can-mdma-lead-to-extended-difficulties.

Evans, J. (2025). Are you a psychedelic cultist? *Ecstatic Integration*. URL: ecstaticintegration.org/p/are-you-a-psychedelic-cultist.

Evans, J., Robinson, O. C., Argyri, E. K., Suseelan, S., Murphy-Beiner, A., McAlpine, R., Luke, D., Michelle, K., & Prideaux, E. (2023). Extended difficulties following the use of psychedelic drugs: A mixed methods study. *PLOS One*, *18*(10), e0293349. DOI: 10.1371/journal.pone.0293349.

Evens, R., Schmidt, M. E., Majić, T., & Schmidt, T. T. (2023). The psychedelic afterglow phenomenon: A systematic review of subacute effects of classic serotonergic psychedelics. *Therapeutic Advances in Psychopharmacology*, *13*, 20451253231172254. DOI: 10.1177/20451253231172254.

Farré, M., Tomillero, A., Pérez-Mañá, C., Yubero, S., Papaseit, E., Roset, P.-N., Pujadas, M., Torrens, M., Camí, J., & de la Torre, R. (2015). Human pharmacology of 3,4-methylenedioxymethamphetamine

(MDMA, ecstasy) after repeated doses taken 4 h apart human pharmacology of MDMA after repeated doses taken 4 h apart. *European Neuropsychopharmacology*, *25*(10), 1637–1649. DOI: 10.1016/j.euroneuro.2015.05.007.

Feduccia, A. A., Jerome, L., Mithoefer, M. C., & Holland, J. (2021). RETRACTED ARTICLE: Discontinuation of medications classified as reuptake inhibitors affects treatment response of MDMA-assisted psychotherapy. *Psychopharmacology*, *238*(2), 581–588. DOI: 10.1007/s00213-020-05710-w.

Feduccia, A. A., Jerome, L., Mithoefer, M. C., & Holland, J. (2024). Retraction note: Discontinuation of medications classified as reuptake inhibitors affects treatment response of MDMA-assisted psychotherapy. DOI: 10.1007/s00213-024-06671-0.

Feduccia, A. A., & Mithoefer, M. C. (2018). MDMA-assisted psychotherapy for PTSD: Are memory reconsolidation and fear extinction underlying mechanisms? *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, *84*(A), 221–228. DOI: 10.1016/j.pnpbp.2018.03.003.

feilong420. (2016). *A user friendly purification guide - turn brown oily MDMA into beautiful white crystals!* URL: reddit.com/r/MDMA/comments/4a48iq/a_user_friendly_purification_guide_turn_brown/.

Fiorentini, A., Cantù, F., Crisanti, C., Cereda, G., Oldani, L., & Brambilla, P. (2021). Substance-induced psychoses: An updated literature review. *Frontiers in Psychiatry*, *Volume 12 - 2021*. DOI: 10.3389/fpsyt.2021.694863.

Fireside Project. (2023). *The psychedelic peer support line provides emotional support during and after psychedelic experiences.* URL: firesideproject.org.

First, M. B. (2026). Medical assessment of the patient with psychiatric symptoms. In *Merck manual professional version.* URL: merckmanuals.com/professional/psychiatric-disorders/approach-to-the-patient-with-psychiatric-symptoms/medical-assessment-of-the-patient-with-psychiatric-symptoms.

Firth, N., Saxon, D., Stiles, W. B., & Barkham, M. (2019). Therapist and clinic effects in psychotherapy: A three-level model of outcome variability. *Journal of Consulting and Clinical Psychology*, *87*(4), 345. DOI: 10.1037/ccp0000388.

Flockhard, D. (2025). *Drug interactions Flockhart table.* URL: drug-interactions.medicine.iu.edu/MainTable.aspx.

Flückiger, C., Del Re, A. C., Wampold, B. E., & Horvath, A. O. (2018). The alliance in adult psychotherapy: A meta-analytic synthesis. *Psychotherapy*, *55*(4), 316. DOI: 10.1037/pst0000172.

Foa, E. B., McLean, C. P., Zang, Y., Rosenfield, D., Yadin, E., Yarvis, J. S., Mintz, J., Young-McCaughan, S., Borah, E. V., Dondanville, K. A., Fina, B. A., Hall-Clark, B. N., Lichner, T., Litz, B. T., Roache, J., Wright, E. C., Peterson, A. L., & STRONG STAR Consortium. (2018). Effect of prolonged exposure therapy delivered over 2 weeks vs 8 weeks vs present-centered therapy on PTSD symptom severity in military personnel: A randomized clinical trial. *JAMA*, *319*(4), 354–364. DOI: 10.1001/jama.2017.21242.

Foisy, N. (2024). Is BetterHelp a scam? unpacking the truth. URL: compassitc.com/blog/is-betterhelp-a-scam-unpacking-the-truth.

Forstmann, M., Kettner, H. S., Sagioglou, C., Irvine, A., Gandy, S., Carhart-Harris, R. L., & Luke, D. (2023). Among psychedelic-experienced users, only past use of psilocybin reliably predicts nature relatedness. *Journal of Psychopharmacology*, *37*(1), 93–106. DOI: 10.1177/02698811221146356.

Freidel, N., Kreuder, L., Rabinovitch, B. S., Chen, F. Y., Huang, R. S. T., & Lewis, E. C. (2024). Psychedelics, epilepsy, and seizures: A review. *Frontiers in Pharmacology*, *14*. DOI: 10.3389/fphar.2023.1326815.

Friedwoman, L., Dean, H., Fine, C., Hall, W., Dennis, T. P., Lancelotta, R., Dreisbach, S., Berjot, C., Putnam, N., & Armeni, K. (2025). *Psychedelic safety flags.* URL: docs.google.com/document/d/1lK2Rif24BAmJqqsLfUSkAVCO48IFNrGdysS2nI1EjZA.

Fromson, N. (2023). When chest pain isn't a heart attack. *Health Lab*. URL: michiganmedicine.org/health-lab/when-chest-pain-isnt-heart-attack.

Galef, J. (2021). *The scout mindset*. Penguin.

Girl, B. (2004). *Combined with psych meds: An experience with ecstasy (MDMA), divalproex, venlaxafine, & risperidone (exp27915)*. Erowid. URL: erowid.org/exp/27915.

Godes, M., Lucas, J., & Vermetten, E. (2023). Perceived key change phenomena of MDMA-assisted psychotherapy for the treatment of severe PTSD: An interpretative phenomenological analysis of clinical integration sessions. *Frontiers in Psychiatry*, *14*. DOI: 10.3389/fpsyt.2023.957824.

Goldberg, S. B., Rousmaniere, T., Miller, S. D., Whipple, J., Nielsen, S. L., Hoyt, W. T., & Wampold, B. E. (2016). Do psychotherapists improve with time and experience? a longitudinal analysis of outcomes in a clinical setting. *Journal of Counseling Psychology*, *63*(1), 1–11. DOI: 10.1037/cou0000131.

Goodman, D. A. (2003). *MDMA in a severely disturbed man with psychosis, administered by his brother*. URL: maps.org/research-archive/mdma/edsstory.html.

Gottman, J. M. (2011). *The science of trust: Emotional attunement for couples*. WW Norton & Company.

Goyal, M., Singh, S., Sibinga, E. M. S., Gould, N. F., Rowland-Seymour, A., Sharma, R., Berger, Z., Sleicher, D., Maron, D. D., Shihab, H. M., Ranasinghe, P. D., Linn, S., Saha, S., Bass, E. B., & Haythornthwaite, J. A. (2014). Meditation programs for psychological stress and well-being: A systematic review and meta-analysis. *JAMA Internal Medicine*, *174*(3), 357–368. DOI: 10.1001/jamainternmed.2013.13018.

Grajek, M., Krupa-Kotara, K., Białek-Dratwa, A., Sobczyk, K., Grot, M., Kowalski, O., & Staśkiewicz, W. (2022). Nutrition and mental health: A review of current knowledge about the impact of diet on mental health. *Frontiers in Nutrition*, *Volume 9 - 2022*. DOI: 10.3389/fnut.2022.943998.

Greater Good Science Center. (2025). *Walking meditation.* URL: ggia.berkeley.edu/practice/walking_meditation.

Greenspace. (2023). *Brief revised working alliance inventory.* URL: greenspacehealth.com/en-us/br-wai.

Grenny, J., Patterson, K., McMillan, R., Switzler, A., & Gregory, E. (2022). *Crucial conversations: Tools for talking when stakes are high.* McGraw Hill.

Gumpper, R. H., & Nichols, D. E. (2024). Chemistry/structural biology of psychedelic drugs and their receptor(s). *British Journal of Pharmacology.* DOI: 10.1111/bph.17361.

Gutheil, T. G. (1991). Patients involved in sexual misconduct with therapists: Is a victim profile possible? *Psychiatric Annals, 21*(11), 661–667. DOI: 10.3928/0048-5713-19911101-08.

Hall, W. (2021). Ending the silence around psychedelic therapy abuse. *Mad in America.* URL: madinamerica.com/2021/09/ending-silence-psychedelic-therapy-abuse/.

Halpern, J. H., Lerner, A. G., & Passie, T. (2018). A review of hallucinogen persisting perception disorder (HPPD) and an exploratory study of subjects claiming symptoms of HPPD. In A. L. Halberstadt, F. X. Vollenweider, & D. E. Nichols (Eds.), *Behavioral neurobiology of psychedelic drugs* (pp. 333–360). Springer Berlin Heidelberg. DOI: 10.1007/7854_2016_457.

Halpern, J. H., Sherwood, A. R., Hudson, J. I., Gruber, S., Kozin, D., & Pope Jr, H. G. (2011). Residual neurocognitive features of long-term ecstasy users with minimal exposure to other drugs. *Addiction, 106*(4), 777–786. DOI: 10.1111/j.1360-0443.2010.03252.x.

Hayes, A. M., & Andrews, L. A. (2020). A complex systems approach to the study of change in psychotherapy. *BMC Medicine, 18*(197). DOI: 10.1186/s12916-020-01662-2.

Hew-Butler, T., Ayus, J. C., Kipps, C., Maughan, R. J., Mettler, S., Meeuwisse, W. H., Page, A. J., Reid, S. A., Rehrer, N. J., Roberts, W. O., Roger, I. R., Rosner, M. H., Siegel, A. J., Speedy, D. B., Stuempfle, K. J., Verbalis, J. G., Weschler, L. B., & Wharam,

P. (2008). Statement of the second international exercise-associated hyponatremia consensus development conference, New Zealand, 2007. *Clinical Journal of Sport Medicine*, *18*(2), 111–121. DOI: 10.1097/JSM.0b013e318168ff31.

Hills, J. (2023). *Phenomenology of MDMA solo sessions* [Doctoral dissertation, Antioch University]. URL: etd.ohiolink.edu/acprod/odb_etd/ws/send_file/send?accession=antioch1690227460998206.

Hoel, E. (2024). Neuroscience is pre-paradigmatic. consciousness is why. *The Intrinsic Perspective*. DOI: 10.32388/substack.2mwu0v.

Hogeveen, J., & Grafman, J. (2021). Chapter 3 - alexithymia. In K. M. Heilman & S. E. Nadeau (Eds.), *Disorders of emotion in neurologic disease* (pp. 47–62, Vol. 183). Elsevier. DOI: 10.1016/B978-0-12-822290-4.00004-9.

Højlund, M., Kafali, H. Y., Kırmızı, B., Fusar-Poli, P., Correll, C. U., Cortese, S., Sabé, M., Fiedorowicz, J., Saraf, G., Zein, J., Berk, M., Husain, M. I., Rosenblat, J. D., Rubaiyat, R., Corace, K., Wong, S., Hatcher, S., Kaluzienski, M., Yatham, L. N., … Solmi, M. (2025). Efficacy, all-cause discontinuation, and safety of serotonergic psychedelics and MDMA to treat mental disorders: A living systematic review with meta-analysis. *European Neuropsychopharmacology*, *101*, 41–55. DOI: 10.1016/j.euroneuro.2025.09.011.

Holze, F., Vizeli, P., Müller, F., Ley, L., Duerig, R., Varghese, N., Eckert, A., Borgwardt, S., & Liechti, M. E. (2020). Distinct acute effects of LSD, MDMA, and D-amphetamine in healthy subjects. *Neuropsychopharmacology*, *45*(3), 462–471. DOI: 10.1038/s41386-019-0569-3.

Hook, J., & Devereux, D. (2018). Boundary violations in therapy: The patient's experience of harm. *BJPsych Advances*, *24*(6), 366–373. DOI: 10.1192/bja.2018.26.

Horvath, A. O., Del Re, A., Flückiger, C., & Symonds, D. (2011). Alliance in individual psychotherapy. *Psychotherapy*, *48*(1), 9. DOI: 10.1037/a0022186.

Humphry osmond. (2004). *British Medical Journal*, *328*, 713. DOI: 10.1136/bmj.328.7441.713.

Huneke, N. T. M., Fusetto Veronesi, G., Garner, M., Baldwin, D. S., & Cortese, S. (2025). Expectancy effects, failure of blinding integrity, and placebo response in trials of treatments for psychiatric disorders: A narrative review. *JAMA Psychiatry*. DOI: 10.1001/jamapsychiatry.2025.0085.

Hunt, G. P., & Evans, K. (2008a). "The Great Unmentionable": Exploring the pleasures and benefits of ecstacy from the perspectives of drug users. *Drugs: Education, Prevention and Policy*, *15*(4), 329–349. DOI: 10.1080/09687630701726841.

Hunt, G. P., & Evans, K. (2008b). "The Great Unmentionable": Exploring the pleasures and benefits of ecstasy from the perspectives of drug users. *Drugs: Education, Prevention and Policy*, *15*(4), 329–349. DOI: 10.1080/09687630701726841.

I<3Hallucinogens. (2013). URL: bluelight.org/community/threads/mdma-and-bipolar-2-mood-disorder-nos.675952/#post-11552378.

Ingram, D. (2018). *Mastering the core teachings of the Buddha: An unusually hardcore dharma book-revised and expanded edition*. Red Wheel/Weiser.

Ingram, D. (2024). URL: ecstaticintegration.org/p/can-mdma-lead-to-extended-difficulties/comment/56645807.

Integrative Mental Health University. (2025). *Integrative mental health university*. URL: imhu.org.

Jacobs, A. (2024). Three studies of MDMA treatment retracted by scientific journal. *The New York Times*. URL: nytimes.com/2024/08/12/health/mdma-ptsd-retractions.html.

Jacobs, A., & Nuwer, R. (2025). How a leftist activist group helped torpedo a psychedelic therapy. *The New York Times*. URL: nytimes.com/2025/02/04/health/fda-mdma-psychedelic-therapy-psymposia.html.

John Hopkins Medicine. (2024). *Personalized psychiatry*. URL: hopkinsmedicine.org/personalized-care/personalized-psychiatry.

Jomova, K., Alomar, S. Y., Alwasel, S. H., Nepovimova, E., Kuca, K., & Valko, M. (2024). Several lines of antioxidant defense against oxidative stress: Antioxidant enzymes, nanomaterials with multiple enzyme-mimicking activities, and low-molecular-weight antioxidants. *Archives of Toxicology*, *98*(5), 1323–1367. DOI: 10.1007/s00204-024-03696-4.

Jonas, E., & Kording, K. P. (2017). Could a neuroscientist understand a microprocessor? *PLOS Computational Biology*, *13*(1), e1005268. DOI: 10.1371/journal.pcbi.1005268.

Jones, A., Warner-Schmidt, J., Kwak, H., Stogniew, M., Mandell, B., Ching, T. H., Stein, M. B., & Kelmendi, B. (2026). Efficacy and safety of the neuroplastogen TSND-201 for the treatment of PTSD: A randomized clinical trial. *JAMA Psychiatry*. DOI: 10.1001/jamapsychiatry.2025.4625.

Joseph, J. (2022). *The role of coaching in mental health*. URL: healthygamer.gg/blog/the-role-of-coaching-in-mental-health.

Kabat-Zinn, J. (2023). *This loving-kindness meditation is a radical act of love*. URL: mindful.org/this-loving-kindness-meditation-is-a-radical-act-of-love.

KankaanpÄÄ, A., Meririnne, E., Lillsunde, P., & SeppÄlÄ, T. (1998). The acute effects of amphetamine derivatives on extracellular serotonin and dopamine levels in rat nucleus accumbens. *Pharmacology Biochemistry and Behavior*, *59*(4), 1003–1009. DOI: doi.org/10.1016/S0091-3057(97)00527-3.

Kaspian, P. (2024). *MDMA solo*. URL: castaliafoundation.com/kali/MDMA-solo.pdf.

Kennedy, B. (2022). *Good inside: A guide to becoming the parent you want to be*. HarperCollins.

Klaas, B. (2025). The crisis of zombie social science. *The Garden of Forking Paths*. URL: forkingpaths.co/p/the-crisis-of-zombie-social-science.

Klein, E. (2020). *Why we're polarized*. Simon; Schuster.

Klein, E. (2021). A top mental health expert on where America went wrong. *The Ezra Klein Show*. URL: podcasts.apple.com/us/

podcast/a-top-mental-health-expert-on-where-america-went-wrong/id1548604447.

Kondo, M. (2026). *What is the KonMari method?* URL: konmari.com/about-the-konmari-method.

Kornfield, J. (1993). *A path with heart: A guide through the perils and promises of spiritual life*. Bantam.

Kotov, R., Krueger, R. F., Watson, D., Achenbach, T. M., Althoff, R. R., Bagby, R. M., Brown, T. A., Carpenter, W. T., Caspi, A., Clark, L. A., Eaton, N. R., Forbes, M. K., Forbush, K. T., Goldberg, D., Hasin, D., Hyman, S. E., Ivanova, M. Y., Lynam, D. R., Markon, K., … Zimmerman, M. (2017). The hierarchical taxonomy of psychopathology (hitop): A dimensional alternative to traditional nosologies. *Journal of Abnormal Psychology*, *126*(4), 454. DOI: 10.1037/abn0000258.

Kozlowska, K., Walker, P., McLean, L., & Carrive, P. (2015). Fear and the defense cascade: Clinical implications and management. *Harvard Review of Psychiatry*, *23*(4), 263. DOI: 10.1097/hrp.0000000000000065.

Kranenburg, R. F., Ramaker, H.-J., Weesepoel, Y., Arisz, P. W., Keizers, P. H., van Esch, A., Zieltjens - van Uxem, C., van den Berg, J. D., Hulshof, J. W., Bakels, S., Rijs, A. M., & van Asten, A. C. (2023). The influence of water of crystallization in NIR-based MDMA·HCl detection. *Forensic Chemistry*, *32*, 100464. DOI: 10.1016/j.forc.2022.100464.

Kraus, E., Suter, S., Proescholdt, M., Müller, F., Liechti, M. E., Heim, M., Lang, U., & Vogel, M. (2025). Case report: Well-tolerated MDMA-assisted therapy in a 32-year old female patient with advanced alcohol-induced liver cirrhosis. *Psychiatry Research Case Reports*, *4*(1), 100252. DOI: 10.1016/j.psycr.2025.100252.

Lambert, M. J., Whipple, J. L., Smart, D. W., Vermeersch, D. A., Nielsen, S. L., & Hawkins, E. J. (2001). The effects of providing therapists with feedback on patient progress during psychotherapy: Are outcomes enhanced? *Psychotherapy Research*, *11*(1), 49–68. DOI: 10.1080/713663852.

Lane, R. D., Ryan, L., Nadel, L., & Greenberg, L. (2015). Memory reconsolidation, emotional arousal, and the process of change in psychotherapy: New insights from brain science. *Behavioral and Brain Sciences*, *38*, e1. DOI: 10.1017/s0140525x14000041.

Lanius, R. A., Boyd, J. E., McKinnon, M. C., Nicholson, A. A., Frewen, P., Vermetten, E., Jetly, R., & Spiegel, D. (2018). A review of the neurobiological basis of trauma-related dissociation and its relation to cannabinoid-and opioid-mediated stress response: A transdiagnostic, translational approach. *Current Psychiatry Reports*, *20*(118). DOI: 10.1007/s11920-018-0983-y.

Lanius, U. F. (2014). Neurobiology and treatment of traumatic dissociation: Towards an embodied self. In S. L. P. Ulrich F. Lanius & F. M. Corrigan (Eds.). Springer Publishing Company.

Lara, D. R. (2010). Caffeine, mental health, and psychiatric disorders. *Journal of Alzheimer's Disease*, *20*(s1), S239–S248. DOI: 10.3233/JAD-2010-1378.

Lauersen, J. B., Bertelsen, D. M., & Andersen, L. B. (2014). The effectiveness of exercise interventions to prevent sports injuries: A systematic review and meta-analysis of randomised controlled trials. *British Journal of Sports Medicine*, *48*(11), 871–877. DOI: 10.1136/bjsports-2013-092538.

Leucht, S., Hierl, S., Kissling, W., Dold, M., & Davis, J. M. (2012). Putting the efficacy of psychiatric and general medicine medication into perspective: Review of meta-analyses. *British Journal of Psychiatry*, *200*(2), 97–106. DOI: 10.1192/bjp.bp.111.096594.

Lieberman, D. (2021). *Exercised: Why something we never evolved to do is healthy and rewarding*. Vintage.

Liechti, M. (2025). *Swiss limited use program*. URL: youtu.be/VIY3SGO2i-A?t=1565.

Liechti, M., & Schmid, Y. (2026). *Interactions with psychedelics and MDMA*. URL: saept.ch/wp-content/uploads/2026/03/interactions-with-Psychedelics-and-MDMA-V8-23.03.2026.pdf.

Liechti, M. E. (2014). Effects of mdma on body temperature in humans. *Temperature*, *1*(3), 192–200. DOI: 10.4161/23328940.2014.955433.

Liechti, M. E., Gamma, A., & Vollenweider, F. X. (2001). Gender differences in the subjective effects of MDMA. *Psychopharmacology*, *154*, 161–168. DOI: 10.1007/s002130000648.

Liknaitzky, P. (2024). *Mental health professionals' views and experiences.* URL: community.open-foundation.org/c/icpr-2024-recordings/sections/272721/lessons/1015845.

Liknaitzky, P. (2026). *Mdma-assisted therapy for ptsd in veterans and first responders: Safety and efficacy in randomized, waitlist-controlled clinical trial.* URL: community.open-foundation.org/c/icpr-2026/.

Lindahl, J. R., Cooper, D. J., Fisher, N. E., Kirmayer, L. J., & Britton, W. B. (2020). Progress or pathology? differential diagnosis and intervention criteria for meditation-related challenges: Perspectives from Buddhist meditation teachers and practitioners. *Frontiers in Psychology, Volume 11 - 2020.* DOI: 10.3389/fpsyg.2020.01905.

Lindahl, J. R., Fisher, N. E., Cooper, D. J., Rosen, R. K., & Britton, W. B. (2017). The varieties of contemplative experience: A mixed-methods study of meditation-related challenges in western Buddhists. *PLOS One*, *12*(5), e0176239. DOI: 10.1371/journal.pone.0176239.

Litjens, R., Brunt, T., Alderliefste, G.-J., & Westerink, R. (2014). Hallucinogen persisting perception disorder and the serotonergic system: A comprehensive review including new MDMA-related clinical cases. *European Neuropsychopharmacology*, *24*(8), 1309–1323. DOI: 10.1016/j.euroneuro.2014.05.008.

Low, S. (2023). *R/bodyweightfitness recommended routine.* URL: reddit.com/r/bodyweightfitness/wiki/kb/recommended_routine.

Lu, O. D., White, K., Raymond, K., Liu, C., Klein, A. S., Green, N., Vaillancourt, S., Gallagher, A., Shindy, L., Li, A., Li, R., Zou, M., Wallquist, K., Casey, A. B., Cameron, L. P., Pomrenze,

M. B., Sohal, V., Kheirbek, M. A., Gomez, A. M., … Malenka, R. C. (2025). *A multi-institutional investigation of psilocybin's effects on mouse behavior*. DOI: 10.1101/2025.04.08.647810.

Luck, A. (2023). *Polyvagal theory: A critical appraisal*. URL: alyssaluck.com/polyvagal-theory-a-critical-appraisal/.

Lutkajtis, A. (2021). *The dark side of dharma: Meditation, madness and other maladies on the contemplative path*. Aeon Books.

Macdonald, J., & Mellor-Clark, J. (2015). Correcting psychotherapists' blindsidedness: Formal feedback as a means of overcoming the natural limitations of therapists. *Clinical Psychology & Psychotherapy*, *22*(3), 249–257. DOI: 10.1002/cpp.1887.

Magyari, T. (2016). Teaching individuals with traumatic stress. In D. McCown, D. Reibel, & M. S. Micozzi (Eds.), *Resources for teaching mindfulness: An international handbook* (pp. 339–358). Springer. DOI: 10.1007/978-3-319-30100-6_18.

Makunts, T., Dahill, D., Jerome, L., de Boer, A., & Abagyan, R. (2023). Concomitant medications associated with ischemic, hypertensive, and arrhythmic events in MDMA users in FDA adverse event reporting system. *Frontiers in Psychiatry*, *14*. DOI: 10.3389/fpsyt.2023.1149766.

Malcolm, B. (2023). *Ayahuasca and drug interaction: The good, the bad, and the soul*. URL: pharmacy.uconn.edu/wp-content/uploads/sites/2740/2022/01/HO-2-slides-per-page-Ayahuosca-1.pdf.

Malcolm, B. (2024). Healing states or serotonin toxicity? [Malcolm updated the temperature threshold to 38.5 °C (101.3 °F) in a non-public document.]. URL: spiritpharmacist.com/blog/serotoninhealing.

Malcolm, B. (2025a). *Antidepressant and psychedelic drug interaction and taper planning guide*. URL: spiritpharmacist.com/opt-in-8bacfbd2-97c7-47e9-a1e9-372d74f12d02.

Malcolm, B. (2025b). *Spirit pharmacist member resource and support program*. URL: spiritpharmacist.com.

Malcolm, B., & Thomas, K. (2022). Serotonin toxicity of serotonergic psychedelics. *Psychopharmacology*, *239*(6), 1881–1891. DOI: 10.1007/s00213-021-05876-x.

Martin, D. (2020). The ones who dream of tigers. In J. Evans & T. Read (Eds.), *Breaking open*. Aeon Books.

Mattick, R., Breen, C., Kimber, J., & Davoli, M. (2014). Buprenorphine maintenance versus placebo or methadone maintenance for opioid dependence. *Cochrane Database of Systematic Reviews*, (2). DOI: 10.1002/14651858.CD002207.pub4.

McElrath, K. (2005). MDMA and sexual behavior: Ecstasy users' perceptions about sexuality and sexual risk. *Substance Use & Misuse*, *40*(9-10), 1461–1477. DOI: 10.1081/JA-200066814.

McGuire, P., & Fahy, T. (1991). Chronic paranoid psychosis after misuse of MDMA ("Ecstasy"). *BMJ: British Medical Journal*, *302*(6778), 697. DOI: 10.1136/bmj.302.6778.697.

McRaney, D. (2022). *How minds change: The surprising science of belief, opinion, and persuasion*. Portfolio.

Meikle, S., Carter, O., & Bedi, G. (2024). Psychedelic-assisted psychotherapy, patient vulnerability and abuses of power. *Australian & New Zealand Journal of Psychiatry*, *58*(2), 104–106. DOI: 10.1177/00048674231200164.

Mickelson, K. D., Kessler, R. C., & Shaver, P. R. (1997). Adult attachment in a nationally representative sample. *Journal of Personality and Social Psychology*, *73*(5), 1092. DOI: 10.1037/0022-3514.73.5.1092.

Miličević, M. P., Belić, S., Vraneš, M., Tot, A., Stevanović, N. R., Rakić, D., & Gadžurić, S. (2020). Volumetric properties, viscosity and taste behavior of MDMA-HCl in aqueous binary and (water+ D-lactose) ternary mixtures at different temperatures. *The Journal of Chemical Thermodynamics*, *142*, 106027. DOI: 10.1016/j.jct.2019.106027.

Miller, J. (2024). When therapists lose their licenses, some turn to the unregulated life coaching industry instead. *Propublica*.

URL: propublica.org/article/utah-therapists-life-coaches-regulation.

Mingus, M. (2016). *Pods and pod mapping worksheet.* URL: batjc.wordpress.com/resources/pods-and-pod-mapping-worksheet.

Mingus, M. (2019). The four parts of accountability & how to give a genuine apology. *Leaving Evidence.* URL: leavingevidence.wordpress.com/2019/12/18/how-to-give-a-good-apology-part-1-the-four-parts-of-accountability.

Mitchell, J. M., Bogenschutz, M., Lilienstein, A., Harrison, C., Kleiman, S., Parker-Guilbert, K., Ot'alora G., M., Garas, W., Paleos, C., Gorman, I., Nicholas, C., Mithoefer, M., Carlin, S., Poulter, B., Mithoefer, A., Quevedo, S., Wells, G., Klaire, S. S., van der Kolk, B., ... Doblin, R. (2021). MDMA-assisted therapy for severe PTSD: A randomized, double-blind, placebo-controlled phase 3 study. *Nature Medicine*, *27*(6), 1025–1033. DOI: 10.1038/s41591-021-01336-3.

Mitchell, J. M., Ot'alora G., M., van der Kolk, B., Shannon, S., Bogenschutz, M., Gelfand, Y., Paleos, C., Nicholas, C. R., Quevedo, S., Balliett, B., Hamilton, S., Mithoefer, M., Kleiman, S., Parker-Guilbert, K., Tzarfaty, K., Harrison, C., de Boer, A., Doblin, R., Yazar-Klosinski, B., ... Group, M. S. C. (2023). MDMA-assisted therapy for moderate to severe PTSD: A randomized, placebo-controlled phase 3 trial. *Nature Medicine.* DOI: 10.1038/s41591-023-02565-4.

Mithoefer, M., & Mithoefer, A. (2021). MDMA. In C. S. Grob & J. Grigsby (Eds.), *Handbook of medical hallucinogens* (pp. 233–263). The Guilford Press.

Mithoefer, M. (2017). *A manual for MDMA-assisted psychotherapy in the treatment of posttraumatic stress disorder.* Multidisciplinary Association for Psychedelic Studies. URL: maps.org/2014/01/27/a-manual-for-mdma-assisted-therapy-in-the-treatment-of-ptsd.

Mithoefer, M. C., Wagner, M. T., Mithoefer, A. T., Jerome, L., & Doblin, R. (2011). The safety and efficacy of ±3,4-methylenedioxymethamphetamine-assisted psychotherapy in subjects with chronic, treatment-resistant posttraumatic stress disorder: The first randomized controlled pilot study. *Journal of Psychopharmacology*, *25*(4), 439–452. DOI: 10.1177/0269881110378371.

Moncrieff, J. (2006). Does antipsychotic withdrawal provoke psychosis? review of the literature on rapid onset psychosis (supersensitivity psychosis) and withdrawal-related relapse. *Acta Psychiatrica Scandinavica*, *114*(1), 3–13. DOI: 10.1111/j.1600-0447.2006.00787.x.

Moon, E., Kim, K., Partonen, T., & Linnaranta, O. (2022). Role of melatonin in the management of sleep and circadian disorders in the context of psychiatric illness. *Current Psychiatry Reports*, *24*(11), 623–634. DOI: 10.1007/s11920-022-01369-6.

Moonzwe, L. S., Schensul, J. J., & Kostick, K. M. (2011). The role of MDMA (ecstasy) in coping with negative life situations among urban young adults. *Journal of Psychoactive Drugs*, *43*(3), 199–210. DOI: 10.1080/02791072.2011.605671.

Multidisciplinary Association for Psychedelic Studies. (2022). *MDMA investigator's brochure*. Multidisciplinary Association for Psychedelic Studies. URL: maps.org/wp-content/uploads/2022/03/MDMA-IB-14th-Edition-FINAL-18MAR2022.pdf.

Mustafa, R., McQueen, B., Nikiti, D., Nhan, E., Zemplenyi, A., DiStefano, M., Kayali, Y., Richardson, M., & Rind, D. (2024). *MDMA-assisted psychotherapy for post-traumatic stress disorder: Effectiveness and value; final evidence report*. Institute for Clinical and Economic Review. URL: icer.org/assessment/ptsd-2024/#overview.

Nair, J. B., Hakes, L., Yazar-Klosinski, B., & Paisner, K. (2022). Fully validated, multi-kilogram cgmp synthesis of mdma. *ACS Omega*, *7*(1), 900–907. DOI: 10.1021/acsomega.1c05520.

Nardou, R., Lewis, E. M., Rothhaas, R., Xu, R., Yang, A., Boyden, E., & Dölen, G. (2019). Oxytocin-dependent reopening of a social reward learning critical period with MDMA. *Nature*, *569*(7754), 116–120. DOI: 10.1038/s41586-019-1075-9.

Nicholas, C. R., Wang, J. B., Coker, A., Mitchell, J. M., Klaire, S. S., Yazar-Klosinski, B., Emerson, A., Brown, R. T., & Doblin, R. (2022). The effects of MDMA-assisted therapy on alcohol and substance use in a phase 3 trial for treatment of severe PTSD. *Drug and Alcohol Dependence*, *233*, 109356. DOI: 10.1016/j.drugalcdep.2022.109356.

Nickles, D., & Ross, L. K. (Eds.). (2021). Cover story [Power Trip]. URL: thecut.com/2021/11/cover-story-podcast-goes-into-world-of-psychedelic-therapy.html.

Nostrand, E. V. (2022). *Epistemic legibility*. URL: lesswrong.com/posts/jbE85wCkRr9z7tqmD/epistemic-legibility.

Nutt, D., King, L., & Phillips, L. (2010). Drug harms in the UK: A multi-criterion decision analysis. *Lancet*, *376*, 1558–1565. DOI: 10.1016/S0140-6736(10)61462-6.

Nutt, D. J., Lingford-Hughes, A., Erritzoe, D., & Stokes, P. R. (2015). The dopamine theory of addiction: 40 years of highs and lows. *Nature Reviews Neuroscience*, *16*(5), 305–312.

Oanes, C. J., Anderssen, N., Karlsson, B., & Borg, M. (2015). How do therapists respond to client feedback? a critical review of the research literature. *Scandinavian Psychologist*, *2*. DOI: 10.15714/scandpsychol.2.e17.

Ogden, P., Pain, C., & Fisher, J. (2006). A sensorimotor approach to the treatment of trauma and dissociation. *Psychiatric Clinics*, *29*(1), 263–279.

Olthof, M., Hasselman, F., Strunk, G., Aas, B., Schiepek, G., & Lichtwarck-Aschoff, A. (2020). Destabilization in self-ratings of the psychotherapeutic process is associated with better treatment outcome in patients with mood disorders. *Psychotherapy Research*, *30*(4), 520–531. DOI: 10.1080/10503307.2019.1633484.

O'Mathúna, B., Farré, M., Rostami-Hodjegan, A., Yang, J., Cuyàs, E., Torrens, M., Pardo, R., Abanades, S., Maluf, S., Tucker, G. T., & de la Torre, R. (2008). The consequences of 3,4-methylenedioxymethamphetami induced CYP2D6 inhibition in humans. *Journal of Clinical Psy-*

chopharmacology, *28*(5), 523–529. DOI: 10.1097/JCP.0b013e318184ff6e.

Oprel, D. A. C., Hoeboer, C. M., Schoorl, M., de Kleine, R. A., Cloitre, M., Wigard, I. G., van Minnen, A., & van der Does, W. (2021). Effect of prolonged exposure, intensified prolonged exposure and STAIR+prolonged exposure in patients with PTSD related to childhood abuse: A randomized controlled trial. *European Journal of Psychotraumatology*, *12*(1), 1851511. DOI: 10.1080/20008198.2020.1851511.

Otgaar, H., Mangiulli, I., Li, C., Jelicic, M., & Muris, P. (2025). The recovery and retraction of memories of abuse: A scoping review. *Frontiers in Psychology*, *16*. DOI: 10.3389/fpsyg.2025.1498258.

Padawer-Curry, J. A., Krentzman, O. J., Kuo, C.-C., Wang, X., Bice, A. R., Nicol, G. E., Snyder, A. Z., Siegel, J. S., McCall, J. G., & Bauer, A. Q. (2025). Psychedelic $5 - HT_{2A}$ receptor agonism alters neurovascular coupling and differentially affects neuronal and hemodynamic measures of brain function. *Nature Neuroscience*, 1–14. DOI: 10.1038/s41593-025-02069-z.

Parrott, A. (2005). Chronic tolerance to recreational MDMA (3,4-methylenedioxymethamphetamine) or ecstasy. *Journal of Psychopharmacology*, *19*(1), 71–83. DOI: 10.1177/0269881105048900.

Passie, T. (2023). *The history of MDMA*. Oxford University Press.

Patel, A., Moreland, T., Haq, F., Siddiqui, F., Mikul, M., Qadir, H., & Raza, S. (2011). Persistent psychosis after a single ingestion of “Ecstasy”(MDMA). *The Primary Care Companion for CNS Disorders*, *13*(6), 27095. DOI: 10.4088/PCC.11l01200.

Pinsof, D. (2024). The truth about self-deception. *Optimally Irrational*. URL: optimallyirrational.com/p/the-truth-about-self-deception.

Poulter, B., & Ot’Alora, M. (2023). *The art of simmering in MDMA-assisted therapy*. URL: virtualtrip.maps.org/video/the-art-of-simmering-in-mdma-assisted-therapy/.

Priebe, H. (Ed.). (2025). Heidi priebe. URL: youtube.com/@heidipriebe1.
PsyAware. (2024). *Need support?* URL: psyaware.org/need-support.
Psychedelic Alpha. (2025). Unpacking FDA's MDMA rejection letter and the road ahead for lykos. *Psychedelic Alpha.* URL: psychedelicalpha.com/news/unpacking-fdas-mdma-rejection-letter-and-the-road-ahead-for-lykos.
Psychedelic Passage. (2026). *Mushrooms macrodose program.* URL: psychedelicpassage.com/macrodosing/.
Psychedelic Somatic Institute. (2024). *PSIP therapist directory.* URL: psychedelicsomatic.org/find-a-psychedelic-therapist.
Psychedelics in Recovery. (2025). *Psychedelics in recovery.* URL: psychedelicsinrecovery.org.
Quinn, D., Martinez, R., Cuentas, C., Hernandez-Wolfe, P., & Fine, C. (2021). *Addressing abuse and repair: An open letter to the psychedelic community.* URL: psychedeliccommunity.medium.com/addressing-abuse-and-repair-an-open-letter-to-the-psychedelic-community-ccf677dd92b9.
Razvi, S. (2024). *Why MDMA & other psychedelic therapy may not work for you (part 1).* URL: psychedelicsomatic.org/post/why-mdma-psychedelic-therapy-may-not-work-for-you.
Razvi, S., & Elfrink, S. (2020). The PSIP model. an introduction to a novel method of therapy: Psychedelic somatic interactional psychotherapy. *Journal of Psychedelic Psychiatry*, *2*(3), 1–24. URL: journalofpsychedelicpsychiatry.org/_files/ugd/e07c59_d4d1db6fc0174f27bef58a6124aba50e.pdf.
Rigg, K. K., & Sharp, A. (2018). Deaths related to MDMA (ecstasy/molly): Prevalence, root causes, and harm reduction interventions. *Journal of Substance Use*, *23*(4), 345–352. DOI: 10.1080/14659891.2018.1436607.
Ritchie, H. (2024). *Not the end of the world: How we can be the first generation to build a sustainable planet.* Hachette UK.
/r/mdmatherapy. (2025a). *Can meditation ever be safe again after psychosis?* URL: reddit.com/r/mdmatherapy/comments/

1mw061y/can_meditation_ever_be_safe_again_after_psychosis/.

/r/mdmatherapy. (2025b). *Has anyone else had a psychotic break after using MDMA?* URL: reddit.com/r/mdmatherapy/comments/1n1pqw9/has_anyone_else_had_a_psychotic_break_after_using/.

Robinson, O. C., Evans, J., Luke, D., McAlpine, R., Sahely, A., Fisher, A., Sundeman, S., Ketzitzidou Argyri, E., Murphy-Beiner, A., Michelle, K., & Prideaux, E. (2024). Coming back together: A qualitative survey study of coping and support strategies used by people to cope with extended difficulties after the use of psychedelic drugs. *Frontiers in Psychology*, *15*. DOI: 10.3389/fpsyg.2024.1369715.

Robledo, P., Balerio, G., Berrendero, F., & Maldonado, R. (2004). Study of the behavioural responses related to the potential addictive properties of MDMA in mice. *Naunyn-Schmiedeberg's Archives of Pharmacology*, *369*, 338–349. DOI: 10.1007/s00210-003-0862-9.

Rodriguez, L., & Harris, K. (2024). Jonathan Birch on the edge cases of sentience and why they matter. *80,000 Hours Podcast*, (196). URL: 80000hours.org/podcast/episodes/jonathan-birch-edge-sentience-uncertainty/.

Ronca, K. E. (2018). *The impact of complex post-traumatic stress disorder and structural violence on children in impoverished urban communities* [Doctoral dissertation, Temple University]. URL: proquest.com/docview/2046909618.

Rosenbaum, S., Tiedemann, A., Sherrington, C., Curtis, J., & Ward, P. B. (2014). Physical activity interventions for people with mental illness: A systematic review and meta-analysis. *The Journal of Clinical Psychiatry*, *75*(9), 14465. DOI: 10.4088/JCP.13r08765.

Rosenberg, M. (2015). *Nonviolent communication: A language of life*. Puddledancer.

/r/ReagentTesting. (2025a). *Test kit suppliers*. URL: reddit.com/r/ReagentTesting/wiki/test_kit_suppliers.

/r/ReagentTesting. (2025b). *Testing labs.* URL: reddit.com/r/ReagentTesting/wiki/labs.

Ruggeri, A. (2025). Natural doesn't always mean better: How to spot if someone is trying to convince you with an "Appeal to Nature". *British Broadcasting Corporation.* URL: bbc.com/future/article/20250210-the-appeal-to-nature-fallacy-why-natural-doesnt-always-mean-better.

Sabé, M., Sulstarova, A., Glangetas, A., De Pieri, M., Mallet, L., Curtis, L., Richard-Lepouriel, H., Penzenstadler, L., Seragnoli, F., Thorens, G., et al. (2025). Reconsidering evidence for psychedelic-induced psychosis: An overview of reviews, a systematic review, and meta-analysis of human studies. *Molecular psychiatry*, *30*(3), 1223–1255. DOI: 10.1038/s41380-024-02800-5.

Salvadore, G., Quiroz, J. A., Machado-Vieira, R., Henter, I. D., Manji, H. K., & Zarate Jr, C. A. (2010). The neurobiology of the switch process in bipolar disorder: A review. *The Journal of Clinical Psychiatry*, *71*(11), 12633. DOI: 10.4088/JCP.09r05259gre.

Sarparast, A., Thomas, K., Malcolm, B., & Stauffer, C. S. (2022). Drug-drug interactions between psychiatric medications and MDMA or psilocybin: A systematic review. *Psychopharmacology*, *239*(6), 1945–1976. DOI: 10.1007/s00213-022-06083-y.

Sayadaw, M. (2016). *Manual of insight.* Wisdom Publications.

Sayre-McCord, G. (2023). Metaethics. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Spring 2023). Metaphysics Research Lab, Stanford University. URL: plato.stanford.edu/archives/spr2023/entries/metaethics/.

Schenberg, E. (2024). *Evidence-based medicine is inadequate to develop evidence-based psychedelic therapies.* DOI: 10.31234/osf.io/rzdpm.

Schmid, Y., Vizeli, P., Hysek, C. M., Prestin, K., Zu Schwabedissen, H. E. M., & Liechti, M. E. (2016). Cyp2d6 function moderates the pharmacokinetics and pharmacodynamics of 3,4-methylenedioxymethamphetamine in a controlled study in healthy individuals. *Pharmacogenetics and Genomics*, *26*(8), 397–401. DOI: 10.1097/fpc.0000000000000231.

Scoboria, A., Wade, K. A., Lindsay, D. S., Azad, T., Strange, D., Ost, J., & and, I. E. H. (2017). A mega-analysis of memory reports from eight peer-reviewed false memory implantation studies. *Memory*, *25*(2), 146–163. DOI: 10.1080/09658211.2016.1260747.

Scott, A. J., Webb, T. L., Martyn-St James, M., Rowse, G., & Weich, S. (2021). Improving sleep quality leads to better mental health: A meta-analysis of randomised controlled trials. *Sleep Medicine Reviews*, *60*, 101556. DOI: 10.1016/j.smrv.2021.101556.

Scully, M. (2003). *Dominion: The power of man, the suffering of animals, and the call to mercy*. macmillan.

Sele, P., Hoffart, A., Cloitre, M., Hembree, E., & Øktedalen, T. (2023). Comparing phase-based treatment, prolonged exposure, and skills-training for complex posttraumatic stress disorder: A randomized controlled trial. *Journal of Anxiety Disorders*, *100*, 102786. DOI: 10.1016/j.janxdis.2023.102786.

Shankaran, M., Yamamoto, B. K., & Gudelsky, G. A. (2001). Ascorbic acid prevents 3,4-methylenedioxymethamphetamine (MDMA)-induced hydroxyl radical formation and the behavioral and neurochemical consequences of the depletion of brain 5-HT. *Synapse*, *40*(1), 55–64. DOI: 10.1002/1098-2396(200104)40:1%3C55::aid-syn1026%3E3.0.co;2-o.

SHINE Collective. (2024). *Peer-led survivor support groups*. URL: shinesupport.org/shine-support.

Shokry, I., Desuza, K., Ma, Z., Rao, R., Callanan, S., & Shim, G. (2018). Individuals with hyperthyroidism are more susceptible to having a serious serotonin syndrome following MDMA (ecstasy) administration. *Annals of forensic research and analysis*, *5*, 1052.

Sinback, L. (2024a). *Business values*. URL: libbysinback.com/values.

Sinback, L. (2024b). How to get help. *Making Polyamory Work*. URL: makingpolyamorywork.com/episodes/how-to-get-help.

Singer, P. (2017). Famine, affluence, and morality. In *Applied ethics* (pp. 132–142). Routledge.

Singer, P. (2023). *Animal liberation: The definitive classic renewed.* HarperCollins.

Siskind, S. (2020). *Melatonin.* URL: lorienpsych.com/2020/12/20/melatonin/.

Sitko, K., Bewick, B. M., Owens, D., & Masterson, C. (2020). Meta-analysis and meta-regression of cognitive behavioral therapy for psychosis (CBTp) across time: The effectiveness of CBTp has improved for delusions. *Schizophrenia Bulletin Open*, *1*(1), sgaa023. DOI: 10.1093/schizbullopen/sgaa023.

Slime Mold Time Mold. (2025). The mind in the wheel - prologue: Everybody wants a rock. *Slime Mold Time Mold.* URL: slimemoldtimemold.com/2025/02/06/the-mind-in-the-wheel-prologue-everybody-wants-a-rock/.

Smookler, E. (2023). *Beginner's body scan meditation.* URL: mindful.org/beginners-body-scan-meditation.

Sørensen, A. (2025). *Crossing zero: The art and science of coming off — and staying off — psychiatric drugs.* Anders Sørensen.

Sotala, K. (2019). *Book summary: Unlocking the emotional brain.* URL: lesswrong.com/posts/i9xyZBS3qzA8nFXNQ/book-summary-unlocking-the-emotional-brain.

Spiritual Emergence Network. (2025). *Spiritual emergence network.* URL: spiritualemergence.org.

Stanford Medicine. (2024). *Cognitive behavioral therapy for insomnia.* URL: stanfordhealthcare.org/medical-treatments/c/cognitive-behavioral-therapy-insomnia/procedures.html.

Stocker, K., & Liechti, M. E. (2024). Methylenedioxymethamphetamine is a connectogen with empathogenic, entactogenic, and still further connective properties: It is time to reconcile "the great entactogen—empathogen debate". *Journal of Psychopharmacology*, *38*(8), 685–689. DOI: 10.1177/02698811241265352.

Straumann, I., Avedisian, I., Klaiber, A., Varghese, N., Eckert, A., Rudin, D., Luethi, D., & Liechti, M. E. (2024). Acute effects of *r*-mdma, *s*-mdma, and racemic MDMA in a randomized double-blind cross-over trial in healthy participants. *Neuropsychopharmacology*, *50*, 362–371. DOI: 10.1038/s41386-024-01972-6.

Strom-Gottfried, K. (1999). Professional boundaries: An analysis of violations by social workers. *Families in Society*, *80*(5), 439–449. DOI: 10.1606/1044-3894.1473.

Studerus, E., Gamma, A., & Vollenweider, F. X. (2010). Psychometric evaluation of the altered states of consciousness rating scale (OAV). *PLOS One*, *5*(8), e12412. DOI: 10.1371/journal.pone.0012412.

Studerus, E., Vizeli, P., Harder, S., Ley, L., & Liechti, M. E. (2021). Prediction of MDMA response in healthy humans: A pooled analysis of placebo-controlled studies. *Journal of Psychopharmacology*, *35*(5), 556–565. DOI: 10.1177/0269881121998322.

Sulstarova, A., Scheuerlein, L., Monari, S., Seragnoli, F., Gabriel, T., Preller, K., Böge, K., Sentissi, O., Kaiser, S., Solmi, M., Kirschner, M., & Sabé, M. (2025). Treatment approaches and efficacy in psychedelic-induced psychosis: A systematic review. *Asian Journal of Psychiatry*, 104604. DOI: 10.1016/j.ajp.2025.104604.

Tackett, J. L., Brandes, C. M., King, K. M., & Markon, K. E. (2019). Psychology's replication crisis and clinical psychological science. *Annual Review of Clinical Psychology*, *15*(Volume 15, 2019), 579–604. DOI: 10.1146/annurev-clinpsy-050718-095710.

Tagen, M., Mantuani, D., van Heerden, L., Holstein, A., Klumpers, L. E., & Knowles, R. (2023). The risk of chronic psychedelic and MDMA microdosing for valvular heart disease. *Journal of Psychopharmacology*, *37*(9), 876–890. DOI: 10.1177/02698811231190865.

Thal, S., Engel, L. B., & Bright, S. J. (2022). Presence, trust, and empathy: Preferred characteristics of psychedelic carers. *Journal of Humanistic Psychology*. DOI: 10.1177/00221678221081380.

The Bump Editors. (2023). *What is a birth plan and why is it important?* URL: thebump.com/a/tool-birth-plan#3.

The Challenging Psychedelic Experiences Project. (2024). *Online peer support group*. URL: challengingpsychedelicexperiences.com.

The International Center for Ethnobotanical Education, Research, and Service. (2024). *Assistance with challenging experiences*. URL: iceers.org/support-center-2.

The Psychedelic Experience Clinic. (2024). *Psychoanalytic psychotherapy after psychedelic experiences*. URL: thepsychedelicexperienceclinic.co.uk.

The Psychedelics and Recovered Memories Project. (2024). *The psychedelics and recovered memories project*. URL: psychedelicsandrecoveredmemories.com.

Therapevo. (2020). *The 5 pillars of attachment*. URL: therapevo.com/podcasts/the-5-pillars-of-attachment/.

TheraPsil. (2026). *Understanding costs*. URL: therapsil.ca/understanding-costs/.

Tippett, K. (2020). "Notice the Rage; Notice the Silence": Interview with Resmaa Menakem. *On Being with Krista Tippett*.

Toebes, B., van den Brink, W., Gresnigt, F., de Jonge, M., Kolthoff, E., & Vermetten, E. (2024a). *MDMA state commission reports to Dutch parliament*. URL: youtu.be/Ul-J45YRMcU?t=793.

Toebes, B., van den Brink, W., Gresnigt, F., de Jonge, M., Kolthoff, E., & Vermetten, E. (2024b). *MDMA. beyond the ecstasy*. State Commission on MDMA. URL: government.nl/binaries/government/documenten/reports/2024/05/31/mdma-beyond-ecstasy/MDMA+Beyond+Ecstasy.pdf.

Trauer, J. M., Qian, M. Y., Doyle, J. S., Rajaratnam, S. M., & Cunnington, D. (2015). Cognitive behavioral therapy for chronic insomnia: A systematic review and meta-analysis. *Annals of Internal Medicine*, *163*(3), 191–204. DOI: 10.7326/M14-2841.

Treleaven, D. A. (2018). *Trauma-sensitive mindfulness: Practices for safe and transformative healing*. WW Norton & Company.

Tripsit. (2025). *Volumetric dosing tool*. URL: volume.tripsit.me.

United Nations Office on Drugs and Crime. (2014). *World drug report*. United Nations Office on Drugs and Crime. URL: unodc.org/documents/wdr2014/World_Drug_Report_2014_web.pdf.

Vaiva, G., Boss, V., Bailly, D., Thomas, P., Lestavel, P., & Goudemand, M. (2001). An "Accidental" acute psychosis with ecstacy use. *Journal of Psychoactive Drugs*, *33*(1), 95. DOI: 10.1080/02791072.2001.10400473.

Valtin, H. a. (2002). "drink at least eight glasses of water a day." really? is there scientific evidence for "8 x 8"? *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology*, *283*(5), R993–R1004. DOI: 10.1152/ajpregu.00365.2002.

van Amsterdam, J., Brunt, T. M., Pierce, M., & van den Brink, W. (2021). Hard boiled: Alcohol use as a risk factor for MDMA-induced hyperthermia: A systematic review. *Neurotoxicity Research*, *39*(6), 2120–2133. DOI: 10.1007/s12640-021-00416-z.

van Vliet, N. I., Huntjens, R. J. C., van Dijk, M. K., Bachrach, N., Meewisse, M.-L., & de Jongh, A. (2021). Phase-based treatment versus immediate trauma-focused treatment for post-traumatic stress disorder due to childhood abuse: Randomised clinical trial. *BJPsych Open*, *7*(6), e211. DOI: 10.1192/bjo.2021.1057.

van Winkel, R., Stefanis, N. C., & Myin-Germeys, I. (2008). Psychosocial stress and psychosis. a review of the neurobiological mechanisms and the evidence for gene-stress interaction. *Schizophrenia Bulletin*, *34*(6), 1095–1105. DOI: 10.1093/schbul/sbn101.

Van den Bergh, O., Brosschot, J., Critchley, H., Thayer, J. F., & Ottaviani, C. (2021). Better safe than sorry: A common signature of general vulnerability for psychopathology. *Perspectives on Psychological Science*, *16*(2), 225–246. DOI: 10.1177/1745691620950690.

Van den Bergh, O., Witthöft, M., Petersen, S., & Brown, R. J. (2017). Symptoms and the body: Taking the inferential leap. *Neuro-*

science & Biobehavioral Reviews, *74*, 185–203. DOI: 10.1016/j.neubiorev.2017.01.015.

van der Kolk, B. (2015). *The body keeps the score*. Penguin Books.

van der Kolk, B. A., Wang, J. B., Yehuda, R., Bedrosian, L., Coker, A. R., Harrison, C., Mithoefer, M., Yazar-Klosinki, B., Emerson, A., & Doblin, R. (2024). Effects of MDMA-assisted therapy for PTSD on self-experience. *PLOS One*, *19*(1), e0295926. DOI: 10.1371/journal.pone.0295926.

Vanattou-Saifoudine, N., McNamara, R., & Harkin, A. (2012). Caffeine provokes adverse interactions with 3,4-methylenedioxymethamphetamine (MDMA,"Ecstacy") and related psychostimulants: Mechanisms and mediators. *British Journal of Pharmacology*, *167*(5), 946–959. DOI: 10.1111/j.1476-5381.2012.02065.x.

Vinson, J. S. (1987). Use of complaint procedures in cases of therapist-patient sexual contact. *Professional Psychology: Research and Practice*, *18*(2), 159. DOI: 10.1037/0735-7028.18.2.159.

Vizeli, P., & Liechti, M. E. (2017). Safety pharmacology of acute MDMA administration in healthy subjects. *Journal of Psychopharmacology*, *31*(5), 576–588. DOI: 10.1177/0269881117691569.

Weisberg, J. (2024). The hard problem of consciousness. *The Internet Encyclopedia of Philosophy*. URL: iep.utm.edu/hard-problem-of-conciousness.

Willett, W., Rockström, J., Loken, B., Springmann, M., Lang, T., Vermeulen, S., Garnett, T., Tilman, D., DeClerck, F., Wood, A., Jonell, M., Clark, M., Gordon, L. J., Fanzo, J., Hawkes, C., Zurayk, R., Rivera, J. A., De Vries, W., Majele Sibanda, L., ... Murray, C. J. L. (2019). Food in the anthropocene: The EAT-Lancet commission on healthy diets from sustainable food systems. *Lancet*, *393*(10170), 447–492. DOI: 10.1016/S0140-6736(18)31788-4.

Williams, M. H. (2000). Victimized by victims: A taxonomy of antecedents of false complaints against psychotherapists. *Professional Psychology: Research and Practice*, *31*(1), 75. DOI: 10.1037/0735-7028.31.1.75.

Wolfgang, A. S., Fonzo, G. A., Gray, J. C., Krystal, J. H., Grzenda, A., Widge, A. S., Kraguljac, N. V., McDonald, W. M., Rodriguez, C. I., & Nemeroff, C. B. (2025). MDMA and MDMA-assisted therapy. *American Journal of Psychiatry*, *182*(1), 79–103. DOI: 10.1176/appi.ajp.20230681.

Woodhouse, S., Ayers, S., & Field, A. P. (2015). The relationship between adult attachment style and post-traumatic stress symptoms: A meta-analysis. *Journal of Anxiety Disorders*, *35*, 103–117. DOI: 10.1016/j.janxdis.2015.07.002.

World Health Organization. (2019). *International classification of diseases, eleventh revision (ICD-11)*. URL: icd.who.int/browse11.

Yazar-Klosinski, B. (2024). *Midomafetamine-assisted therapy (MDMA-AT) for treatment of post-traumatic stress disorder (PTSD)*. URL: fda.gov/media/179062/download.

Yeager-Cordial, E., Ison, J., Becker, R., Boyd, C., & Weinstock, M. (2022). Psychiatric manifestations of organic disease: Don't get fooled! *Journal of Urgent Care Medicine*, *16*(11), 11–15. URL: jucm.com/psychiatric-manifestations-of-organic-disease-dont-get-fooled/.

Yudkowsky, E. (2007). *Avoiding your belief's real weak points*. URL: lesswrong.com/posts/dHQkDNMhj692ayx78/avoiding-your-belief-s-real-weak-points.